Skateboard form (after 3 weeks)

- Working prototype with minimal functionality: demonstrable, runnable build implementing core MVP features.
- Backlog implementation traceability: explicit mapping of implemented backlog items to team members (who implemented what) and commits/pull requests or timestamps.
- Definition of done (DoD) applied: documented DoD used for each completed backlog item (e.g., code compiled, unit tests passing, peer review, updated docs); evidence that DoD was met.
- Sprint demo & acceptance evidence: stakeholder acceptance notes for the sprint producing the skateboard.

Skateboard reporting form			
*Note your initial minimal viable prototype (MVP) can either be a 'skateboard' (complete program) or a 'wheel' (only some modules implemented). In either case the code **must compile without error and run**. The run need not be interactive per sé, you can read in input files and write to logs as long as an observer can clearly see what the program is doing. The skateboard code and executable must be findable on your git repo.* *This prototype must meet the definition of done; i.e.: code compiles and runs without errors, unit tests with decent coverage pass. Security issues are either addressed or a plan to address them is in place.*			
Team details			
Project name:	MatchPoint		
Team nickname	TLDR		
Team members (first/last name)	Thomas Kabalin	Luke Bryant	Danny Gebhard
Student numbers	KBLTHO001	BRYLUK001	GBHDAN001
Project description 1 or 2 lines	MatchPoint is a tennis statistics tracking that lets coaches capture live match data and turn it into player performance analytics.		
Planned programming language(s)	Languages: TypeScript, Python, SQL Platform: Progressive Web App (React)		
Git repo url	https://gitlab.cs.uct.ac.za/tldr/matchpoint		
Demo session			
Fill in the blue fields prior to the demo, fill in the green fields during or shortly after the demo, make sure to assign a member to take written notes during the demo			

Link to executable	https://matchpoint.thomaskabalin.com/
Link to last commit	***Link to the last commit used in the demo, make sure your commits are up to date - TBD***
List of backlog items implemented	• Data Model • Player, Match, and Point APIs • Accounts and Access • Scoring Engine • Match Data Capture • Statistics and Analytics • Offline Capture • Home Page & App Navigation Flow • Test Data Generation
Demo date	04/09/2026

Backlog items implemented

For each implemented backlog item, copy-paste the black fields from the planning document. Do not change anything from that doc. Add implementation details in the blue fields.

Backlog item 1						
Name	**Data model**					
Description	Shippable increment: migrated PostgreSQL schema plus SQLAlchemy models and test factories					
Priority	High					
Main developer	Thomas Kabalin					
Pair programming?	0%					
Time spend	KBLTHO001	100%	BRYLUK001	0%	GBHDAN001	0%
Estimated time	0.5 weeks					
Acceptance criteria (DoD):						
Functionality	Schema and models must accurately represent coach, player, match, and point data, structured for use by the API layer. Must support input validation and access control at the database level to guard against SQL injection attacks and unauthorized data access					
Security issues	• Direct external access to the database bypassing the application's own access controls and business logic • Exposure of the database port to the internet or wider network, allowing unauthorised connection attempts • Credential reuse across environments, where a leak in one environment compromises others • Hardcoded credentials being exposed via source control history • SQL injection					

Test cases	• Given a migration is run, expect it to complete cleanly with no errors or data loss • Given the test factories are run, expect them to generate valid records for all entities (coach, player, match, point) • Given invalid or malformed input is submitted, expect it to be rejected at the model/DB layer with no injection-style input reaching raw queries • Given a request attempts direct/unauthenticated access to the database, expect it to be denied, with access restricted to the application layer only
Client remarks	
For **implemented** backlog items fill in the below fields prior to the demo	
Actual time	0.5 weeks
Link to last git commit	https://gitlab.cs.uct.ac.za/tldr/matchpoint/-/commit/e85a7ab17192a889f5c331b1e5181304f9d64ef0
Commit history	See Appendix C, Backlog Item 1
Acceptance criteria	Working PostgreSQL schema created through versioned Alembic migrations, covering all entities needed by the API layer (coaches, players, matches, match sets, points, coach sessions, email verifications, password resets, player transfer codes, player sessions, deleted-match tombstones). SQLAlchemy models mapping every table, with input validation and access constraints enforced at the database level. Test factories that generate valid records for all entities, alongside a schema-head test confirming migrations and models stay in sync and a round-trip test verifying every migration applies and rolls back cleanly.
DoD checklist	
Compile compiles	**Y**/N
Unit test coverage	All
Unit tests succeed	All
Unit tests report	See Appendix T, Backlog Item 1
Security done?	**Y**/N
Security actions?	• Access is restricted to the application layer • The database port is not exposed outside the Docker network

	• Credentials are unique per environment and never committed to the repository • By using an ORM, the possibility of SQL injection is largely reduced. Care has been taken to never bypass the ORM and allow raw SQL • Lint rules that run in CI enforce the "never bypass the ORM for raw SQL" convention
Changes made	The schema was fleshed out into a full 13 table data model. Additional features like session management, OTP verification, password resets, transfer codes, photo storage, and player accounts were built.
Backlog item 2	
Name	**Player, match and point APIs**
Description	As a coach, I want to manage players, create matches with their metadata, and have captured points accepted reliably.
Priority	High
Main developer	Thomas Kabalin
Pair programming?	0%
Time spend	KBLTHO001 100% BRYLUK001 0% GBHDAN001 0%
Estimated time	0.7 weeks
Acceptance criteria (DoD):	
Functionality	Must reliably create/read/update players and matches and accept point data with server-side validation. Every endpoint must require authentication and only expose a coach's own data.
Security issues	• Cross-site request forgery (a malicious site could cause a logged-in user's browser to issue requests) • Cross-site scripting (XSS) via injected or unauthorised scripts • Protocol downgrade and SSL-stripping attacks that force a connection back to plain HTTP • MIME-type sniffing that causes a browser to misinterpret and execute an uploaded file's content • Leakage of sensitive URL parameters to third-party sites via the Referrer header • Unauthorised access to browser features (camera, microphone, geolocation) by embedded or malicious scripts • Denial-of-service via oversized payloads, decompression bombs, or memory exhaustion during image processing • Secrets and credentials leaking into logs or error responses, and difficulty tracing an incident back to a specific request

	• Stack traces exposing file paths, library versions, or internal implementation details to an attacker • Brute-force login attempts, credential stuffing, and request-flooding denial-of-service • Time-of-check-to-time-of-use race conditions, where concurrent requests slip past a check meant to block them • Mass-assignment, where unexpected fields in a request body modify data the client shouldn't be able to touch • Theft, replay, or double redemption of transfer codes from a database breach or concurrent requests • Enumeration of other users' private photos by guessing or incrementing predictable filenames
Test cases	• Given valid player/match data is submitted, expect CRUD operations to succeed • Given an invalid point submission is made, expect the API to return a clear validation error • Given an unauthenticated request or a request for another coach's data, expect the API to reject it with a 401/403 response
Client remarks	
For **implemented** backlog items fill in the below fields prior to the demo	
actual time	1.5 weeks
Link to last git commit	https://gitlab.cs.uct.ac.za/tldr/matchpoint/-/commit/e85a7ab17192a889f5c331b1e5181304f9d64ef0
Commit history	See Appendix C, Backlog Item 2
Acceptance criteria	Working CRUD endpoints for players and matches, accessible only to the authenticated coach who owns them, alongside photo upload and player transfer management. Batch point ingestion endpoint accepting up to 500 points per request, with idempotent duplicate rejection so retried submissions are safe. Match lifecycle management including capture claims and deletion with tombstones, alongside a point read endpoint for retrieving captured data. Account closure endpoint that cascades deletes across all of a coach's data. Security hardening across all endpoints: CSRF origin checking, security headers (CSP, HSTS), request body size limits, rate

	limiting, structured logging with request IDs, and generic error responses that never expose internals.
DoD checklist	
Compile compiles	**Y**/N
Unit test coverage	All
Unit tests succeed	All
Unit tests report	See Appendix T, Backlog Item 2
Security done?	Y
Security actions?	• CSRF origin-check middleware refuses state-changing requests from foreign origins • Security headers on every response (CSP, HSTS, X-Content-Type-Options, Referrer-Policy, Permissions-Policy) • Request body size capped at 1 MB; image uploads pixel-count bounded • Structured request logging with request IDs; secret redaction in 422 error bodies and log lines • Unhandled exception handler returns a generic JSON body, never a stack trace • Rate limiting: per-IP, per-account, and per-endpoint token buckets • Row-level locking before credential checks (TOCTOU defence) • Request validation refuses unknown keys in request bodies • Transfer codes hashed with SHA-256, single-use, atomic redemption • Photo filenames are unguessable 128-bit random keys
Changes made	The original 0.7-week estimate covered basic CRUD. Actual work grew to ~1.5 weeks as security hardening (CSRF, headers, rate limiting, logging), photo management, player transfer, account closure, player transfers, and player accounts, were built.
Backlog item 3	
Name	**Accounts And Access**
Description	As a coach, I want my own account and sign-in so that my players and match data are private to me and only reachable by me.
Priority	Middle
Main developer	Danny Gebhard

Pair programming?	*0%*					
Time spend	KBLTHO001	40%	BRYLUK001	0%	GBHDAN001	60%
Estimated time	0.7 weeks					
Acceptance criteria (DoD):						
Functionality	Must let a coach register, log in, and log out, with passwords hashed, sessions correctly isolating each coach's data, and multi-factor authentication supported to reduce the risk of broken authentication.					
Security issues	• Plaintext password exposure on database compromise • Brute-force cracking of stolen hashes • Credential-stuffing attacks using passwords already known to be compromised elsewhere • Weak, guessable passwords vulnerable to dictionary and pattern-based attacks • Account enumeration, where an attacker maps which emails are registered for phishing or targeted credential stuffing • A timing side-channel that reveals whether an email exists based on response time • Session hijacking via reuse of tokens leaked in a database breach • OTP brute-forcing, OTP reuse from a leaked database, and stale codes remaining usable long after issue • Unauthorised account deletion via a hijacked or unattended session • Residual sensitive data being readable by the next user of a shared or public device					
Test cases	• Given valid registration/login details are submitted, expect the account to be created/logged in successfully; given invalid credentials, expect them to be rejected • Given a password is stored, expect it to be hashed and never stored in plaintext • Given Coach A attempts to access Coach B's data, expect the request to be denied					
Client remarks						
For **implemented** backlog items fill in the below fields prior to the demo						
Actual time	1.5 weeks (Danny ~0.75 weeks frontend, Thomas ~0.75 weeks backend)					
Link to last git commit	https://gitlab.cs.uct.ac.za/tldr/matchpoint/-/commit/ed348c7919697334bfe10a3678483deef0b2c5aa					
Commit history	See Appendix C backlog item 3					

Acceptance criteria	• Coach sign-up with email verification (6-digit OTP), sign-in, sign-out, and forgot-password flows, all rate-limited with 429 and Retry-After when limits are exceeded. • Passwords hashed with Argon2id, screened against the HIBP breach list, and validated by a NIST-compliant policy (repetition, sequence, common-word, and context checks). A client-side strength meter blocks weak passwords before submission. • Sessions use opaque tokens stored in HttpOnly cookies with renewal on use, keeping the coach signed in across visits without exposing the token to JavaScript. • Route guards redirect unauthenticated users to sign-in. Account enumeration is defended: all unauthenticated endpoints handle known and unknown identities with identical status codes, bodies, and rate limits (except sign-up OTP request returning a 409 error for already registered emails). • Settings pages for changing password, signing out other sessions, configuring match defaults, toggling appearance, and closing the account (with password re-entry). Profile editing covers display name and profile photo.
DoD checklist	
Compile compiles	**Y**
Unit test coverage	All (API), Some (web - auth screen covered, settings partially)
Unit tests succeed	All
Unit tests report	Refer to appendix T backlog item 3
Security done?	**Y**
Security actions?	• Passwords hashed with Argon2id (t=3, m=64 MiB, p=1), never stored in plaintext • HIBP breach-list screening rejects compromised passwords • Password policy: repetition, sequence, common-word, and context checks (NIST SP 800-63B) • Account enumeration defense: all unauthenticated endpoints respond identically for known and unknown emails • Dummy Argon2 verification on unknown-email login (timing defense) • Session tokens SHA-256 hashed in the database, never stored in plaintext

	• OTP codes SHA-256 hashed, attempt-capped (5 attempts), 10-minute expiry • Account closure requires password re-entry; sign-out purges local data
Changes made	Danny was planned as the sole developer; in practice Thomas built the entire auth backend (password hashing, sessions, OTP, rate limiting, password policy, email sending) while Danny built the frontend UI. MFA (mentioned in the original acceptance criteria) was dropped. The team decided that OTP email verification at sign-up, rate-limited login, and the NIST-compliant password policy provided sufficient security for the use case without the UX friction of per-login MFA.

Backlog item 4						
Name	**Scoring Engine**					
Description	As a coach, I want the app to work out the score from the points I record so that I never track games and sets by hand.					
Priority	High					
Main developer	Luke Bryant					
Pair programming?	0%					
Time spend	KBLTHO001	0%	BRYLUK001	100%	GBHDAN001	0%
Estimated time	0.5 weeks					
Acceptance criteria (DoD):						
Functionality	Must correctly calculate game/set/match scores from recorded points, matching standard scoring rules exactly (verified against known scenarios like deuce/tiebreak).					
Security issues	• *A client-computer score being trusted as truth* • *Malformed or hostile data reaching the fold* • *Cross-match contamination* • *Injection through free-text carried alongside the score*					
Test cases	• Given a standard sequence of points is recorded, expect the game/set/match score to be calculated correctly • Given an edge case scenario (deuce, tiebreak, advantage) occurs, expect the score to resolve correctly per standard rules					
Client remarks						
For **implemented** backlog items fill in the below fields prior to the demo						
actual time	*1.5*					

Link to last git commit	*https://gitlab.cs.uct.ac.za/tldr/matchpoint/-/commit/b53f8e6c688f456c7773612768ab44fb8a88baf9*
Commit history	*See Appendix C backlog item 4*
Acceptance criteria	• *Game, set and match scores are calculated from the recorded point, with deuce, advantage, tiebreak and match-tiebreak situations resolving correctly* • *The engine supports seven scoring formats: traditional men's Grand slam, no-ad, short sets, 10-point match tiebreak, pro set and Legace* • *The score is reconstructed from the match's recorded events each time, ensuring that undoing actions or reopening a match produces the same score without maintaining a separate score state* • *A coach can enter a score directly for a passage of play they did not observe, without the app fabricating the individual points that produced it* • *Matches ending through retirement walkover or default are handled correctly including case where the winner is not player leading on sets* • *Service rotation is calculated by the app*
DoD checklist	
Compile compiles	*Y*
Unit test coverage	*All*
Unit tests succeed	*All*
Unit tests report	*Refer to appendix T backlog item 4*
Security done?	*y*
Security actions?	• *UI enforces input so that a user does not do something the system cant handle* • *Every point and match checks its own fields as it constructed. Anything that fails is thrown out rather than scored.* • *Each match has a unique match ID and the points are related to the match ID so that they can be differentiated* • *Notes render through Reacts escaping, getting treated as a markup instead of text, and text input is reduced to only necessary prompts*
Changes made	• *The plan assumed the score could always be derived by replaying a complete, contiguous list of points. This is*

	not possible in reality as there are always distractions court side • *An earlier "skip ahead" feature synthesised the points behind a jumped-to score, and was removed for a more flexible system* • *Matches that end outside the scoring rules were unrepresentable in the planned design* • *Flat point and match classes for ease of editing*

Backlog item 5						
Name	**Match Data Capture**					
Description	As a coach at the court, I want to set up a match in a few taps and then record every point in no more than three taps so that the whole courtside workflow keeps up with live play.					
Priority	Middle					
Main developer	Luke Bryant					
Pair programming?	0%					
Time spend	KBLTHO001	0%	BRYLUK001	100%	GBHDAN001	0%
Estimated time	0.9 weeks					
Acceptance criteria (DoD):						
Functionality	Must let a coach set up a match easily and record each point in no more than 3 taps accurately and in real time. Access must be restricted to the authenticated coach's own match					
Security issues	• *Data loss through a crash in the scoring path* • *Unauthorised access to match data* • *The note field is uncontrolled text input*					
Test cases	• Given a coach sets up a new match, expect setup to be completed in ≤3 taps • Given a coach records a point, expect it to be completed in ≤3 taps • Given points are entered rapidly in succession, expect each point to be correctly attributed with no duplicates • The UI should not allow the coach to enter a state that is impossible as per the rules of tennis (e.g. asking for a forced / unforced error classification when the player has not yet served)					
Client remarks						
For **implemented** backlog items fill in the below fields prior to the demo						
actual time	*1 week*					

Link to last git commit	*https://gitlab.cs.uct.ac.za/tldr/matchpoint/-/commit/525c00b774d1b0bf81f25db9d64811c5d9e972ce*
Commit history	*Refer to appendix T backlog item 5*
Acceptance criteria	• *A match can be set up and started in few taps using the coach's saved scoring-system and surface defaults* • *Points can be entered rapidly without creating duplicates or assigning points to the wrong match* • *The app does not ask questions that are irrelevant to the rules and the state of the game* • *Skips are available for un seen data* • *Coach notes can be dictated or typed directly from the capture header without leaving the match* • *Back buttons are used available to be able to change mis clicks*
DoD checklist	
Compile compiles	*Y*
Unit test coverage	*All*
Unit tests succeed	*All*
Unit tests report	*Refer to appendix T backlog item 5*
Security done?	*y*
Security actions?	• *Every point is stored locally so if the application crashes before syncing to server the match data is rebuilt from the local log before data capture is continued* • *Only approved coaches have access to their own students* • *Not gets stored as mark up through reacts auto safety feature*
Changes made	• *Changed the strict following of recording 3 clicks per point* • *Coach notes where added* • *Changed from recording every shot to recording the beginning and end of a point*
Backlog item 6	
Name	**Statistics And Analytics**
Description	As a coach, I want the statistics computed from captured points, shown as a post-match summary and as a dashboard of trends across matches and opponents.
Priority	Low

Main developer	Danny Gebhard					
Pair programming?	0%					
Time spend	KBLTHO001	30%	BRYLUK001	0%	GBHDAN001	70%
Estimated time	0.7 weeks					
Acceptance criteria (DoD):						
Functionality	Must accurately compute post-match summaries and cross-match trend dashboards from captured data, visible only to the coach who owns that data.					
Security issues	• Unauthorised access to other coaches' player data • Unauthorised creation, modification, or deletion of player data through these endpoints					
Test cases	• Given match data has been captured, expect the post-match summary to match manually calculated values • Given data exists across multiple matches/opponents, expect the dashboard to correctly display and aggregate trends • Given a coach views their statistics, expect them to see only their own data and not another coach's • For each statistic, there should be unit tests to ensure that it is computed correctly, even with edge-case data					
Client remarks						
For **implemented** backlog items fill in the below fields prior to the demo						
actual time	0.75 weeks (Danny ~0.5 weeks frontend, Thomas ~0.25 weeks backend)					
Link to last git commit	https://gitlab.cs.uct.ac.za/tldr/matchpoint/-/commit/f72a0c43ead6cbe03e9f3882f44e7747bdbe741a					
Commit history	See Appendix C backlog item 6					
Acceptance criteria	• Player profile with serve and rally statistics tabs, stat cards with percentage formatting, and breakdown bars visualising categorical distributions. Coach notes accessible per player via a dedicated view. • Squad stats page with a headline card, leaderboard, and trend chart across the roster. • A consistent colour palette differentiating stat categories in both light and dark mode. • Read-only API endpoints computing analytics from captured points scoped to the authenticated coach's data.					

	• Frontend components wired to the live analytics endpoints - current data is test data generated by Claude.
DoD checklist	
Compile compiles	**Y**
Unit test coverage	Some (frontend stats tabs partially covered; API endpoints fully covered)
Unit tests succeed	All
Unit tests report	Refer to appendix C backlog item 6
Security done?	Y
Security actions?	• Endpoints scoped to authenticated coach's own data • Analytics endpoints are read-only
Changes made	A visual redesign of the drill-down layout was done on Aug 23 after stakeholder feedback. Thomas built the backend analytics endpoints (not in the original plan, which had Danny as the sole developer).
Backlog item 7	
Name	**Offline Capture**
Description	As a coach on a court with no signal, I want capture to keep working and sync when connectivity returns so that I never lose a match's data.
Priority	Low
Main developer	Thomas Kabalin
Pair programming?	0%
Time spend	KBLTHO001 90% BRYLUK001 10% GBHDAN001 0%
Estimated time	0.7 weeks
Acceptance criteria (DoD):	
Functionality	Must keep recording points with no connection, then sync accurately with no data loss once connectivity returns, without exposing cached data to other apps/users.
Security issues	• Device fingerprinting through client-generated identifiers • Cross-user data exposure on a shared device after sign-out • A malicious client submitting crafted IDs to collide with or predict another user's resources • Unsynced data remaining on a device after its session is revoked for a security reason

	• Fabricated or tampered point data submitted in bulk during sync
Test cases	• Given the device has no network connection, expect the app to continue recording points successfully • Given connectivity returns after offline recording, expect queued data to sync correctly with no loss or duplication
Client remarks	
For **implemented** backlog items fill in the below fields prior to the demo	
actual time	0.75 weeks
Link to last git commit	https://gitlab.cs.uct.ac.za/tldr/matchpoint/-/commit/c75d1fd2cb63fa4064ae549123f3a2a54850efdb
Commit history	See Appendix C, Backlog Item 7
Acceptance criteria	Working offline point recording using a local IndexedDB outbox that persists across page reloads and app restarts, so capture keeps working with no network connection. Automatic sync to the server when connectivity returns, with idempotent ingestion ensuring no data loss or duplication. A capture claim mechanism preventing two devices from recording the same match simultaneously. Tombstone handling so that points queued for a match deleted elsewhere are discarded cleanly on sync. The outbox isolated in its own IndexedDB database, separate from the read cache, so that clearing or evicting cached data can never lose unsynced points.
DoD checklist	
Compile compiles	**Y**/N
Unit test coverage	All
Unit tests succeed	All
Unit tests report	See Appendix T, Backlog Item 7
Security done?	Y
Security actions?	• Authenticated, coach-scoped access: every sync endpoint requires a session cookie; matches are filtered

	by coach ownership so one coach can never read or write another's data. • CSRF origin checking: non-safe requests are refused unless the Origin header matches the server's Host, blocking cross-site request forgery against the sync endpoints. • Idempotent ingestion: point upserts use ON CONFLICT DO NOTHING on client-minted UUIDs and all-or-nothing batches, so retried or overlapping offline flushes are safe by design. • Row-level locking against race conditions: every read-then-write path (capture claim, ingest, delete) takes a FOR UPDATE lock, preventing TOCTOU races between concurrent devices. • Capture claim with time-bounded locking: only one device can hold a match's capture claim at a time (4-hour TTL, FOR UPDATE arbitration), preventing conflicting live entry. • Request size and batch limits: 1 MB body cap enforced at both proxy and application level; batches capped at 500 points; the client splits flushes under the limit. • Client-side outbox isolation: unsynced points live in a separate IndexedDB database from the read cache, so clearing or evicting cached data can never lose unsynced points; 401 halts the drain without deleting the outbox.
Changes made	Scope expanded beyond the planned "record and sync" to include: a two-database architecture (outbox vs read cache), a capture claim mechanism to prevent concurrent device conflicts, tombstone handling for deleted matches, and full data lifecycle management

Backlog item 8						
Name	**Home Page & App Navigation Flow**					
Description	As a coach, I want a home page that gives me a clear overview when I log in, with easy navigation to my players, matches, and stats, so that I can quickly get to what I need without hunting through menus.					
Priority	Middle					
Main developer	Danny Gebhard					
Pair programming?	0%					
Time spend	KBLTHO001	0%	BRYLUK001	0%	GBHDAN001	100%
Estimated time	0.5 weeks					
Acceptance criteria (DoD):						

Functionality	Must present an authenticated coach with a home page summarizing key info (e.g., recent matches, quick actions) and provide clear, consistent navigation to all core app pages (players, matches, stats, account), restricted to that coach's own data only.
Security issues	• Unauthorised or unauthenticated access to data via the API • Unauthenticated users reaching protected pages or views directly by URL • Enumeration of valid application routes or resources via differing 404 versus authenticated-page behaviour
Test cases	• Given an authenticated coach logs in, expect the home page to load and display only that coach's own data • Given a coach with recent matches recorded, expect the home page to display a summary of those recent matches • Given a coach taps/clicks a navigation link (e.g., Players, Matches, Stats), expect the app to route to the correct corresponding page • Given an unauthenticated user attempt to access the home page directly (e.g., via URL), expect the app to redirect them to the login page • Given a coach navigates between core pages (Players, Matches, Stats, Settings), expect all links to work with no dead ends or broken routes • Given a coach on a typical mobile connection loads the home page, expect it to fully load within an acceptable time (e.g., ≤2 seconds)
Client remarks	
For **implemented** backlog items fill in the below fields prior to the demo	
actual time	1 week
Link to last git commit	https://gitlab.cs.uct.ac.za/tldr/matchpoint/-/commit/84495dc3fa016c03cd0a62316e29c994254cfe1d
Commit history	See Appendix C backlog item 8
Acceptance criteria	• Home page showing the coach's display name, squad snapshot (player and match counts), a last-viewed player card, a player bubble row with avatars, a player search bar, and an add-player tile. • Bottom navigation with four tabs (Home, Players, Matches, Stats) providing one-tap access to every area of the app. • Player roster page with list, add, edit, and delete actions. Match list page with add and delete actions.

	• Profile editing for display name and profile photo, with the MatchPoint logo mark in the app header. • Dark mode toggle, a custom 404 page for unmatched routes, and a PWA install guide with platform-specific instructions for iOS, Android, and desktop. • Route guards redirect unauthenticated users to sign-in; all data fetched through authenticated API calls.
DoD checklist	
Compile compiles	**Y**
Unit test coverage	Some
Unit tests succeed	All
Unit tests report	Refer to appendix T backlog item 8
Security done?	Y
Security actions?	• All data fetched through authenticated API calls. • Route guards redirect unauthenticated users to sign-in. • The 404 page does not leak information about valid routes.
Changes made	Scope expanded to include player management (add/edit/delete), match list management, profile editing for coaches and players, dark mode, PWA install guide, and a 404 page. These are navigation-adjacent features that naturally live in the app shell layer. The home page was rebuilt visually to match updated wireframes. Danny was the sole developer on this item.
Backlog item 9	
Name	**Test data generation**
Description	As a developer, I want realistic seed/test data for coaches, players, matches, and points so that the team can properly test functionality, edge cases, and performance without relying on manual data entry.
Priority	Middle
Main developer	Luke Bryant
Pair programming?	0%
Time spend	KBLTHO001 20% BRYLUK001 80% GBHDAN001 0%
Estimated time	0.4 weeks
Acceptance criteria (DoD):	
Functionality	

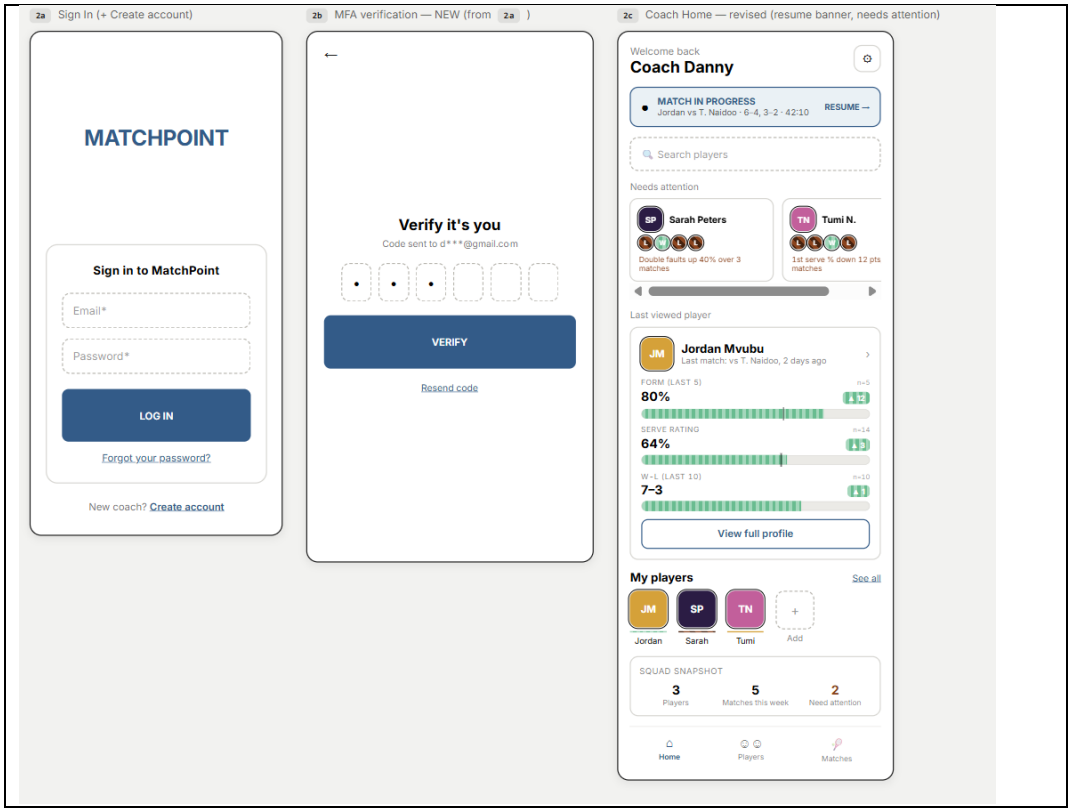
Security issues	• Exposure of a privileged or destructive operation to unauthenticated or unauthorised users via a public API • Accidental or malicious invocation of the test data injection script against a production environment, corrupting or polluting real data
Test cases	• Given the test data script is run, expect it to populate the database with valid coach, player, match, and point records with no errors • Given generated match data, expect it to include edge case scenarios (e.g. deuce, tiebreak, double faults) so scoring logic can be fully tested • Given generated match data, expect it to include edge case scenarios (e.g. deuce, tiebreak, double faults) so scoring logic can be fully tested • Given the test data script is run in a non-production environment, expect it to have no effect on production data • Given the test data script is run multiple times, expect it to not create duplicate or conflicting records
Client remarks	
For **implemented** backlog items fill in the below fields prior to the demo	
actual time	0.10 weeks (Thomas) and 0.10 weeks (Luke)
Link to last git commit	*https://gitlab.cs.uct.ac.za/tldr/matchpoint/-/commit/83709b28c0bd551866cd8ef468533b80a6ad9230*
Commit history	See Appendix C, Backlog Item 9
Acceptance criteria	• *No points are manually constructed, every point is generated by driving the real capture flow through simulated taps, while scorelines are calculated by the real scoring engine* • *Every generated point is validated against the API's request schema before being written, ensuring that generated data can be accepted by the import process* • *the generator produces two output formats: the capture model's own columns for inspection, and a four-file format that can be consumed directly by the API import script* • *players are distributed amongst multiple coaches* • *the import script is not exposed as an API endpoint and can only be executed with server access*
DoD checklist	
Compile compiles	**Y**

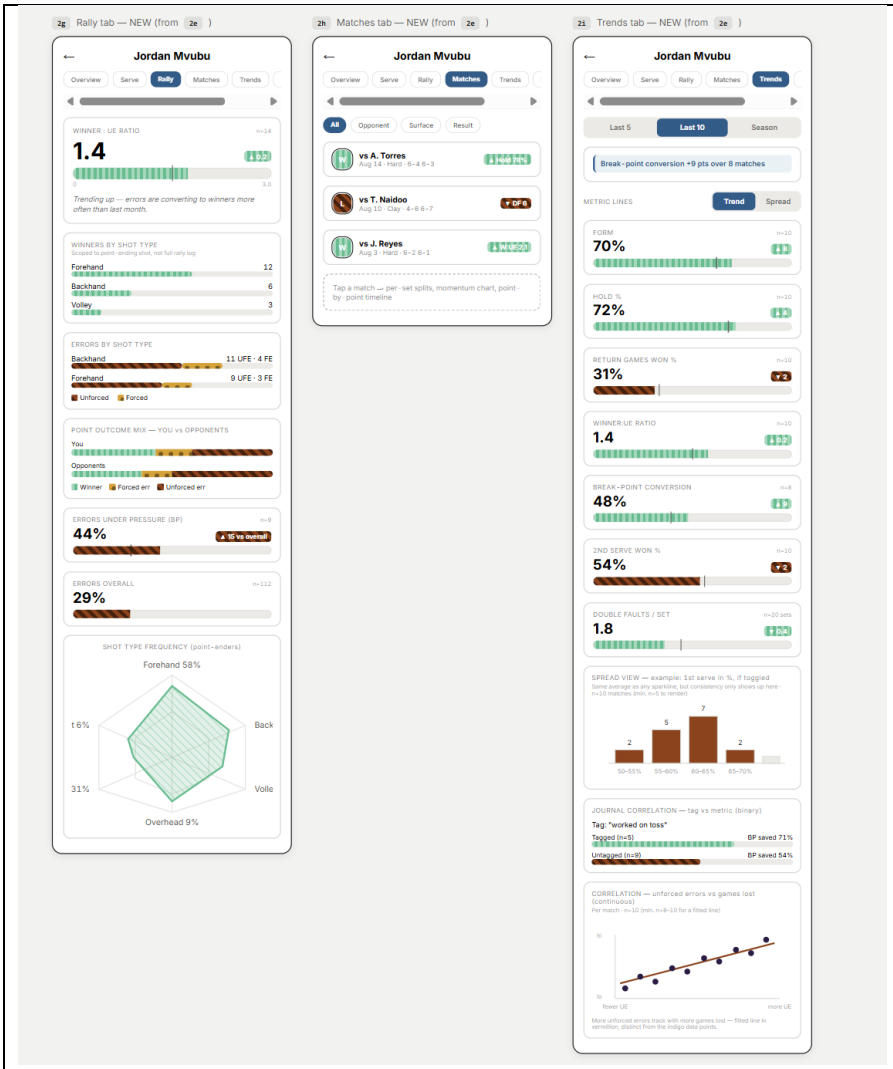
Unit test coverage	All
Unit tests succeed	All
Unit tests report	See Appendix T, Backlog Item 9
Security done?	**Y**
Security actions?	• The script is not exposed as an API endpoint • Running it in production requires SSH access to the VPS, which only Thomas, the developer in charge of DevOps has, adhering to the principle of least privilege
Changes made	*What was actually in the data was determined by the final match capture system so change according to changes on what data we captured*
GenAI declaration	
Did you use GenAI	**Y**/N
Describe use	**Danny:** Claude Sonnet 5 was used to create all code (and test cases) in backlog items 3, 6, and 8. Claude Opus 5 was used for more complex planning to deal with issues that were run into during the building process for these same backlog items. This was allowed under instruction from the client. **Thomas:** Various Claude models were used within Claude Code to assist with planning and writing the code for my backlog items (1, 2, and 7), as well as my parts of other backlog items (3, 6, and 9). Planning was an iterative process conducted in collaboration with GenAI (Claude). All code in my sections was written by GenAI (as allowed by the client), under varying levels of autonomy depending on the task. This ranged from Claude implementing a GitLab issue independently from start to finish, to closely supervised collaboration in which I directed each step of the implementation. Every AI-generated change was submitted as a GitLab merge request. I manually reviewed each merge request before it was

	merged, and no code was merged into the codebase without my review and explicit approval. Below are counts of how many commits each model was used for: Claude Opus 5 171 Claude Fable 5 82 Claude Sonnet 5 24 Claude Opus 4.6 14 Claude Opus 4.8 2 In this document, GenAI was used to analyse the codebase and surface actions that have been taken in order to prevent security issues, and list places where the implementation has deviated from the original plan. **Luke:** Claude Sonnet 5 and Claude Opus 5 were used for assistance in coding and generating all test cases. They were both also used in Claude Design to create initial protypes for the capture UI. Claude Opus 5 was also used to help make the fake session dataset. Claude Fable 5 was used once in the generating of a solution plan for a bug we ran into which was then implemented by Claude Opus 5. **General** Claude Sonnet 5 was used to assist in the structuring and formatting of this document, as well as appendix C and T.

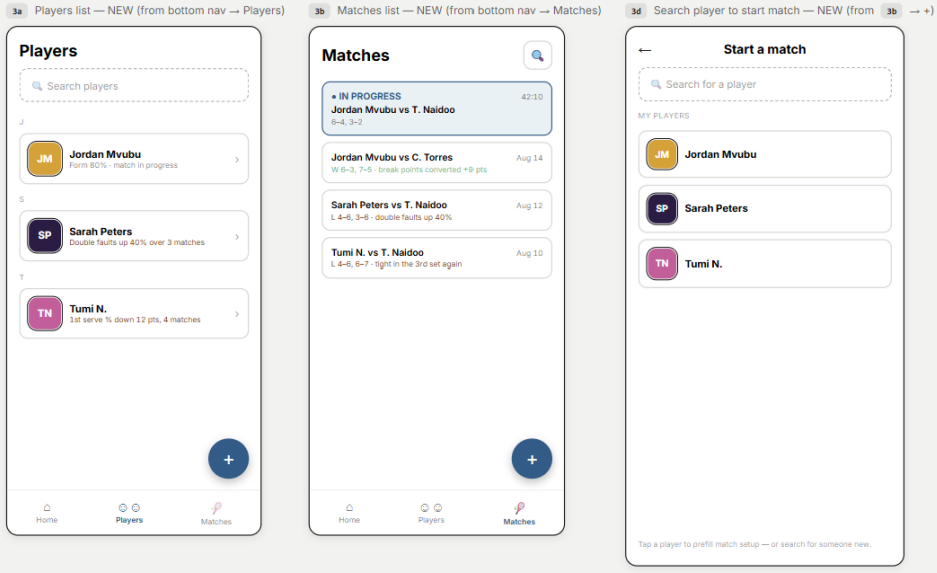
The second iteration of the app layout. Advice taken from client to make this from first iteration:

(1) colours just have distinct purpose throughout the app;
(2) colours should have varied luminance to each other to ensure readability in black and white;
(3) colours should be chosen to avoid colour blind issues ;
(4) colours should include unique patterns inside to assist in differentiation;
(5) varied chart designs for displaying data rather than just numbers.


2a Sign In (+ Create account)
MATCHPOINT
Sign in to MatchPoint
Email*
Password*
LOG IN
Forgot your password?
New coach? Create account
2b MFA verification — NEW (from 2a)
Verify it's you
Code sent to d***@gmail.com
VERIFY
Resend code
2c Coach Home — revised (resume banner, needs attention)
Welcome back
Coach Danny
MATCH IN PROGRESS
Jordan vs T. Naidoo · 6-4, 3-2 · 42:10
RESUME →
Search players
Needs attention
Sarah Peters
Double faults up 40% over 3 matches
Tumi N.
1st serve % down 12 pts matches
Last viewed player
Jordan Mvubu
Last match: vs T. Naidoo, 2 days ago
FORM (LAST 5)
80%
SERVE RATING
64%
W-L (LAST 10)
7–3
View full profile
My players
See all
Jordan
Sarah
Tumi
Add
SQUAD SNAPSHOT
3 Players
5 Matches this week
2 Need attention
Home
Players
Matches



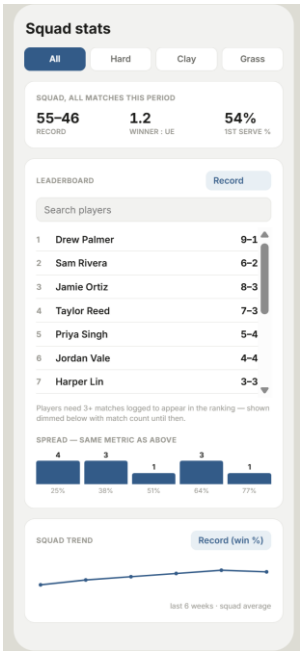
3 MatchPoint — iteration 3: Players list, Matches list, player-tap popup

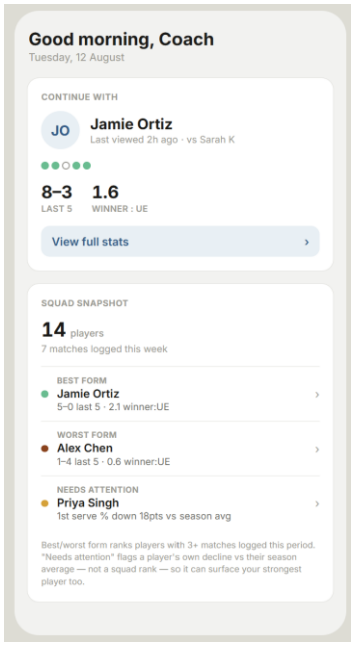


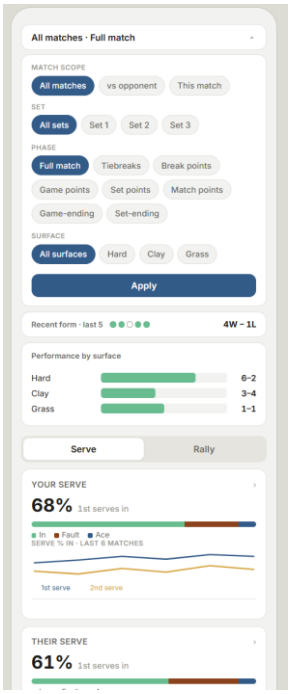
The third iteration of the app layout. Advice taken from client to make this from second iteration:

(1) too many pages is distracting to users - fixed with filters and drill down menus ;
(2) flow should always start with high level summary and then expands through more app exploration.

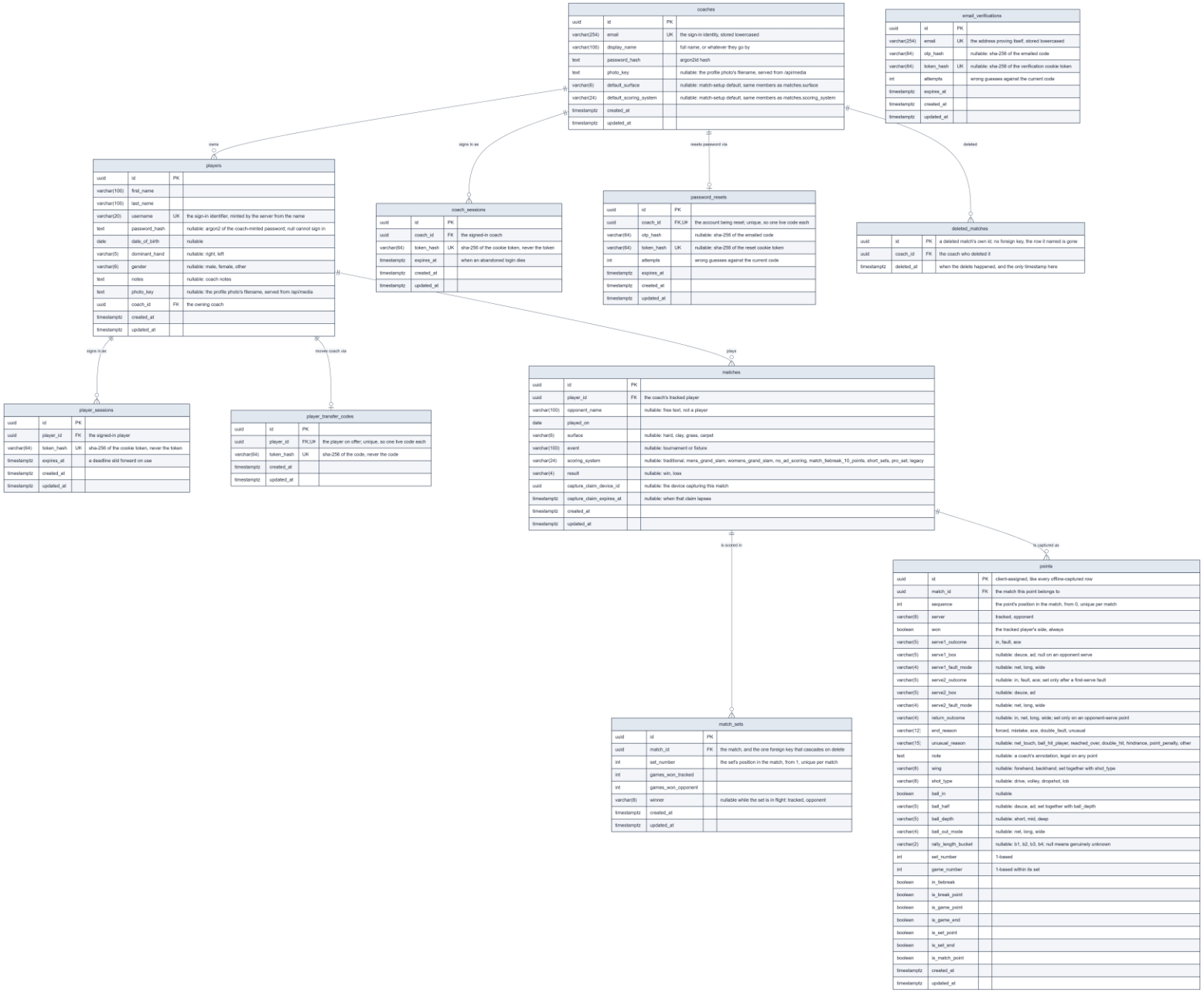
Note - these diagrams were not required by the client but were made after advice from the second iteration to assist in production.







ERD Diagram


coaches
uuid | id | PK |
varchar(254) | email | UK | the sign-in identity, stored lowercased
varchar(100) | display_name | | full name, or whatever they go by
text | password_hash | | argon2id hash
text | photo_key | | nullable: the profile photo's filename, served from /api/media
varchar(6) | default_surface | | nullable: match-setup default, same members as matches.surface
varchar(24) | default_scoring_system | | nullable: match-setup default, same members as matches.scoring_system
timestamptz | created_at | |
timestamptz | updated_at | |
email_verifications
uuid | id | PK |
varchar(254) | email | UK | the address proving itself, stored lowercased
varchar(64) | otp_hash | | nullable: sha-256 of the emailed code
varchar(64) | token_hash | UK | nullable: sha-256 of the verification cookie token
int | attempts | | wrong guesses against the current code
timestamptz | expires_at | |
timestamptz | created_at | |
timestamptz | updated_at | |
owns
signs in as
resets password via
deleted
players
uuid | id | PK |
varchar(100) | first_name | |
varchar(100) | last_name | |
varchar(20) | username | UK | the sign-in identifier, minted by the server from the name
text | password_hash | | nullable: argon2 of the coach-minted password; null cannot sign in
date | date_of_birth | | nullable
varchar(5) | dominant_hand | | nullable: right, left
varchar(6) | gender | | nullable: male, female, other
text | notes | | nullable: coach notes
text | photo_key | | nullable: the profile photo's filename, served from /api/media
uuid | coach_id | FK | the owning coach
timestamptz | created_at | |
timestamptz | updated_at | |
coach_sessions
uuid | id | PK |
uuid | coach_id | FK | the signed-in coach
varchar(64) | token_hash | UK | sha-256 of the cookie token, never the token
timestamptz | expires_at | | when an abandoned login dies
timestamptz | created_at | |
timestamptz | updated_at | |
password_resets
uuid | id | PK |
uuid | coach_id | FK,UK | the account being reset; unique, so one live code each
varchar(64) | otp_hash | | nullable: sha-256 of the emailed code
varchar(64) | token_hash | UK | nullable: sha-256 of the reset cookie token
int | attempts | | wrong guesses against the current code
timestamptz | expires_at | |
timestamptz | created_at | |
timestamptz | updated_at | |
deleted_matches
uuid | id | PK | a deleted match's own id; no foreign key, the row it named is gone
uuid | coach_id | FK | the coach who deleted it
timestamptz | deleted_at | | when the delete happened, and the only timestamp here
signs in as
moves coach via
plays
player_sessions
uuid | id | PK |
uuid | player_id | FK | the signed-in player
varchar(64) | token_hash | UK | sha-256 of the cookie token, never the token
timestamptz | expires_at | | a deadline slid forward on use
timestamptz | created_at | |
timestamptz | updated_at | |
player_transfer_codes
uuid | id | PK |
uuid | player_id | FK,UK | the player on offer; unique, so one live code each
varchar(64) | token_hash | UK | sha-256 of the code, never the code
timestamptz | created_at | |
timestamptz | updated_at | |
matches
uuid | id | PK |
uuid | player_id | FK | the coach's tracked player
varchar(100) | opponent_name | | nullable: free text, not a player
date | played_on | |
varchar(6) | surface | | nullable: hard, clay, grass, carpet
varchar(100) | event | | nullable: tournament or fixture
varchar(24) | scoring_system | | nullable: traditional, mens_grand_slam, womens_grand_slam, no_ad_scoring, match_tiebreak_10_points, short_sets, pro_set, legacy
varchar(4) | result | | nullable: win, loss
uuid | capture_claim_device_id | | nullable: the device capturing this match
timestamptz | capture_claim_expires_at | | nullable: when that claim lapses
timestamptz | created_at | |
timestamptz | updated_at | |
is scored in
is captured as
match_sets
uuid | id | PK |
uuid | match_id | FK | the match, and the one foreign key that cascades on delete
int | set_number | | the set's position in the match, from 1, unique per match
int | games_won_tracked | |
int | games_won_opponent | |
varchar(8) | winner | | nullable while the set is in flight: tracked, opponent
timestamptz | created_at | |
timestamptz | updated_at | |
points
uuid | id | PK | client-assigned, like every offline-captured row
uuid | match_id | FK | the match this point belongs to
int | sequence | | the point's position in the match, from 0, unique per match
varchar(8) | server | | tracked, opponent
boolean | won | | the tracked player's side, always
varchar(5) | serve1_outcome | | in, fault, ace
varchar(5) | serve1_box | | nullable: deuce, ad; null on an opponent serve
varchar(4) | serve1_fault_mode | | nullable: net, long, wide
varchar(5) | serve2_outcome | | nullable: in, fault, ace; set only after a first-serve fault
varchar(5) | serve2_box | | nullable: deuce, ad
varchar(4) | serve2_fault_mode | | nullable: net, long, wide
varchar(4) | return_outcome | | nullable: in, net, long, wide; set only on an opponent-serve point
varchar(12) | end_reason | | forced, mistake, ace, double_fault, unusual
varchar(15) | unusual_reason | | nullable: net_touch, ball_hit_player, reached_over, double_hit, hindrance, point_penalty, other
text | note | | nullable: a coach's annotation, legal on any point
varchar(8) | wing | | nullable: forehand, backhand; set together with shot_type
varchar(8) | shot_type | | nullable: drive, volley, dropshot, lob
boolean | ball_in | | nullable
varchar(5) | ball_half | | nullable: deuce, ad; set together with ball_depth
varchar(5) | ball_depth | | nullable: short, mid, deep
varchar(4) | ball_out_mode | | nullable: net, long, wide
varchar(2) | rally_length_bucket | | nullable: b1, b2, b3, b4; null means genuinely unknown
int | set_number | | 1-based
int | game_number | | 1-based within its set
boolean | in_tiebreak | |
boolean | is_break_point | |
boolean | is_game_point | |
boolean | is_game_end | |
boolean | is_set_point | |
boolean | is_set_end | |
boolean | is_match_point | |
timestamptz | created_at | |
timestamptz | updated_at | |

Appendix C

Backlog Item 1

Date	Hash	Author	MR	Commit message
2026-08-08	bb8f8fb	Thomas Kabalin	!5	feat(api): add player model and initial migration
2026-08-08	1eb9983	Thomas Kabalin	!5	test(api): add transaction-rollback postgres fixtures
2026-08-08	9e04f42	Thomas Kabalin	!6	feat(api): add match model and migration
2026-08-10	3a8f0cc	Thomas Kabalin	!5	refactor(api): mirror the test tree onto the source layout
2026-08-10	8c4cf6d	Thomas Kabalin	!6	test(api): pin the match model's design decisions
2026-08-10	f9f4974	Thomas Kabalin	!6	docs: correct why str_enum avoids native Postgres enums
2026-08-04	17bc19d	Thomas Kabalin		feat(api): add the sqlalchemy engine and alembic setup
2026-08-11	b46daa9	Thomas Kabalin	!7	feat(api): add coach model and migration
2026-08-11	00b2c79	Thomas Kabalin	!7	test(api): pin the coach model's design decisions
2026-08-11	126f00d	Thomas Kabalin	!7	docs: add an ERD and the schema conventions behind it
2026-08-11	98c6132	Thomas Kabalin	!7	feat(api): store coach emails lowercased
2026-08-11	f71b898	Thomas Kabalin	!6	refactor(api): scope the match format field to the sets axis
2026-08-11	beec14a	Thomas Kabalin	!7	chore(api): drop the migrations versions placeholder
2026-08-11	3dd8a79	Thomas Kabalin	!9	feat(api): add the coach sign-up endpoint
2026-08-11	0be97bf	Thomas Kabalin	!9	test(api): cover coach sign-up
2026-08-11	a669583	Thomas Kabalin	!9	docs: record the password hashing choice
2026-08-11	00ec089	Thomas Kabalin	!10	feat(api): add the coach sessions table
2026-08-11	da724cf	Thomas Kabalin	!10	test(api): cover the coach sessions table
2026-08-11	8854143	Thomas Kabalin	!10	feat(api): add coach login and logout
2026-08-11	45c01be	Thomas Kabalin	!13	feat(api): scope players and matches to the authenticated coach
2026-08-11	8aa0791	Thomas Kabalin	!13	test(api): cover coach scoping on players and matches
2026-08-12	9d781b6	Thomas Kabalin	!24	feat(api): add the email verifications table
2026-08-12	90453eb	Thomas Kabalin	!24	test(api): cover the email verifications table

2026-08-13	bd9cc14	Thomas Kabalin	!28	feat(api): require a verified email to create a coach
2026-08-13	b9307e6	Thomas Kabalin	!32	feat(api): reset a forgotten password with an emailed code
2026-08-13	ed9d01f	Thomas Kabalin	!32	test(api): cover password reset end to end
2026-08-13	8435303	Thomas Kabalin	!36	docs: record the match deletion policy
2026-08-13	cc55be5	Thomas Kabalin	!35	docs: record that list endpoints do not paginate
2026-08-15	89412e7	Thomas Kabalin	!35	docs: record the timestamp source and archiving's reach
2026-08-15	863e1f4	Thomas Kabalin	!62	feat(api): record a match result
2026-08-15	d81c398	Thomas Kabalin	!77	feat(api): store per-coach match-setup defaults
2026-08-17	a35602c	Thomas Kabalin	!102	feat(api): store a scoring system instead of a sets format
2026-08-17	c64ab8b	Thomas Kabalin	!103	feat(api): verify a reset code without spending it
2026-08-17	de5f334	Thomas Kabalin	!111	feat(api): let a coach close their account
2026-08-17	922ec97	Thomas Kabalin	!111	docs: record account closure as the exception to soft deletion
2026-08-18	8881991	Thomas Kabalin	!111	feat(api): refuse an account closure while players are still active
2026-08-18	d1711bd	Thomas Kabalin	!111	docs: narrow the archiving rule to data inside an account
2026-08-18	b29487a	Thomas Kabalin	!131	feat(api): transfer a player to another coach via a transfer code
2026-08-18	577acd2	Thomas Kabalin	!132	feat(api): store which device is capturing a match
2026-08-19	5abefd2	Thomas Kabalin	!131	fix(api): refuse account closure while a player has a live transfer code
2026-08-19	f972828	Thomas Kabalin	!149	refactor(api): delete players outright instead of archiving them
2026-08-19	889d473	Thomas Kabalin	!150	feat(api): report whether a player has a live transfer code
2026-08-19	22f62c9	Thomas Kabalin	!156	feat(api): add photo_key to coaches and players
2026-08-19	5bf5165	Thomas Kabalin	!156	feat(api): upload and clear a player photo
2026-08-19	caebf03	Thomas Kabalin	!160	feat(api): add the derived match_sets table
2026-08-20	5ca0b57	Thomas Kabalin	!164	docs(data-model): record capture claim, transfer re-mint, and coach defaults rationale
2026-08-20	ab73ed4	Thomas Kabalin	!160	docs(api): name the set-closing stamp is_set_end, as #175 settled it

2026-08-20	95cd0f1	Thomas Kabalin	!170	feat(api): add the points table
2026-08-20	d211f74	Thomas Kabalin	!171	feat(api): validate a batch of captured points
2026-08-20	bed36a3	Thomas Kabalin	!173	docs: record the capture-claim refusal in the deletion convention
2026-08-20	0879923	Thomas Kabalin	!175	feat(api): give every player a unique generated username
2026-08-20	0760071	Thomas Kabalin	!175	feat(api): mint a one-time password when a player is created
2026-08-20	ed348c7	Thomas Kabalin	!175	feat(api): player_sessions table, and the deletes that must clear it
2026-08-20	c92951d	Thomas Kabalin	!175	feat(api): player login, logout, and their own /me on a player session
2026-08-21	bc1b942	Thomas Kabalin	!182	feat(api): delete a match along with its points
2026-08-21	30e7c8e	Thomas Kabalin	!182	feat(api): refuse to re-create a deleted match
2026-08-21	bea0def	Thomas Kabalin	!182	docs: record the new match-delete policy and its tombstone
2026-08-21	59b8396	Thomas Kabalin	!182	feat(api): erase a coach's deleted-match tombstones on account closure
2026-08-21	bc7f33b	Thomas Kabalin		docs: shorten data-model prose
2026-08-23	674280d	Thomas Kabalin	!188	refactor(api): drop serve1_target/serve2_target from points
2026-08-24	73b6295	Thomas Kabalin	!199	docs(data-model): what completing a match freezes
2026-08-24	2de8cca	Thomas Kabalin	!205	docs: draw the coach relationship on deleted matches in the ERD
2026-08-24	da8cb84	Thomas Kabalin	!207	refactor(api): split the sample-data import into run_import and main
2026-08-24	4d6d7e9	Thomas Kabalin	!207	fix(api): mint player usernames in the sample-data import
2026-08-24	5a1187d	Thomas Kabalin	!207	fix(api): report a bad matches.csv row by its line number
2026-08-24	707ed2d	Thomas Kabalin	!207	feat(api): make the sample-data import re-runnable
2026-08-24	4dbed59	Thomas Kabalin	!207	chore(api): measure and type-check the scripts directory
2026-08-24	2a8cb96	Thomas Kabalin	!207	fix(api): name the CSV row when a join between files finds nothing
2026-08-31	e85a7ab	Thomas Kabalin	!221	feat(api): store coach notes against a player
2026-08-31	aab06cd	Thomas Kabalin	!218	refactor(api): drop mid from the court depth enum

Backlog Item 2

Date	Hash	Author	MR	Commit message
2026-08-08	55e0c28	Thomas Kabalin	!5	feat(api): add players create/list/get endpoints
2026-08-10	3a8f0cc	Thomas Kabalin	!5	refactor(api): mirror the test tree onto the source layout
2026-08-10	2bc6dfa	Thomas Kabalin	!6	feat(api): add matches create/list/get endpoints
2026-08-10	563c304	Thomas Kabalin	!6	test(api): add unit tests for the match schema rules
2026-08-11	f71b898	Thomas Kabalin	!6	refactor(api): scope the match format field to the sets axis
2026-08-11	3dd8a79	Thomas Kabalin	!9	feat(api): add the coach sign-up endpoint
2026-08-11	7596f58	Thomas Kabalin	!9	test(api): cover 422 password redaction
2026-08-11	ea3d30a	Thomas Kabalin	!9	feat(api): redact submitted passwords from 422 bodies
2026-08-11	24453d3	Thomas Kabalin	!11	feat(api): rate limit coach sign-up
2026-08-11	83c0b3f	Thomas Kabalin	!11	test(api): cover the sign-up rate limit
2026-08-11	45c01be	Thomas Kabalin	!13	feat(api): scope players and matches to the authenticated coach
2026-08-11	8aa0791	Thomas Kabalin	!13	test(api): cover coach scoping on players and matches
2026-08-12	060abc9	Thomas Kabalin	!24	feat(api): add OTP email verification for sign-up
2026-08-13	52809ec	Thomas Kabalin	!27	feat(api): send sign-up codes through Resend
2026-08-13	138893a	Thomas Kabalin	!27	test(api): cover the Resend sender and its 502
2026-08-13	cd94dc5	Thomas Kabalin	!27	docs: record Resend in the tech stack and the VPS key step
2026-08-13	b9307e6	Thomas Kabalin	!32	feat(api): reset a forgotten password with an emailed code
2026-08-13	9a2162a	Thomas Kabalin	!36	feat(api): delete a match
2026-08-13	e21484c	Thomas Kabalin	!35	feat(api): archive and restore a player

2026-08-13	4f85b6f	Thomas Kabalin	!38	feat(api): update players and matches with PATCH
2026-08-14	e4a433a	Thomas Kabalin	!39	feat(api): refuse state-changing requests from a foreign origin
2026-08-15	12fdf95	Thomas Kabalin	!35	refactor(api): take the session first in owned_player
2026-08-15	ec8b3ca	Thomas Kabalin	!35	refactor(api): reuse owned_player in update_player
2026-08-15	5a1a857	Thomas Kabalin	!35	feat(api): refuse a new match against an archived player
2026-08-15	863e1f4	Thomas Kabalin	!62	feat(api): record a match result
2026-08-15	98ef7ff	Thomas Kabalin	!63	feat(api): set security headers on every API response
2026-08-15	d308e2e	Thomas Kabalin	!64	feat(api): bound request body size at 1 MB
2026-08-15	d27a7cc	Thomas Kabalin	!65	feat(api): log every request as a structured line with a request id
2026-08-15	ca4cb4e	Thomas Kabalin	!66	feat(api): bound an authenticated coach's request rate
2026-08-15	dfa604b	Thomas Kabalin	!67	feat(api): refuse unknown keys in request bodies
2026-08-15	0505578	Thomas Kabalin	!67	refactor(api): share the update loop as UpdateModel.apply_to
2026-08-15	0b52471	Thomas Kabalin	!67	refactor(api): reuse match_for_coach in update_match
2026-08-15	62c09fd	Thomas Kabalin	!78	test(api): freeze the limiter clock in bucket-draining tests
2026-08-15	629a7ac	Thomas Kabalin		style: drop the "load-bearing" phrase from comments and docstrings
2026-08-17	a35602c	Thomas Kabalin	!102	feat(api): store a scoring system instead of a sets format
2026-08-17	c64ab8b	Thomas Kabalin	!103	feat(api): verify a reset code without spending it
2026-08-17	4c6f2fa	Thomas Kabalin	!108	fix(api): answer an unhandled exception with a generic JSON body
2026-08-18	e4c6d59	Thomas Kabalin	!130	feat(api): accept a client-supplied match id on create
2026-08-18	b29487a	Thomas Kabalin	!131	feat(api): transfer a player to another coach via a transfer code

2026-08-18	9b297c0	Thomas Kabalin	!132	feat(api): claim a match for capture, and lock it once complete
2026-08-18	cfa0308	Thomas Kabalin	!143	feat(api): let a test run scale the rate-limit buckets up
2026-08-19	f972828	Thomas Kabalin	!149	refactor(api): delete players outright instead of archiving them
2026-08-19	889d473	Thomas Kabalin	!150	feat(api): report whether a player has a live transfer code
2026-08-19	7523acd	Thomas Kabalin	!156	feat(api): add a photo store served at /media
2026-08-19	22f62c9	Thomas Kabalin	!156	feat(api): add photo_key to coaches and players
2026-08-19	5bf5165	Thomas Kabalin	!156	feat(api): upload and clear a player photo
2026-08-19	b31f2e3	Thomas Kabalin	!156	refactor(api): move shared photo-upload helpers out of coaches router
2026-08-19	caebf03	Thomas Kabalin	!160	feat(api): add the derived match_sets table
2026-08-20	14c8adc	Thomas Kabalin	!170	feat(api): refuse deleting a match that has points
2026-08-20	d211f74	Thomas Kabalin	!171	feat(api): validate a batch of captured points
2026-08-20	b9842b9	Thomas Kabalin	!171	feat(api): derive a match's set scores from its points
2026-08-20	8ff1bdc	Thomas Kabalin	!171	feat(api): ingest a batch of captured points
2026-08-20	79221b7	Thomas Kabalin	!172	feat(api): read a match's points
2026-08-20	a7ba608	Thomas Kabalin	!173	feat(api): refuse deleting a match another device is capturing
2026-08-20	0879923	Thomas Kabalin	!175	feat(api): give every player a unique generated username
2026-08-20	0760071	Thomas Kabalin	!175	feat(api): mint a one-time password when a player is created
2026-08-20	ed348c7	Thomas Kabalin	!175	feat(api): player_sessions table, and the deletes that must clear it
2026-08-20	15c3eb3	Thomas Kabalin	!174	feat(api): accept only version-4 client match ids
2026-08-20	c92951d	Thomas Kabalin	!175	feat(api): player login, logout, and their own /me on a player session

2026-08-20	970324d	Thomas Kabalin	!175	feat(api): a coach regenerates a player's password
2026-08-21	bc1b942	Thomas Kabalin	!182	feat(api): delete a match along with its points
2026-08-21	30e7c8e	Thomas Kabalin	!182	feat(api): refuse to re-create a deleted match
2026-08-21	05df2c9	Thomas Kabalin	!174	fix(api): tolerate clock_timestamp() drift between created_at and updated_at
2026-08-23	674280d	Thomas Kabalin	!188	refactor(api): drop serve1_target/serve2_target from points
2026-08-24	8c5933a	Thomas Kabalin	!199	feat(api): metadata edits on a completed match
2026-08-28	d29fa03	Luke Bryant	!208	fix(api): tell a client its queued edit names a deleted match
2026-08-28	a8f0c4a	Thomas Kabalin	!208	fix(api): answer 410 where a match is deleted while a write waits on its row
2026-08-28	126b5fc	Thomas Kabalin	!211	fix(api): remember a deleted player's matches, so a queued flush is discarded
2026-08-28	ede74dd	Thomas Kabalin	!211	docs: say a player delete leaves tombstones too
2026-08-31	7e62292	Thomas Kabalin	!219	docs: record what makes a match in progress
2026-08-31	9b98efa	Thomas Kabalin	!219	feat(api): answer which of a coach's matches capture is under way on
2026-08-31	e85a7ab	Thomas Kabalin	!221	feat(api): store coach notes against a player

Backlog item 3

Date	Hash	Author	MR	Commit message
2026-08-12	fe61a8ae	Danny Gebhard	!15	feat(web): add sign-in/sign-up/home route placeholders
2026-08-12	27d4147e	Danny Gebhard	!15	fix(web): update e2e smoke test for the new sign-in placeholder
2026-08-12	3542815c	Danny Gebhard	!16	feat(web): move the player-roster demo from / to /players
2026-08-12	bb18d3f4	Danny Gebhard	!17	feat(web): add sign-in form UI
2026-08-12	84ae52f8	Danny Gebhard	!18	feat(web): wire the sign-in form to POST /auth/login
2026-08-12	b5e9a8f6	Danny Gebhard	!19	feat(web): add a sign-up link to the sign-in screen
2026-08-12	a26ed1e0	Danny Gebhard	!20	feat(web): add sign-up email step UI
2026-08-12	6d0c3b5e	Danny Gebhard	!21	feat(web): wire sign-up to request-otp and add the OTP entry step
2026-08-12	a6695834	Thomas Kabalin	!9	docs: record the password hashing choice
2026-08-12	7596f583	Thomas Kabalin	!9	test(api): cover 422 password redaction
2026-08-12	ea3d30ab	Thomas Kabalin	!9	feat(api): redact submitted passwords from 422 bodies
2026-08-12	0be97bfe	Thomas Kabalin	!9	test(api): cover coach sign-up
2026-08-12	3dd8a79e	Thomas Kabalin	!9	feat(api): add the coach sign-up endpoint
2026-08-12	422b9f13	Thomas Kabalin	!9	test(api): cover password hashing
2026-08-12	a155c252	Thomas Kabalin	!9	feat(api): hash passwords with argon2id
2026-08-13	892db62b	Danny Gebhard	!22	feat(web): add password + confirm step to sign-up
2026-08-13	86afe4a2	Danny Gebhard	!23	feat(web): wire the password step to POST /coaches
2026-08-13	97e825fa	Thomas Kabalin	!10	style(docs): trim definitions and obvious explanations

Date	Hash	Author	MR	Commit message
2026-08-13	07457674	Thomas Kabalin	!10	style(docs): remove AI writing patterns from docs
2026-08-13	db9a3932	Thomas Kabalin	!10	docs: record the session mechanism
2026-08-13	eabd43cd	Thomas Kabalin	!10	test(api): cover GET /coaches/me
2026-08-13	4cada64f	Thomas Kabalin	!10	feat(api): identify the caller and turn the coach by-id route into /me
2026-08-13	49008237	Thomas Kabalin	!10	test(api): cover login and logout
2026-08-13	88541437	Thomas Kabalin	!10	feat(api): add coach login and logout
2026-08-13	da724cff	Thomas Kabalin	!10	test(api): cover the coach sessions table
2026-08-13	00ec0895	Thomas Kabalin	!10	feat(api): add the coach sessions table
2026-08-13	7d49f111	Thomas Kabalin	!10	fix(scripts): refuse to run dev.sh from a git worktree
2026-08-13	22d6418d	Thomas Kabalin	!11	test(api): pin the login limit's cost claims and forged-cookie logout
2026-08-13	8f0e3b42	Thomas Kabalin	!11	docs: record the rate limiting choice
2026-08-13	dfeff4d4	Thomas Kabalin	!11	test(api): cover the login rate limit
2026-08-13	54462c1c	Thomas Kabalin	!11	feat(api): rate limit coach login
2026-08-13	83c0b3f0	Thomas Kabalin	!11	test(api): cover the sign-up rate limit
2026-08-13	24453d33	Thomas Kabalin	!11	feat(api): rate limit coach sign-up
2026-08-13	90453eb2	Thomas Kabalin	!24	test(api): cover the email verifications table
2026-08-13	9d781b6d	Thomas Kabalin	!24	feat(api): add the email verifications table
2026-08-13	060abc98	Thomas Kabalin	!24	feat(api): add OTP email verification for sign-up
2026-08-13	04691e9d	Thomas Kabalin	!24	docs: shorten Argon2, sessions, and rate-limiting rationale
2026-08-13	4c78e1e3	Thomas Kabalin	!24	chore(claude): add Resend skills

Date	Hash	Author	MR	Commit message
2026-08-13	c6d56346	Thomas Kabalin	!24	test(api): cover requesting and verifying sign-up codes
2026-08-13	23fe5175	Thomas Kabalin	!27	test(api): keep the suite off a real email provider
2026-08-13	ff2252cc	Thomas Kabalin	!27	chore(infra): pass the Resend key to the api container
2026-08-13	1fea21c0	Thomas Kabalin	!27	ci: check a real send on every production deploy
2026-08-13	138893a9	Thomas Kabalin	!27	test(api): cover the Resend sender and its 502
2026-08-13	52809ecf	Thomas Kabalin	!27	feat(api): send sign-up codes through Resend
2026-08-13	cd94dc59	Thomas Kabalin	!27	docs: record Resend in the tech stack and the VPS key step
2026-08-13	72f26696	Thomas Kabalin	!27 !28	docs(api): trim the comments in the coaches router
2026-08-13	a0ee6451	Thomas Kabalin	!27 !28	fix(web): send one sign-up request per click
2026-08-13	bd9cc14f	Thomas Kabalin	!27 !28	feat(api): require a verified email to create a coach
2026-08-13	9de6ca9c	Thomas Kabalin	!27 !28	docs: bound comment length in CLAUDE.md
2026-08-13	45c01bee	Thomas Kabalin	!13	feat(api): scope players and matches to the authenticated coach
2026-08-13	361c7632	Thomas Kabalin	!13	test(web): pin the player fixture to the generated contract
2026-08-13	8aa0791f	Thomas Kabalin	!13	test(api): cover coach scoping on players and matches
2026-08-13	44ee2b10	Danny Gebhard	!30	feat(web): wire the OTP step to POST /auth/verify-otp
2026-08-13	97041e04	Thomas Kabalin	!32	docs: record the account-enumeration policy
2026-08-13	36a21946	Thomas Kabalin	!32	test(api): pin that rate limits answer known and unknown emails alike

Date	Hash	Author	MR	Commit message
2026-08-13	ed9d01ff	Thomas Kabalin	!32	test(api): cover password reset end to end
2026-08-13	b9307e6f	Thomas Kabalin	!32	feat(api): reset a forgotten password with an emailed code
2026-08-14	a21227a9	Danny Gebhard	!41	feat(web): add the forgot-password route and email step UI
2026-08-14	236caadc	Danny Gebhard	!42	feat(web): wire forgot-password to POST /auth/forgot-password, add reset step UI
2026-08-14	cbce7020	Danny Gebhard	!43	feat(web): wire reset step to POST /auth/reset-password, add confirmation view
2026-08-14	5ab46eb0	Danny Gebhard	!45	feat(web): add a request-a-new-code affordance to the reset step
2026-08-14	36353095	Danny Gebhard	!44	feat(web): add a forgot-password link to the sign-in screen
2026-08-15	48c65fd7	Thomas Kabalin	!46	docs(security): record the re-authentication a password change demands
2026-08-15	ba853d98	Thomas Kabalin	!46	feat(api): change a password with the current one
2026-08-16	6d2b26ce	Danny Gebhard	!80	feat(web): add a "back to sign in" link to sign-up and forgot-password
2026-08-16	646340db	Danny Gebhard	!81	feat(web): make the sign-in failure message more explicit
2026-08-16	22bbcdf0	Danny Gebhard	!82	feat(web): make the sign-up duplicate-email message more explicit
2026-08-16	cbb0ec40	Danny Gebhard	!83	feat(web): add a sign-out control to the settings page
2026-08-16	d3dfbb91	Danny Gebhard	!84	feat(web): show a live countdown on 429 responses
2026-08-16	bfa393ce	Danny Gebhard	!84	feat(web): add recovery paths to the sign-up OTP and password steps
2026-08-16	cd2cebdc	Danny Gebhard	!94	feat(web): add a show/hide toggle to every password field

Date	Hash	Author	MR	Commit message
2026-08-16	54bbd5bd	Danny Gebhard	!95	feat(web): real-time password strength meter with submit-gating
2026-08-16	0ad73b49	Danny Gebhard	!95	test(web): paste the too-long password fixture instead of typing it
2026-08-16	43a3d8eb	Danny Gebhard	!96	feat(web): build the shared auth-page shell and styled form primitives
2026-08-16	d9414707	Danny Gebhard	!97 !99	feat(web): style the sign-in page per the wireframe
2026-08-16	6a2a72ab	Danny Gebhard	!98	feat(web): style the sign-up page across its three steps
2026-08-16	602c4118	Danny Gebhard	!100	feat(web): style the forgot-password page across its three steps
2026-08-18	1ff4128e	Thomas Kabalin	!104	docs(security): record the password validation policy
2026-08-18	da450878	Thomas Kabalin	!104	feat(api): refuse repeated, sequential, common and self-naming passwords
2026-08-18	ab925ed5	Thomas Kabalin	!107	docs(security): record the breach-list check and its fail-open
2026-08-18	c16c4bb8	Thomas Kabalin	!107	feat(api): screen new passwords against the Pwned Passwords corpus
2026-08-18	131c749b	Danny Gebhard	!134	feat(web): replace sign-up's plain OTP field with the 6-box input
2026-08-18	10a5e1cb	Danny Gebhard	!134	feat(web): split forgot-password's reset step into OTP and password steps
2026-08-19	6413622d	Thomas Kabalin	!145	docs(tech-stack): record the hand-authored email bodies
2026-08-19	8ece27ad	Thomas Kabalin	!145	feat(api): send the code emails as designed HTML with a text alternative
2026-08-20	21a6d730	Danny Gebhard	!165	feat(web): rebuild Settings as a drill-down menu (#178)

Date	Hash	Author	MR	Commit message
2026-08-20	192da1ac	Danny Gebhard	!168	feat(web): close an account and set match defaults (#184, #104)

Backlog Item 4

Date	Hash	Author	MR	Commit message
2026-08-12	8d07ad0	Luke Bryant	!25	feat(web): add Point scoring model
2026-08-12	29812e4	Luke Bryant	!25	refactor(web): simplify Point model
2026-08-12	a128079	Luke Bryant	!25	feat(web): rework Point outcome model, exact rally length
2026-08-12	6b8cfb7	Luke Bryant	!25	test(web): add Point unit tests; fix mutable-reference leak
2026-08-14	c302841	Luke Bryant	!69	feat(web): add Match class
2026-08-15	e89ba14	Luke Bryant	!69	feat(web): add createMatch factory with ScoringSystem-driven sets
2026-08-15	bf23554	Luke Bryant	!71	refactor(web): stop pre-populating Match.sets, extract maxSetsFor
2026-08-15	5561dae	Luke Bryant	!71	fix(web): silence no-empty-object-type on the MatchSet placeholder
2026-08-15	f0cd192	Luke Bryant	!72	fix(web): rename Match.timestamp to playedOn, store date-only
2026-08-15	cc12a1f	Luke Bryant	!73	fix(web): add Match.opponentName, generate a stand-in opponentId
2026-08-15	fd63123	Luke Bryant	!74	fix(web): default scoringSystem/surfaceType instead of requiring them
2026-08-15	8e15476	Luke Bryant	!75	fix(web): make Match.firstServer an id, not a role label
2026-08-17	914a6f0	Luke Bryant	!71	feat(web): add MatchSet, a flat record referencing Match by matchId
2026-08-17	302c048	Luke Bryant	!117	feat(web): add gameStateFor, deriving a game's score from its Points
2026-08-17	b10daa0	Luke Bryant	!117	refactor(web): move maxSetsFor to state.ts, add isAdScoringFor
2026-08-17	b5da17c	Luke Bryant	!118	feat(web): add Point.isTiebreakPoint
2026-08-17	1c3233b	Luke Bryant	!118	feat(web): add MatchSet.tiebreak
2026-08-17	559fb55	Luke Bryant	!118	feat(web): add setRulesFor and setStateFor
2026-08-17	d49f987	Luke Bryant	!119	feat(web): add matchRulesFor
2026-08-17	5a86636	Luke Bryant	!119	feat(web): add matchStateFor
2026-08-18	a5fcd6d	Luke Bryant	!120	feat(web): add Match.outcomeOverride for retirement/walkover/default
2026-08-19	3f42970	Luke Bryant	!146	refactor(web): rework Point to the perspective-based capture model
2026-08-19	32a9133	Luke Bryant	!148	feat(web): add the single-point capture flow state machine
2026-08-19	c7f43d1	Luke Bryant	!155	feat(web): close the state.ts gaps between committed points and live match state
2026-08-19	bdaadab	Luke Bryant	!159	feat(web): add the match-level flow layer over capture-flow and state
2026-08-20	2b1212d	Luke Bryant		docs: confirm the four scoring-system values docs/scoring-systems.md left open

2026-08-20	29b1117	Luke Bryant	!166	feat(web): derive which service box a point is served from
2026-08-20	52bfcb8	Luke Bryant	!166	feat(web): stop asking the coach to tap the service box
2026-08-20	4e1bfd8	Luke Bryant		docs: confirm the four scoring-system values docs/scoring-systems.md left open
2026-08-20	1c38750	Luke Bryant	!167	feat(web): drop shot detail on a return error, place it in the outcome itself
2026-08-21	5f2fbdd	Luke Bryant	!179	feat(web): derive the current score as a two-row scoreboard
2026-08-21	8512ef3	Luke Bryant	!183	feat(web): record the deuce/advantage state a point was played under
2026-08-21	19c6859	Luke Bryant	!183	refactor(web): promote courtSide to Point's only deuce/ad field
2026-08-23	02425ac	Luke Bryant	!208	refactor(web): let a landed serve record no placement
2026-08-24	04bc754	Luke Bryant	!208	feat(web): record a serve nobody watched
2026-08-24	c6b63fd	Luke Bryant	!208	feat(web): skip a point from the serve screen
2026-08-24	3a8f9a6	Luke Bryant	!208	refactor(web): skip the shot screen on an unforced error
2026-08-24	151b28b	Luke Bryant	!208	refactor(web): the final ball is four zones, and always one that landed
2026-08-24	5e00035	Luke Bryant	!208	feat(web): synthesise the points behind a skipped-to score
2026-08-26	3a6db6f	Luke Bryant	!208	fix(web): stop a let from carrying the rallied length into the replay
2026-08-26	6f27402	Luke Bryant	!208	feat(web): let the rally screen's two taps land in either order
2026-08-27	cb8d0cf	Luke Bryant	!208	feat(web): fold the score from an event log, so a coach can state it
2026-08-27	b53f8e6	Luke Bryant	!208	refactor(scoring): drop womens_grand_slam as a distinct scoring system
2026-08-27	b66c86e	Luke Bryant	!208	feat(web): explain each scoring system on Match Defaults

Backlog Item 5

Date	Hash	Author	MR	Commit message
2026-08-22	6a85b0b	Luke Bryant	!186	feat(web): add carpet and unspecified surface options to match setup
2026-08-23	83b0c7d	Luke Bryant	!208	feat(web): court diagram shared by the capture screens
2026-08-23	6398bce	Luke Bryant	!208	feat(web): serve screen
2026-08-24	51ea023	Luke Bryant	!208	feat(web): match header shared by the capture screens
2026-08-24	0b0d9c0	Luke Bryant	!208	feat(web): return screen
2026-08-24	a98745d	Luke Bryant	!208	feat(web): rally screen
2026-08-24	1222b0f	Luke Bryant	!208	feat(web): final shot screen
2026-08-24	be8047a	Luke Bryant	!208	feat(web): final ball screen
2026-08-24	8fcf4fd	Luke Bryant	!208	feat(web): match setup screen
2026-08-24	630f093	Luke Bryant	!208	feat(web): result screen
2026-08-24	12fcd0b	Luke Bryant	!208	feat(web): unusual point end sheet
2026-08-24	ba0393c	Luke Bryant	!208	feat(web): skip ahead sheet
2026-08-24	03d7753	Luke Bryant	!208	feat(web): run the capture flow
2026-08-24	06c4580	Luke Bryant	!208	feat(web): paint the court its own surface
2026-08-25	f9d028a	Luke Bryant	!208	feat(web): back steps out of a match it hasn't started
2026-08-26	07894fc	Luke Bryant	!208	chore: keep design-tool handoff bundles out of the repo
2026-08-26	e0717f9	Luke Bryant	!208	fix(web): queue a match's outcome even when ending it queued nothing else
2026-08-26	80d6b7b	Luke Bryant	!208	fix(web): keep real points when skipping ahead to a score
2026-08-26	19382cc	Luke Bryant	!208	fix(web): never let skip ahead rewrite who served first
2026-08-26	6f27402	Luke Bryant	!208	feat(web): let the rally screen's two taps land in either order
2026-08-26	9507441	Luke Bryant	!208	fix(web): open skip ahead after the hold releases, not mid-press
2026-08-27	f12ef38	Luke Bryant	!208	fix(web): disable serve/return buttons while a court tap's flash is pending
2026-08-27	65911a5	Luke Bryant	!208	fix(web): stop back from reopening setup after the match is queued
2026-08-27	236a7d9	Luke Bryant	!208	feat(web): add a coach note button to the live capture header
2026-08-27	5051cfc	Luke Bryant	!208	feat(web): back button on match setup
2026-08-27	f6c2abe	Luke Bryant	!208	fix(web): stop live-match reading a ref during render, and a flaky select click
2026-08-27	41c073d	Luke Bryant	!208	feat(web): flash the point, game or set a captured point won
2026-08-27	a41f25a	Luke Bryant	!208	fix(web): stop a long-press stranding the next keyboard activation
2026-08-27	cb8d0cf	Luke Bryant	!208	feat(web): fold the score from an event log, so a coach can state it

2026-08-27	9e7b24c	Luke Bryant	!208	feat(web): let a coach say "I didn't see that" without losing the point
2026-08-27	a16639c	Luke Bryant	!208	feat(web): keep the match in a durable log, so a dead phone doesn't lose it
2026-08-28	9f7fab9	Luke Bryant	!208	fix(web): hold the court at one size, so a tap lands where the last one did
2026-08-28	11454ee	Luke Bryant	!208	fix(web): start a match without waiting on the coach, and honour their defaults
2026-08-28	a1e728f	Luke Bryant	!208	fix(web): sit each capture card inside a court region, not near one
2026-08-28	2839951	Luke Bryant	!208	fix(web): let back leave a match queued before anything was captured
2026-08-28	4ec792b	Luke Bryant	!208	fix(web): the five defects the branch review turned up
2026-08-28	d29fa03	Luke Bryant	!208	fix(api): tell a client its queued edit names a deleted match
2026-08-28	68edf7e	Luke Bryant	!208	fix(web): count only the queued rows a drain can actually send
2026-08-28	3bb3f0c	Luke Bryant	!208	fix(web): keep a voice note whole when the speech service ends it
2026-08-28	8d181e6	Luke Bryant	!208	chore: keep the schema-gaps note in docs/scratch, not the source tree
2026-08-28	525c00b	Luke Bryant	!208	feat(web): let a coach hold the note button to type a note

Backlog item 6

Date	Hash	Author	MR	Commit message
2026-08-18	0cb45c8f	Danny Gebhard	!126	fix(web): address Thomas's review on !126
2026-08-18	75b23c82	Danny Gebhard	!126	feat(web): build the seven player-stats tabs
2026-08-18	035a5cdc	Danny Gebhard	!126	feat(web): build the wireframe's home, roster and match-list screens
2026-08-18	613c4a44	Danny Gebhard	!126	feat(web): add the provisional analytics layer behind the stats screens
2026-08-18	264c2d63	Danny Gebhard	!126	feat(web): add the categorical palette tokens and the shared stat primitives
2026-08-18	3ab1b382	Danny Gebhard	!126	chore(web): install the shadcn components the analytics screens need
2026-08-23	a118d618	Thomas Kabalin	!177	refactor(web): collapse the winner/forced-error split capture never records
2026-08-23	1a4ee0e9	Thomas Kabalin	!177	feat(api): player analytics endpoints scoped to the capture record
2026-08-23	09b2944d	Danny Gebhard	!189	feat(web): redesign the player statistics drill-down
2026-08-23	36442d40	Danny Gebhard	!189	feat(web): restore season record, player details and notes to the stats page
2026-08-31	3131022	Thomas Kabalin	!217	refactor(api): drop the analytics endpoints no screen reads
2026-08-31	aab06cd	Thomas Kabalin	!218	refactor(api): drop mid from the court depth enum
2026-08-31	15b00b7	Thomas Kabalin	!220	feat(api): scope the player analytics tabs to one opponent
2026-08-31	c5b3248f	Danny Gebhard	!216	fix(web): harden Serve/Rally tabs and the court heatmap for real analytics payloads
2026-08-31	ca09ca13	Danny Gebhard	!222	feat(web): wire the squad snapshot and roster to real analytics data
2026-08-31	31487a4e	Danny Gebhard	!223	feat(web): wire the player profile and per-match stats to real analytics data

Date	Hash	Author	MR	Commit message
2026-08-31	f72a0c43	Danny Gebhard	!224	feat(web): wire the opponent axis to real data (MR 4 of the stats-wiring program)

Backlog Item 7

Date	Hash	Author	MR	Commit message
2026-08-15	70ff86b	Thomas Kabalin	!70	feat(web): map a captured Match onto the API's create payload
2026-08-17	a35602c	Thomas Kabalin	!102	feat(api): store a scoring system instead of a sets format
2026-08-17	f0177fa	Thomas Kabalin	!109	chore(web): add dexie for the offline outbox
2026-08-17	6dac6aa	Thomas Kabalin	!109	feat(web): add the Dexie outbox storage layer
2026-08-17	90e8c5b	Thomas Kabalin	!109	docs(testing): note the fake-indexeddb shim in the setup file
2026-08-17	38c932a	Thomas Kabalin	!109	docs(web): carve out features/outbox from the no-cross-imports rule
2026-08-18	cbc49ff	Thomas Kabalin	!109	docs(web): point the outbox comments at the settled sync decisions
2026-08-18	22255bd	Thomas Kabalin	!140	feat(web): send the client-generated match id when syncing a match
2026-08-18	df72239	Thomas Kabalin	!141	feat(web): purge local data on sign-out and a coach switch
2026-08-19	2248029	Thomas Kabalin	!152	feat(web): mint and persist a device id
2026-08-20	a7ba608	Thomas Kabalin	!173	feat(api): refuse deleting a match another device is capturing
2026-08-20	2526607	Thomas Kabalin	!173	feat(web): name this device when deleting a match
2026-08-20	bed36a3	Thomas Kabalin	!173	docs: record the capture-claim refusal in the deletion convention
2026-08-20	53c2229	Thomas Kabalin	!174	feat(web): record why an outbox row was parked
2026-08-20	1e40e15	Thomas Kabalin	!174	feat(web): drain the outbox into the API
2026-08-20	15c3eb3	Thomas Kabalin	!174	feat(api): accept only version-4 client match ids
2026-08-20	e36cb92	Thomas Kabalin	!174	feat(web): wire the drain to every flush opportunity
2026-08-21	e5e2464	Thomas Kabalin	!176	test(web): e2e offline round trip against a production build
2026-08-21	6516675	Thomas Kabalin	!181	feat(web): discard outbox rows for a deleted match
2026-08-21	12ee9aa	Thomas Kabalin	!181	feat(web): purge one match's rows from the outbox
2026-08-21	05df2c9	Thomas Kabalin	!174	fix(api): tolerate clock_timestamp() drift between created_at and updated_at
2026-08-21	9dc84bb	Thomas Kabalin	!174	fix(web): update sync point fixtures for the courtSide/location split

2026-08-21	eb04542	Thomas Kabalin		docs: add the offline sync mechanism doc
2026-08-21	83f2fbc	Thomas Kabalin	!185	test(web): cover the parts of the offline round trip points don't block
2026-08-21	d4667e9	Thomas Kabalin	!185	docs: note the offline read side in the testing layer split
2026-08-24	a011d8f	Thomas Kabalin	!201	test(web): cover the drain's match-side failure verdicts
2026-08-26	e0717f9	Luke Bryant	!208	fix(web): queue a match's outcome even when ending it queued nothing else
2026-08-27	a16639c	Luke Bryant	!208	feat(web): keep the match in a durable log, so a dead phone doesn't lose it
2026-08-28	68edf7e	Luke Bryant	!208	fix(web): count only the queued rows a drain can actually send
2026-08-28	8d181e6	Luke Bryant	!208	chore: keep the schema-gaps note in docs/scratch, not the source tree
2026-08-28	7eb8880	Thomas Kabalin	!208	fix(web): keep a refused match's undrainable rows out of the failed count
2026-08-28	c75d1fd	Thomas Kabalin	!208	fix(web): forget a deleted match's log when the drain hears it is gone
2026-08-28	2a6ab5f	Thomas Kabalin	!208	docs: drop broken scratch-doc reference from offline-sync

Backlog item 8

Date	Hash	Author	MR	Commit message
2026-08-15	d3b77207	Danny Gebhard	!47	feat(web): move QueryClientProvider to the root layout, retire the players App scaffold
2026-08-15	07319d47	Danny Gebhard	!51	feat(web): add the bottom nav component
2026-08-15	ede16155	Danny Gebhard	!52	feat(web): add profile/settings buttons and placeholder routes to home
2026-08-15	e89c6584	Danny Gebhard	!53	feat(web): show the signed-in coach's name on home
2026-08-15	c233962d	Danny Gebhard	!54	feat(web): add a player search bar to home
2026-08-15	9ddc2ec0	Danny Gebhard	!55	feat(web): add the My Players bubble row to home
2026-08-15	fa9aa665	Danny Gebhard	!56	feat(web): add the last-viewed-player box to home
2026-08-15	07755c12	Danny Gebhard	!57	feat(web): add the squad snapshot box to home
2026-08-15	a5e8ad26	Danny Gebhard	!58	feat(web): add the bottom nav to home
2026-08-15	c7dfbcd9	Danny Gebhard	!59	feat(web): add the player profile route and page
2026-08-15	9ae05f1b	Danny Gebhard	!60	feat(web): link players to their profile page, track last-viewed
2026-08-15	07794b02	Danny Gebhard	!61	feat(web): add the bottom nav to the players roster page
2026-08-15	b5c8dbff	Danny Gebhard	!48	feat(web): give home its own feature slice
2026-08-15	d56f3814	Danny Gebhard	!68	feat(web): add the bottom nav to the players roster page
2026-08-15	a814c561	Danny Gebhard	!68	feat(web): link players to their profile page, track last-viewed
2026-08-15	cfe4c65b	Danny Gebhard	!68	feat(web): add the player profile route and page

Date	Hash	Author	MR	Commit message
2026-08-15	c048b7de	Danny Gebhard	!68	feat(web): add the bottom nav to home
2026-08-15	c4a0acfa	Danny Gebhard	!68	feat(web): add the squad snapshot box to home
2026-08-15	462f3db3	Danny Gebhard	!68	feat(web): add the last-viewed-player box to home
2026-08-15	1a454dbb	Danny Gebhard	!68	feat(web): add the My Players bubble row to home
2026-08-15	26804b5d	Danny Gebhard	!68	feat(web): add a player search bar to home
2026-08-15	5d57e862	Danny Gebhard	!68	feat(web): show the signed-in coach's name on home
2026-08-15	9a0ce3c5	Danny Gebhard	!68	feat(web): add profile/settings buttons and placeholder routes to home
2026-08-15	4d39408a	Danny Gebhard	!68	feat(web): add the bottom nav component
2026-08-15	c844ad84	Danny Gebhard	!68	feat(web): add the /stats route
2026-08-15	7166d91a	Danny Gebhard	!68	feat(web): add the /matches route
2026-08-15	407952b2	Thomas Kabalin	!68	fix(web): render the player profile at its own route
2026-08-15	0abc65f6	Thomas Kabalin	!68	fix(web): count matches by local calendar date
2026-08-17	f1e290e7	Danny Gebhard	!116	feat(web): player profile tabs and match setup screen
2026-08-18	0cb45c8f	Danny Gebhard	!126	fix(web): address Thomas's review on !126
2026-08-18	75b23c82	Danny Gebhard	!126	feat(web): build the seven player-stats tabs
2026-08-18	035a5cdc	Danny Gebhard	!126	feat(web): build the wireframe's home, roster and match-list screens
2026-08-18	613c4a44	Danny Gebhard	!126	feat(web): add the provisional analytics layer behind the stats screens
2026-08-18	264c2d63	Danny Gebhard	!126	feat(web): add the categorical palette tokens and the shared stat primitives

Date	Hash	Author	MR	Commit message
2026-08-18	3ab1b382	Danny Gebhard	!126	chore(web): install the shadcn components the analytics screens need
2026-08-18	85fb095c	Danny Gebhard	!127	fix(web): restore role="alert" on a failed player load
2026-08-18	1e5a5e1d	Danny Gebhard	!127	fix(web): say "not found" for a player id that isn't the coach's
2026-08-18	1b197c16	Danny Gebhard	!128	feat(web): require date of birth and gender when adding a player
2026-08-18	6eea5a40	Danny Gebhard	!128	fix(web): cap a top/bottom sheet's height to the viewport
2026-08-18	59a559c2	Danny Gebhard	!136	feat(web): add a custom 404 page for unmatched routes
2026-08-18	97427887	Danny Gebhard	!137	fix(web): wire dark mode to the OS preference, fix its brand blue, quiet scrollTo test noise
2026-08-18	251f45e5	Danny Gebhard	!138	feat(web): add a player edit form, fix home's Add tile opening the roster instead
2026-08-19	7e4533dd	Danny Gebhard	!139	feat(web): add a PWA install guide with platform-specific steps
2026-08-19	5a565e64	Danny Gebhard	!153	feat(web): add a delete action for players
2026-08-19	11655ef2	Danny Gebhard	!154	feat(web): add a delete action for matches
2026-08-19	92b64bf6	Danny Gebhard	!161	feat(web): edit display name and profile photo from Profile (#179)
2026-08-19	1e3abdc8	Danny Gebhard	!161	feat(web): nudge a photo-less coach from Home, show their avatar in the header (#179)
2026-08-19	442fe32e	Danny Gebhard	!161	test(web): promote the profile photo e2e check to a real spec
2026-08-20	cd50fd4b	Danny Gebhard	!162	feat(web): add, change, or clear a player photo (#192)
2026-08-20	84495dc3	Danny Gebhard	!163	feat(web): transfer a player to another coach via a one-time code (#185

Date	Hash	Author	MR	Commit message
2026-08-29	e3eeae2	Luke Bryant	!209	feat(web): a fake season, captured the way the app captures one
2026-08-30	1f2cb11	Thomas Kabalin	— (direct push to main)	docs: document the CSV format import_sample_data.py expects
2026-08-30	71412e8	Thomas Kabalin	— (direct push to main)	docs: note the shared demo password in the sample-data README
2026-08-30	9db6a26	Luke Bryant	!215	feat(web): generate the season in the API importer's format too
2026-08-30	83709b2	Thomas Kabalin	!215	docs: point the demo-data docs at where the CSVs actually live

Backlog Item 9

Date	Hash	Author	MR	Commit message
2026-08-21	a4b74a3	Thomas Kabalin	!178	feat(api): add a one-off script to import sample coaches/players/matches/points from CSV
2026-08-23	674280d	Thomas Kabalin	!188	refactor(api): drop serve1_target/serve2_target from points
2026-08-24	da8cb84	Thomas Kabalin	!207	refactor(api): split the sample-data import into run_import and main
2026-08-24	4d6d7e9	Thomas Kabalin	!207	fix(api): mint player usernames in the sample-data import
2026-08-24	5a1187d	Thomas Kabalin	!207	fix(api): report a bad matches.csv row by its line number
2026-08-24	707ed2d	Thomas Kabalin	!207	feat(api): make the sample-data import re-runnable
2026-08-24	4dbed59	Thomas Kabalin	!207	chore(api): measure and type-check the scripts directory
2026-08-24	2a8cb96	Thomas Kabalin	!207	fix(api): name the CSV row when a join between files finds nothing

Date	Hash	Author	MR	Commit message
2026-08-28	e3eeae2	Luke Bryant	!209	feat(web): a fake season, captured the way the app captures one
2026-08-30	ed36d74e	Luke Bryant	!215	feat(web): generate the season in the API importer's format too

Appendix T

Strictly, a unit test checks one function on its own, with no database or network involved. Most of our backend tests instead run real HTTP requests against a real test database, which is technically an integration test.

We chose this deliberately: a database-backed feature (queries, constraints, aggregates) can pass against a fake database and still fail against the real one, so testing against real Postgres gives much stronger confidence than mocking it.

Real unit tests cover pure, self-contained logic: a function that computes or transforms something with no database, network, or UI involved. Anything that has to prove a whole feature actually works uses an integration test instead.

Backlog item 1

Test Results Summary

API (pytest): 207/207 passed across 11 files

API — pytest (207/207)

tests/models/test_coach.py — 29/29

- test_coach_round_trips
- test_created_at_differs_between_rows_in_one_transaction
- test_updated_at_advances_when_a_row_changes
- test_database_requires_every_column [email, display_name, password_hash]
- test_database_rejects_a_duplicate_email
- test_database_rejects_an_email_that_is_not_lowercased
- test_the_orm_lowercases_an_email_on_the_way_in
- test_database_rejects_an_email_over_the_column_width
- test_database_rejects_a_display_name_over_the_column_width
- test_database_rejects_default_surface_outside_the_enum
- test_database_rejects_default_scoring_system_outside_the_enum
- test_database_accepts_every_member_of_the_reused_enums (12 cases)
- test_database_accepts_a_coach_with_no_preferences
- test_database_accepts_a_password_hash_of_any_length
- test_a_coach_without_a_photo_has_no_photo_url
- test_a_stored_photo_key_becomes_a_url

tests/models/test_coach_session.py — 8/8

- test_coach_session_round_trips
- test_database_requires_every_column [coach_id, token_hash, expires_at]
- test_database_rejects_a_duplicate_token_hash
- test_database_rejects_a_token_hash_over_the_column_width
- test_database_requires_the_coach_to_exist
- test_a_coach_with_sessions_cannot_be_hard_deleted

tests/models/test_email_verification.py — 13/13

- test_email_verification_round_trips
- test_database_requires_every_column [email, attempts, expires_at]
- test_database_rejects_a_second_row_for_the_same_address
- test_database_rejects_a_duplicate_token_hash
- test_database_permits_many_rows_at_the_code_stage
- test_database_rejects_a_hash_over_the_column_width [otp_hash, token_hash]
- test_database_rejects_an_email_over_the_column_width
- test_database_rejects_an_email_that_is_not_lowercased
- test_database_rejects_a_row_holding_both_secrets
- test_database_rejects_a_row_holding_neither_secret

tests/models/test_password_reset.py — 13/13

- test_password_reset_round_trips
- test_database_requires_every_column [coach_id, attempts, expires_at]
- test_database_rejects_a_second_row_for_the_same_coach
- test_database_rejects_an_unknown_coach
- test_database_refuses_to_delete_a_coach_with_a_pending_reset
- test_database_rejects_a_hash_over_the_column_width [otp_hash, token_hash]
- test_database_rejects_a_duplicate_token_hash
- test_database_permits_many_rows_at_the_code_stage
- test_database_rejects_a_row_holding_both_secrets
- test_database_rejects_a_row_holding_neither_secret

tests/models/test_player.py — 23/23

- test_player_round_trips
- test_committed_rows_do_not_leak_between_tests_a
- test_committed_rows_do_not_leak_between_tests_b
- test_only_the_name_parts_and_the_owner_are_required
- test_created_at_differs_between_rows_in_one_transaction
- test_updated_at_advances_when_a_row_changes
- test_has_transfer_code_follows_the_code_row
- test_has_transfer_code_answers_for_one_player_at_a_time
- test_database_requires_both_name_parts [first_name, last_name]
- test_database_rejects_dominant_hand_outside_the_enum
- test_database_rejects_gender_outside_the_enum
- test_database_rejects_a_name_over_the_column_width [first_name, last_name]
- test_database_requires_a_username
- test_database_rejects_a_duplicate_username
- test_database_rejects_a_username_over_the_column_width
- test_database_requires_an_owner
- test_database_rejects_an_unknown_coach_id
- test_database_refuses_to_delete_a_coach_who_still_has_players
- test_coach_id_is_indexed
- test_a_player_without_a_photo_has_no_photo_url

- test_a_stored_photo_key_becomes_a_url

tests/models/test_player_session.py — 8/8

- test_player_session_round_trips
- test_database_requires_every_column [player_id, token_hash, expires_at]
- test_database_rejects_a_duplicate_token_hash
- test_database_rejects_a_token_hash_over_the_column_width
- test_database_requires_the_player_to_exist
- test_a_player_with_sessions_cannot_be_hard_deleted

tests/models/test_player_transfer_code.py — 7/7

- test_transfer_code_round_trips
- test_database_requires_every_column [player_id, token_hash]
- test_database_rejects_a_second_code_for_the_same_player
- test_database_rejects_a_reused_token_hash
- test_database_refuses_a_code_for_a_player_that_does_not_exist
- test_database_bounds_the_token_hash_at_the_hash_width

tests/models/test_match.py — 23/23

- test_match_round_trips_with_player
- test_only_the_player_and_date_are_required
- test_match_requires_a_played_on_date
- test_match_requires_an_existing_player
- test_a_player_with_matches_cannot_be_deleted
- test_database_rejects_surface_outside_the_enum
- test_database_rejects_scoring_system_outside_the_enum
- test_database_accepts_every_scoring_system (8 cases)
- test_database_rejects_result_outside_the_enum
- test_database_rejects_free_text_over_the_column_width
- test_database_rejects_half_a_capture_claim [half0, half1]
- test_database_accepts_a_whole_capture_claim
- test_player_id_is_indexed
- test_created_at_differs_between_rows_in_one_transaction
- test_updated_at_advances_when_a_row_changes

tests/models/test_match_set.py — 12/12

- test_match_set_round_trips_with_match
- test_a_set_in_flight_has_a_tally_and_no_winner
- test_match_set_requires_an_existing_match
- test_deleting_a_match_takes_its_derived_sets_with_it
- test_a_set_number_is_unique_within_a_match
- test_the_same_set_number_is_free_under_another_match
- test_database_rejects_a_winner_outside_the_enum
- test_database_accepts_every_side [tracked, opponent]
- test_match_id_leads_an_index
- test_created_at_differs_between_rows_in_one_transaction

- test_updated_at_advances_when_a_row_changes

tests/models/test_point.py — 67/67

- test_point_round_trips_with_the_minimal_shape
- test_point_round_trips_with_a_full_two_serve_opponent_serve_point
- test_point_requires_an_existing_match
- test_deleting_a_match_with_points_is_refused
- test_sequence_is_unique_within_a_match
- test_the_same_sequence_is_free_under_another_match
- test_match_id_leads_an_index
- test_nullable_columns_accept_null
- test_database_rejects_a_half_set_pair [half0, half1, half2, half3]
- test_database_rejects_end_reason_outside_the_enum
- test_database_accepts_every_enum_member (52 cases)
- test_created_at_differs_between_rows_in_one_transaction
- test_updated_at_advances_when_a_row_changes
- test_database_rejects_a_mid_ball_depth

apps/api/tests/models/test_player_note.py — 25/25

- test_player_note_round_trips
- test_a_note_filed_under_nothing_stores_an_empty_list
- test_the_note_date_is_independent_of_when_the_row_was_written
- test_deleting_a_player_with_notes_is_refused
- test_database_requires_every_column
- test_database_refuses_a_note_for_a_player_that_does_not_exist
- test_database_refuses_a_focus_area_outside_the_four
- test_database_accepts_no_focus_areas_at_all
- test_database_bounds_a_focus_area_at_the_widest_member
- test_only_the_body_and_the_date_are_required
- test_rejects_a_missing_required_field
- test_rejects_a_key_the_schema_does_not_carry
- test_strips_surrounding_whitespace_from_the_body
- test_rejects_a_whitespace_only_body
- test_accepts_a_body_at_the_length_limit
- test_rejects_an_overlong_body
- test_accepts_a_note_dated_today
- test_rejects_a_note_dated_in_the_future
- test_accepts_a_note_about_a_session_seasons_ago
- test_rejects_an_implausibly_old_note_date
- test_keeps_focus_areas_in_declaration_order
- test_files_a_note_under_each_area_once
- test_rejects_a_focus_area_outside_the_four
- test_rejects_a_null_focus_areas_list
- test_an_update_may_carry_nothing
- test_an_update_carries_each_mutable_field
- test_an_update_clears_the_focus_areas_with_an_empty_list

- test_an_update_rejects_a_null_over_a_non_nullable_column
- test_an_update_applies_the_same_bounds_as_a_create

tests/test_migrations.py — 5/5

- test_migrations_upgrade_downgrade_and_upgrade_again
- test_requiring_a_player_owner_refuses_orphans_rather_than_deleting_them
- test_the_username_backfill_numbers_existing_players_deterministically
- test_the_schema_at_head_matches_the_models
- test_dropping_mid_clears_the_placement_rather_than_the_point

Backlog item 2

Test Results Summary

API (pytest): 516/516 passed across 14 files

API — pytest (516/516)

tests/routers/test_players.py — 118/118

- test_create_player_persists_and_returns_row
- test_create_player_belongs_to_the_caller
- test_create_player_with_every_optional_field
- test_create_player_without_optional_fields
- test_create_player_cannot_set_coach_id
- test_create_player_mints_the_username_and_numbers_namesakes
- test_create_player_numbers_past_usernames_minted_before_this_request
- test_create_player_returns_a_one_time_password_and_stores_only_its_hash
- test_the_password_never_appears_on_a_read
- test_the_minted_password_never_meets_the_policy
- test_creates_beyond_the_minting_burst_are_refused
- test_create_player_retries_when_a_concurrent_create_takes_the_username
- test_create_player_gives_up_after_three_username_collisions
- test_regenerating_gives_a_backfilled_player_a_password_that_signs_them_in
- test_regenerating_stores_only_a_hash_and_replaces_the_old_password
- test_regenerating_ends_the_player_s_live_sessions_and_no_one_else_s
- test_regenerating_another_coach_s_player_is_the_404_an_unknown_id_gets
- test_a_404_from_the_password_route_spends_no_minting_token
- test_creation_and_regeneration_spend_the_same_minting_budget
- test_create_player_re_raises_an_integrity_error_that_is_not_the_username
- test_create_player_rejects_future_date_of_birth
- test_create_player_rejects_implausibly_old_date_of_birth
- test_create_player_accepts_a_plausibly_old_date_of_birth
- test_create_player_accepts_date_of_birth_of_today
- test_create_player_accepts_an_explicit_null_date_of_birth
- test_create_player_rejects_missing_first_name
- test_create_player_rejects_missing_last_name
- test_create_player_rejects_invalid_dominant_hand
- test_create_player_rejects_invalid_gender
- test_create_player_rejects_overlong_name
- test_create_player_accepts_name_at_the_length_limit
- test_create_player_rejects_overlong_notes
- test_create_player_accepts_notes_at_the_length_limit
- test_create_player_rejects_whitespace_only_name
- test_create_player_strips_surrounding_whitespace
- test_list_players_returns_all_sorted_by_last_then_first_name
- test_list_players_omits_another_coachs_players

- test_get_player_returns_one
- test_get_player_of_another_coach_is_404
- test_list_players_with_none_stored_is_an_empty_list
- test_get_player_unknown_id_is_404
- test_get_player_malformed_id_is_422
- test_update_player_changes_only_the_fields_sent
- test_update_player_clears_a_nullable_field_with_an_explicit_null
- test_update_player_rejects_a_null_name_part [first_name, last_name]
- test_update_player_with_an_empty_body_changes_nothing
- test_update_player_bumps_updated_at
- test_update_player_rejects_coach_id
- test_update_player_rejects_a_future_date_of_birth
- test_update_player_of_another_coach_is_404
- test_update_player_unknown_id_is_404
- test_delete_player_takes_their_matches_with_them
- test_delete_player_takes_their_matches_points_with_them
- test_delete_player_takes_a_match_being_captured_with_it
- test_delete_player_leaves_the_rest_of_the_roster_alone
- test_delete_player_is_refused_while_a_transfer_code_is_outstanding
- test_cancelling_the_code_clears_the_way_for_the_delete
- test_delete_player_leaves_another_coachs_roster_alone
- test_delete_player_unknown_id_is_404
- test_player_routes_require_a_session (8 cases)
- test_a_transfer_code_moves_the_player_and_their_matches
- test_a_pasted_code_may_carry_whitespace
- test_minting_again_replaces_the_previous_code
- test_a_redeemed_code_cannot_be_used_again
- test_redeeming_an_unknown_code_is_400
- test_redeeming_your_own_code_is_409_and_leaves_it_usable
- test_deleting_a_transfer_code_stops_it_working
- test_deleting_a_transfer_code_that_was_never_minted_is_204
- test_transfer_code_for_another_coachs_player_is_404 [post, delete]
- test_a_freshly_created_player_has_no_transfer_code
- test_the_roster_flags_only_the_players_holding_a_code
- test_the_roster_reports_every_flag_in_one_statement
- test_minting_a_code_shows_up_on_the_player
- test_cancelling_a_code_clears_the_flag
- test_redeeming_a_code_clears_the_flag_on_the_response
- test_the_flag_and_the_delete_refusal_agree
- test_uploading_a_photo_answers_with_the_player
- test_an_uploaded_photo_is_served_at_the_url_it_reports
- test_replacing_a_photo_removes_the_one_it_replaced
- test_clearing_a_photo_removes_the_file
- test_clearing_a_photo_that_is_not_there_is_a_no_op
- test_another_coachs_player_is_a_404_and_stores_nothing
- test_a_file_that_is_not_an_image_is_refused

- test_deleting_a_player_removes_their_photo
- test_a_transferred_player_keeps_their_photo
- test_delete_player_records_the_ids_of_the_matches_it_freed
- test_delete_player_with_no_matches_records_nothing
- test_a_note_survives_the_write_and_comes_back_on_the_next_read
- test_create_note_files_it_under_the_areas_sent
- test_create_note_defaults_to_no_focus_areas
- test_create_note_cannot_name_a_player_in_the_body
- test_create_note_against_another_coachs_player_is_404
- test_list_notes_is_newest_session_first
- test_list_notes_breaks_a_tie_on_when_the_note_was_written
- test_list_notes_carries_only_this_players_notes
- test_list_notes_with_none_written_is_an_empty_list
- test_list_notes_of_another_coachs_player_is_404
- test_update_note_changes_only_the_fields_sent
- test_update_note_clears_the_focus_areas_with_an_empty_list
- test_update_note_of_another_coachs_player_is_404
- test_a_note_reached_through_the_wrong_player_is_404
- test_delete_note_removes_it_and_leaves_the_rest
- test_delete_note_unknown_id_is_404
- test_delete_player_takes_their_notes_with_them

tests/routers/test_matches.py — 141/141

- test_create_match_persists_and_returns_row
- test_create_match_with_only_the_required_fields
- test_create_match_rejects_unknown_player
- test_create_match_against_another_coachs_player_is_404
- test_create_match_for_a_transferred_player_is_404
- test_create_match_rejects_missing_player_id
- test_create_match_rejects_a_malformed_player_id
- test_create_match_rejects_missing_played_on
- test_create_match_rejects_a_malformed_played_on
- test_create_match_rejects_a_future_played_on
- test_create_match_accepts_a_played_on_of_today
- test_create_match_rejects_an_implausibly_old_played_on
- test_create_match_rejects_an_invalid_surface
- test_create_match_rejects_an_invalid_scoring_system
- test_create_match_rejects_an_invalid_result
- test_create_match_rejects_a_whitespace_only_opponent_name
- test_create_match_rejects_an_overlong_opponent_name
- test_create_match_accepts_free_text_at_the_length_limit
- test_create_match_strips_surrounding_whitespace
- test_create_match_stores_a_client_supplied_id
- test_create_match_mints_a_fresh_id_each_time_none_is_sent
- test_create_match_repeated_with_the_same_id_returns_the_stored_match
- test_create_match_repeated_under_another_player_returns_the_stored_match

- test_create_match_refuses_an_id_the_coach_deleted
- test_create_match_allows_an_id_whose_delete_has_aged_out
- test_create_match_with_another_coachs_match_id_is_404
- test_create_match_rejects_a_malformed_id
- test_create_match_rejects_an_id_the_client_could_not_have_minted [nil, v1, v5]
- test_list_matches_newest_first
- test_list_matches_breaks_same_day_ties_by_creation_order
- test_list_matches_spans_every_player_of_the_caller
- test_list_matches_omits_another_coachs_matches
- test_list_matches_filtered_by_player
- test_list_matches_filtered_by_an_unknown_player_is_empty
- test_list_matches_filtered_by_another_coachs_player_is_empty
- test_list_matches_rejects_a_malformed_player_id
- test_list_matches_with_none_stored_is_an_empty_list
- test_get_match_returns_one
- test_get_match_of_another_coach_is_404
- test_get_match_unknown_id_is_404
- test_get_match_malformed_id_is_422
- test_update_match_changes_only_the_fields_sent
- test_update_match_clears_a_nullable_field_with_an_explicit_null
- test_update_match_records_a_result
- test_update_match_edits_the_metadata_of_a_completed_match
- test_update_match_with_an_empty_body_leaves_a_completed_match_alone
- test_update_match_cannot_change_a_recorded_result
- test_update_match_cannot_change_a_completed_matchs_scoring_system
- test_update_match_cannot_clear_a_completed_matchs_scoring_system
- test_update_match_fills_in_a_missing_scoring_system_after_completion
- test_update_match_resending_the_stored_values_is_a_no_op
- test_update_match_rejects_a_null_played_on
- test_update_match_with_an_empty_body_changes_nothing
- test_update_match_bumps_updated_at
- test_update_match_cannot_move_the_match_to_another_player
- test_update_match_rejects_a_future_played_on
- test_update_match_of_another_coach_is_404
- test_update_match_unknown_id_is_404
- test_delete_match_removes_it
- test_delete_match_takes_its_derived_sets_with_it
- test_delete_match_takes_its_points_with_it
- test_delete_match_leaves_another_matchs_points_alone
- test_delete_match_records_the_id_it_freed
- test_deleting_a_reused_id_starts_its_window_over
- test_delete_match_twice_is_404
- test_delete_match_of_another_coach_is_404
- test_delete_match_unknown_id_is_404
- test_delete_match_malformed_id_is_422
- test_delete_match_is_allowed_on_a_completed_match

- test_delete_match_under_another_devices_claim_is_refused [False, True]
- test_the_capturing_device_may_delete_its_own_match
- test_a_lapsed_claim_does_not_block_the_delete
- test_releasing_the_claim_clears_the_way_for_the_delete
- test_the_capture_refusal_still_stands_over_a_match_with_points
- test_claim_capture_takes_a_free_match
- test_claim_capture_renews_the_claim_of_the_device_holding_it
- test_claim_capture_refuses_a_second_device
- test_claim_capture_takes_over_a_lapsed_claim
- test_claim_capture_on_a_completed_match_is_409
- test_claim_capture_of_another_coachs_match_is_404
- test_claim_capture_unknown_id_is_404
- test_claim_capture_requires_a_device_id
- test_release_capture_frees_the_match_for_another_device
- test_release_capture_on_an_unclaimed_match_is_a_no_op
- test_release_capture_of_another_coachs_match_is_404
- test_recording_the_result_releases_the_capture_claim
- test_ingest_points_stores_the_batch_and_answers_with_the_derived_sets
- test_ingest_stores_every_field_the_client_stamped
- test_ingesting_the_same_batch_again_lands_on_the_same_answer
- test_a_repeated_point_keeps_the_stored_version
- test_an_overlapping_batch_converges
- test_ingest_recomputes_the_sets_a_late_batch_changes
- test_ingest_into_a_completed_match_is_refused
- test_a_sequence_already_held_by_a_different_point_is_refused
- test_a_point_already_stored_against_another_match_is_refused
- test_one_invalid_point_rejects_the_whole_batch
- test_an_empty_batch_is_refused
- test_a_flush_from_the_claim_holder_renews_the_claim
- test_a_flush_from_another_device_is_still_ingested
- test_a_flush_does_not_claim_a_match_nobody_holds
- test_a_flush_does_not_revive_the_lapsed_claim_it_once_held
- test_ingest_into_another_coachs_match_is_404
- test_ingest_into_an_unknown_match_is_404
- test_ingest_into_a_deleted_match_is_410
- test_ingest_into_a_match_whose_delete_has_aged_out_is_404
- test_list_points_orders_by_sequence_regardless_of_insertion_order
- test_list_points_of_a_match_with_none_recorded_is_an_empty_list
- test_list_points_of_another_coachs_match_is_404
- test_list_points_of_an_unknown_match_is_404
- test_list_points_round_trips_every_field_the_ingest_route_stored
- test_list_points_carries_no_ingestion_timestamps
- test_an_integrity_error_that_is_not_a_conflict_is_not_absorbed
- test_the_routes_that_read_the_claim_lock_the_match_row (5 cases)
- test_match_routes_require_a_session (9 cases)
- test_list_in_progress_matches_returns_the_claimed_match

- test_in_progress_does_not_care_which_device_holds_the_claim
- test_in_progress_puts_the_freshest_claim_first
- test_a_match_nobody_is_capturing_is_not_in_progress
- test_a_lapsed_claim_is_not_in_progress
- test_a_completed_match_is_not_in_progress
- test_in_progress_omits_another_coachs_match
- test_create_match_refuses_an_id_a_player_delete_freed
- test_update_a_deleted_match_is_410
- test_update_a_match_a_player_delete_took_is_410
- test_ingest_into_a_match_a_player_delete_took_is_410

tests/schemas/test_player.py — 48/48

- test_only_the_name_parts_are_required
- test_both_name_parts_are_required [first_name, last_name]
- test_rejects_a_blank_name_part (6 cases)
- test_accepts_a_single_character_name
- test_strips_surrounding_whitespace_from_a_name
- test_strips_a_name_before_measuring_its_length
- test_rejects_a_name_over_the_length_limit
- test_accepts_a_date_of_birth_of_today
- test_rejects_a_date_of_birth_of_tomorrow
- test_accepts_a_date_of_birth_just_inside_the_age_limit
- test_rejects_a_date_of_birth_just_outside_the_age_limit
- test_coerces_a_dominant_hand_string_to_the_enum
- test_rejects_a_dominant_hand_outside_the_enum
- test_coerces_a_gender_string_to_the_enum
- test_rejects_a_gender_outside_the_enum
- test_accepts_notes_at_the_length_limit
- test_rejects_notes_over_the_length_limit
- test_strips_surrounding_whitespace_from_notes
- test_rejects_blank_notes (2 cases)
- test_accepts_an_explicit_null_for_notes
- test_an_empty_update_carries_no_fields
- test_an_update_carries_only_the_fields_sent
- test_an_update_distinguishes_an_absent_field_from_a_null_one
- test_an_update_rejects_a_null_name_part [first_name, last_name]
- test_an_update_accepts_a_null_for_every_nullable_field [date_of_birth, dominant_hand, gender, notes]
- test_an_update_applies_the_name_bounds
- test_an_update_rejects_a_date_of_birth_of_tomorrow
- test_an_update_rejects_a_date_of_birth_outside_the_age_limit
- test_an_update_accepts_a_date_of_birth_inside_the_bounds
- test_a_transfer_code_is_bounded
- test_a_transfer_code_is_a_secret
- test_a_read_carries_the_photo_url_derived_from_the_key
- test_a_created_player_carries_the_password_in_the_clear

- test_neither_create_nor_update_accepts_a_username [PlayerCreate, PlayerUpdate]
- test_neither_create_nor_update_accepts_a_photo_url [PlayerCreate, PlayerUpdate]

tests/schemas/test_match.py — 51/51

- test_only_the_player_and_date_are_required
- test_coerces_an_id_string_to_a_uuid
- test_rejects_a_malformed_id
- test_treats_an_explicitly_null_id_as_an_absent_one
- test_coerces_a_player_id_string_to_a_uuid
- test_rejects_a_malformed_player_id
- test_rejects_a_missing_player_id
- test_rejects_a_missing_played_on
- test_rejects_a_played_on_that_is_not_an_iso_date (4 cases)
- test_accepts_a_played_on_of_today
- test_rejects_a_played_on_of_tomorrow
- test_accepts_a_played_on_just_inside_the_age_limit
- test_rejects_a_played_on_just_outside_the_age_limit
- test_coerces_a_surface_string_to_the_enum
- test_rejects_a_surface_outside_the_enum
- test_coerces_a_scoring_system_string_to_the_enum
- test_rejects_a_scoring_system_outside_the_enum
- test_coerces_a_result_string_to_the_enum
- test_rejects_a_result_outside_the_enum
- test_accepts_an_omitted_or_null_text_field [opponent_name, event]
- test_rejects_a_blank_text_field (6 cases)
- test_accepts_a_single_character_text_field [opponent_name, event]
- test_strips_surrounding_whitespace_from_a_text_field [opponent_name, event]
- test_strips_a_text_field_before_measuring_its_length [opponent_name, event]
- test_rejects_a_text_field_over_the_length_limit [opponent_name, event]
- test_an_empty_update_carries_no_fields
- test_an_update_carries_only_the_fields_sent
- test_an_update_rejects_a_player_id
- test_an_update_rejects_a_null_played_on
- test_an_update_accepts_a_null_for_every_nullable_field [opponent_name, surface, event, scoring_system, result]
- test_an_update_applies_the_text_bounds
- test_an_update_rejects_a_played_on_of_tomorrow
- test_an_update_rejects_a_played_on_outside_the_past_limit
- test_an_update_accepts_a_played_on_inside_the_bounds

tests/schemas/test_point.py — 71/71

- test_the_default_shape_round_trips
- test_shapes_the_contract_allows (15 cases)
- test_shapes_the_contract_refuses (34 cases)
- test_a_missing_required_field_is_refused [sequence, server, won, serve1_outcome, end_reason]

- test_every_stamp_is_required (7 cases)
- test_the_note_is_stripped_and_bounded
- test_the_full_enum_membership_reaches_the_columns
- test_a_batch_carries_a_device_and_its_points
- test_a_batch_without_a_device_is_refused
- test_an_empty_batch_is_refused
- test_a_batch_at_the_ceiling_is_accepted
- test_a_batch_over_the_ceiling_is_refused
- test_a_batch_cannot_repeat_a_point_id
- test_a_batch_cannot_repeat_a_sequence

tests/test_usernames.py — 13/13

- test_the_surname_prefers_consonants_and_the_first_name_reads_plainly
- test_skipped_vowels_fill_in_when_the_consonants_run_short
- test_accents_fold_to_their_bare_letters
- test_punctuation_and_spaces_drop_out_before_folding
- test_a_short_name_is_padded_to_keep_the_prefix_fixed_width
- test_a_name_with_no_latin_letters_folds_to_the_placeholder
- test_casefold_handles_one_to_many_letters
- test_the_prefix_is_always_six_characters
- test_the_first_holder_of_a_name_is_numbered_001
- test_a_namesake_takes_the_next_number
- test_numbering_resumes_past_the_highest_ever_minted_not_the_count
- test_another_name_s_numbers_do_not_interfere
- test_past_999_the_number_widens_rather_than_wrapping

tests/test_photos.py — 23/23

- test_a_jpeg_is_stored_as_a_square_webp
- test_every_accepted_format_round_trips [JPEG, PNG, WEBP]
- test_two_uploads_are_stored_under_different_keys
- test_a_file_that_is_not_an_image_is_refused
- test_an_image_in_an_unlisted_format_is_refused
- test_an_image_claiming_too_many_pixels_is_refused
- test_an_image_at_the_pixel_bound_is_accepted
- test_the_stored_photo_carries_no_exif
- test_a_rotated_photo_is_uprighted_before_it_is_cropped
- test_a_wide_photo_is_centre_cropped_rather_than_squashed
- test_transparency_is_flattened_onto_white
- test_a_palette_image_is_stored_as_colour
- test_an_image_pillow_refuses_to_open_at_all_is_refused
- test_a_truncated_image_is_refused
- test_nothing_is_written_for_an_image_that_is_refused
- test_deleting_a_stored_photo_removes_it
- test_deleting_a_photo_that_is_already_gone_is_not_an_error
- test_deleting_nothing_is_not_an_error
- test_a_key_that_could_escape_the_media_directory_is_refused

- test_a_url_is_built_from_a_key
- test_no_key_is_no_url

tests/test_match_sets.py — 9/9

- test_a_set_in_flight_has_no_winner
- test_the_closing_point_names_the_set_winner
- test_a_set_won_by_the_tracked_player
- test_each_set_gets_its_own_row
- test_points_of_another_match_are_not_counted
- test_a_second_recompute_overwrites_the_row_and_stamps_updated_at
- test_a_set_row_the_points_no_longer_support_is_deleted
- test_a_match_with_no_points_keeps_no_set_rows
- test_an_unknown_match_derives_nothing

tests/test_csrf.py — 18/18

- test_a_write_from_our_own_origin_is_allowed [POST, PUT, PATCH, DELETE]
- test_a_write_from_another_origin_is_refused [POST, PUT, PATCH, DELETE]
- test_a_write_from_a_sibling_host_of_our_own_apex_is_refused
- test_a_write_with_no_origin_at_all_is_refused
- test_a_write_with_an_unusable_origin_is_refused (3 cases)
- test_a_read_from_another_origin_is_allowed [GET, HEAD, OPTIONS]
- test_another_port_on_our_own_host_is_another_origin
- test_the_scheme_is_deliberately_not_compared

tests/test_headers.py — 9/9

- test_a_stored_photo_carries_them_too
- test_json_responses_carry_the_strict_headers
- test_error_responses_carry_them_too
- test_a_refused_origin_gets_them_too
- test_an_unhandled_exception_gets_them_too
- test_docs_page_gets_its_scoped_csp
- test_the_docs_csp_survives_the_deployed_root_path
- test_docs_relaxation_does_not_reach_the_spec_route
- test_the_html_pages_nobody_uses_are_gone

tests/test_body_limit.py — 7/7

- test_oversized_declared_body_is_refused
- test_oversized_chunked_body_is_refused
- test_body_at_the_limit_reaches_the_route
- test_overflow_outside_an_exception_handler_is_still_a_413
- test_overflow_after_a_started_response_propagates
- test_unreadable_declared_length_is_left_to_the_count
- test_non_body_messages_are_not_counted

tests/test_rate_limit.py — 17/17

- test_a_key_is_admitted_up_to_the_burst_and_refused_beyond_it

- test_the_scale_setting_widens_a_bucket
- test_the_refusal_says_when_to_retry
- test_tokens_refill_at_the_sustained_rate
- test_an_idle_key_accrues_no_more_than_the_burst
- test_keys_do_not_share_a_bucket
- test_a_refused_request_spends_nothing
- test_a_refunded_request_leaves_the_bucket_as_it_found_it
- test_a_refund_cannot_bank_more_than_the_burst
- test_refunding_a_key_that_never_spent_is_harmless
- test_taking_from_several_buckets_admits_only_when_all_of_them_can_pay
- test_a_bucket_refused_by_another_is_not_charged_for_it
- test_a_slower_per_key_tier_leaves_the_shared_one_to_other_keys
- test_the_refusal_quotes_the_bucket_furthest_from_a_token
- test_the_furthest_bucket_is_quoted_even_when_another_is_refused_first
- test_a_new_key_evicts_buckets_that_have_refilled_to_full
- test_a_new_key_keeps_buckets_still_refilling

tests/test_logs.py — 16/16

- test_every_response_carries_a_request_id
- test_a_well_formed_inbound_id_is_kept
- test_a_malformed_inbound_id_is_replaced (4 cases)
- test_each_request_logs_one_line_with_the_documented_fields
- test_an_authenticated_request_logs_the_coach_id
- test_an_authenticated_player_request_logs_the_player_id
- test_a_response_no_route_saw_is_logged_too
- test_a_crashed_handler_still_gets_its_line
- test_a_line_logged_inside_a_route_inherits_the_request_id
- test_no_secret_from_the_auth_flows_reaches_the_log
- test_secret_named_fields_are_redacted_at_any_depth
- test_a_tty_gets_console_lines_and_a_pipe_gets_json
- test_uvicorns_loggers_are_rerouted_and_its_access_log_dropped

tests/test_errors.py — 11/11

- test_a_secret_field_is_the_only_thing_redacted
- test_a_password_alongside_another_fields_error_is_redacted
- test_a_password_nested_below_the_reported_field_is_redacted
- test_an_unknown_keys_value_is_redacted
- test_an_error_carrying_no_input_is_passed_through
- test_a_field_that_is_not_secret_keeps_its_input
- test_every_secret_field_in_the_schemas_is_named_in_secret_fields
- test_an_unhandled_exception_answers_a_readable_json_body
- test_an_unhandled_exception_tells_the_client_nothing_about_itself
- test_the_500_carries_the_request_id_that_logged_the_traceback
- test_a_failure_above_the_logging_middleware_is_answered_too

apps/api/tests/schemas/test_player_note.py — 25/25

- test_only_the_body_and_the_date_are_required
- test_rejects_a_missing_required_field
- test_rejects_a_key_the_schema_does_not_carry
- test_strips_surrounding_whitespace_from_the_body
- test_rejects_a_whitespace_only_body
- test_accepts_a_body_at_the_length_limit
- test_rejects_an_overlong_body
- test_accepts_a_note_dated_today
- test_rejects_a_note_dated_in_the_future
- test_accepts_a_note_about_a_session_seasons_ago
- test_rejects_an_implausibly_old_note_date
- test_keeps_focus_areas_in_declaration_order
- test_files_a_note_under_each_area_once
- test_rejects_a_focus_area_outside_the_four
- test_rejects_a_null_focus_areas_list
- test_an_update_may_carry_nothing
- test_an_update_carries_each_mutable_field
- test_an_update_clears_the_focus_areas_with_an_empty_list
- test_an_update_rejects_a_null_over_a_non_nullable_column
- test_an_update_applies_the_same_bounds_as_a_create

Backlog item 3

Test Results Summary

API (pytest): 755/755 passed across 21 files
Web (vitest): 159/159 passed across 17 files
Total: 914/914 passed

API — pytest (755/755)

tests/models/test_coach.py — 29/29

- test_coach_round_trips
- test_created_at_differs_between_rows_in_one_transaction
- test_updated_at_advances_when_a_row_changes
- test_database_requires_every_column [email]
- test_database_requires_every_column [display_name]
- test_database_requires_every_column [password_hash]
- test_database_rejects_a_duplicate_email
- test_database_rejects_an_email_that_is_not_lowercased
- test_the_orm_lowercases_an_email_on_the_way_in
- test_database_rejects_an_email_over_the_column_width
- test_database_rejects_a_display_name_over_the_column_width
- test_database_rejects_default_surface_outside_the_enum
- test_database_rejects_default_scoring_system_outside_the_enum
- test_database_accepts_every_member_of_the_reused_enums [default_surface: hard, clay, grass, carpet]
- test_database_accepts_every_member_of_the_reused_enums [default_scoring_system: traditional, mens_grand_slam, womens_grand_slam, no_ad_scoring, match_tiebreak_10_points, short_sets, pro_set, legacy]
- test_database_accepts_a_coach_with_no_preferences
- test_database_accepts_a_password_hash_of_any_length
- test_a_coach_without_a_photo_has_no_photo_url
- test_a_stored_photo_key_becomes_a_url

tests/models/test_coach_session.py — 8/8

- test_coach_session_round_trips
- test_database_requires_every_column [coach_id, token_hash, expires_at]

- test_database_rejects_a_duplicate_token_hash
- test_database_rejects_a_token_hash_over_the_column_width
- test_database_requires_the_coach_to_exist
- test_a_coach_with_sessions_cannot_be_hard_deleted

tests/models/test_email_verification.py — 13/13

- test_email_verification_round_trips
- test_database_requires_every_column [email, attempts, expires_at]
- test_database_rejects_a_second_row_for_the_same_address
- test_database_rejects_a_duplicate_token_hash
- test_database_permits_many_rows_at_the_code_stage
- test_database_rejects_a_hash_over_the_column_width [otp_hash, token_hash]
- test_database_rejects_an_email_over_the_column_width
- test_database_rejects_an_email_that_is_not_lowercased
- test_database_rejects_a_row_holding_both_secrets
- test_database_rejects_a_row_holding_neither_secret

tests/models/test_password_reset.py — 13/13

- test_password_reset_round_trips
- test_database_requires_every_column [coach_id, attempts, expires_at]
- test_database_rejects_a_second_row_for_the_same_coach
- test_database_rejects_an_unknown_coach
- test_database_refuses_to_delete_a_coach_with_a_pending_reset
- test_database_rejects_a_hash_over_the_column_width [otp_hash, token_hash]
- test_database_rejects_a_duplicate_token_hash
- test_database_permits_many_rows_at_the_code_stage
- test_database_rejects_a_row_holding_both_secrets
- test_database_rejects_a_row_holding_neither_secret

tests/models/test_player.py — 23/23

- test_player_round_trips
- test_committed_rows_do_not_leak_between_tests_a
- test_committed_rows_do_not_leak_between_tests_b
- test_only_the_name_parts_and_the_owner_are_required

- test_created_at_differs_between_rows_in_one_transaction
- test_updated_at_advances_when_a_row_changes
- test_has_transfer_code_follows_the_code_row
- test_has_transfer_code_answers_for_one_player_at_a_time
- test_database_requires_both_name_parts [first_name, last_name]
- test_database_rejects_dominant_hand_outside_the_enum
- test_database_rejects_gender_outside_the_enum
- test_database_rejects_a_name_over_the_column_width [first_name, last_name]
- test_database_requires_a_username
- test_database_rejects_a_duplicate_username
- test_database_rejects_a_username_over_the_column_width
- test_database_requires_an_owner
- test_database_rejects_an_unknown_coach_id
- test_database_refuses_to_delete_a_coach_who_still_has_players
- test_coach_id_is_indexed
- test_a_player_without_a_photo_has_no_photo_url
- test_a_stored_photo_key_becomes_a_url

tests/routers/test_auth.py — 131/131

- test_logging_in_returns_the_coach_and_sets_the_session_cookie
- test_the_cookie_is_locked_down
- test_logging_in_stores_the_hash_of_the_token_and_never_the_token
- test_a_login_expires_a_lifetime_from_now
- test_each_login_is_its_own_session
- test_logging_in_clears_the_coach_s_expired_sessions
- test_logging_in_leaves_other_coaches_expired_sessions_alone
- test_a_wrong_password_is_rejected_without_a_session
- test_an_unknown_email_answers_exactly_like_a_wrong_password
- test_an_unknown_email_still_burns_a_verification
- test_a_mixed_case_email_finds_the_account
- test_a_password_below_the_sign_up_floor_is_a_401_not_a_422
- test_a_stale_hash_is_upgraded_on_login

- test_a_current_hash_is_left_alone_on_login
- test_a_login_that_verified_a_retired_password_mints_no_session
- test_a_login_locks_the_coach_row_before_minting_a_session
- test_login_attempts_beyond_the_account_burst_are_refused
- test_the_login_429_answers_a_known_and_an_unknown_address_alike
- test_a_flooded_account_does_not_lock_out_the_rest
- test_the_account_bucket_folds_case
- test_cycling_accounts_does_not_escape_the_address_burst
- test_login_has_a_total_bound_no_address_escapes
- test_a_body_the_schema_rejects_is_still_counted
- test_an_arrival_beyond_the_address_burst_spends_no_verification
- test_a_correct_password_is_never_refused_by_the_account_tier
- test_logging_out_revokes_the_session_and_clears_the_cookie
- test_logging_out_only_ends_the_presented_session
- test_logging_out_signed_out_is_still_a_204
- test_logging_out_with_a_forged_cookie_revokes_nothing
- test_a_rejected_login_does_not_echo_the_password
- test_revoking_ends_the_coach_s_other_sessions
- test_the_token_the_revocation_was_made_with_stops_working
- test_a_revocation_from_a_session_due_for_renewal_hands_back_the_new_token
- test_revoking_with_nothing_else_to_end_is_still_a_204
- test_revoking_needs_a_session_of_its_own
- test_requesting_a_code_emails_it_to_the_address
- test_a_failed_send_answers_502
- test_a_failed_send_gives_the_address_its_code_budget_back
- test_the_code_is_stored_hashed_and_expiring
- test_a_registered_address_is_refused_with_no_code
- test_a_mixed_case_address_meets_the_409
- test_the_stored_address_is_lowercased
- test_requesting_again_replaces_the_code
- test_requesting_a_code_undoes_a_finished_verification

- test_requesting_sweeps_expired_rows_and_leaves_live_ones
- test_a_fourth_code_for_one_address_is_refused
- test_a_registered_address_meets_the_same_429_as_a_fresh_one
- test_the_code_budget_folds_case
- test_cycling_addresses_meets_the_address_burst
- test_request_otp_has_a_total_bound_no_address_escapes
- test_answering_the_code_sets_the_verification_cookie
- test_the_verification_cookie_is_locked_down
- test_answering_the_code_retires_it_for_the_token
- test_the_whole_flow_ends_in_an_account
- test_a_wrong_code_is_refused_and_charged
- test_an_unknown_address_answers_exactly_like_a_wrong_code
- test_an_expired_code_is_refused
- test_an_answered_code_cannot_be_answered_again
- test_verification_folds_the_address_case
- test_the_attempt_budget_kills_the_code
- test_a_guess_short_of_the_budget_does_not_cost_the_code
- test_guess_arrivals_beyond_the_address_burst_are_refused
- test_a_rejected_verification_does_not_echo_the_code
- test_the_flow_runs_on_the_default_logging_sender
- test_forgetting_a_password_emails_a_code
- test_an_unknown_address_answers_the_same_202_with_no_mail_and_no_row
- test_the_reset_code_is_stored_hashed_and_expiring
- test_a_mixed_case_address_finds_its_account
- test_forgetting_again_replaces_the_code
- test_forgetting_undoes_a_finished_verification
- test_forgetting_sweeps_expired_rows_and_leaves_live_ones
- test_a_fourth_code_for_one_account_is_refused
- test_the_forgot_password_429_answers_a_known_and_an_unknown_address_alike
- test_forgot_password_cycling_addresses_meets_the_address_burst
- test_forgot_password_has_a_total_bound_no_address_escapes

- test_a_failed_send_answers_502_and_keeps_the_row
- test_a_failed_reset_send_gives_the_address_its_code_budget_back
- test_answering_the_reset_code_sets_the_reset_cookie
- test_the_reset_cookie_is_locked_down
- test_answering_the_reset_code_retires_it_for_the_token
- test_a_correct_code_resets_the_password
- test_a_reset_to_a_password_the_policy_refuses_is_a_400
- test_a_reset_to_a_breached_password_is_a_400
- test_a_reset_survives_a_corpus_that_cannot_answer
- test_a_reset_revokes_every_live_session
- test_the_reset_row_is_consumed
- test_a_wrong_reset_code_is_refused_and_charged
- test_an_unknown_address_answers_exactly_like_a_wrong_reset_code
- test_an_expired_reset_code_is_refused
- test_an_answered_reset_code_cannot_be_answered_again
- test_the_attempt_budget_kills_the_reset
- test_a_guess_short_of_the_budget_does_not_cost_the_reset
- test_the_reset_flow_folds_the_address_case
- test_reset_guess_arrivals_beyond_the_address_burst_are_refused
- test_a_rejected_reset_verification_does_not_echo_the_code
- test_a_reset_without_the_cookie_is_refused
- test_a_reset_cookie_proving_one_address_cannot_reset_another
- test_an_unknown_address_answers_exactly_like_a_dead_reset_cookie
- test_an_expired_reset_proof_is_refused
- test_reset_arrivals_beyond_the_address_burst_are_refused
- test_a_rejected_reset_does_not_echo_the_new_password
- test_the_current_password_buys_a_new_one
- test_the_token_the_change_was_made_with_stops_working
- test_a_change_from_a_session_due_for_renewal_hands_back_the_new_token
- test_the_new_password_is_what_signs_in_afterwards
- test_a_wrong_current_password_changes_nothing

- test_a_change_to_a_password_the_policy_refuses_is_a_400
- test_a_change_to_a_breached_password_is_a_400
- test_the_policy_takes_the_coach_s_own_words_from_their_row
- test_a_wrong_current_password_answers_before_the_policy_does
- test_changing_a_password_needs_a_session
- test_the_change_ends_the_coach_s_other_sessions
- test_a_pending_reset_code_dies_with_a_deliberate_change
- test_wrong_guesses_beyond_the_coach_s_budget_are_refused
- test_a_correct_password_is_never_refused_by_the_coach_tier
- test_change_arrivals_beyond_the_address_burst_are_refused
- test_an_unauthenticated_flood_spends_none_of_the_change_budget
- test_change_password_has_a_total_bound_no_address_escapes
- test_a_rejected_change_echoes_neither_password
- test_a_change_verified_before_another_write_overwrites_neither_hash_nor_revocation
- test_a_change_locks_the_coach_row_before_writing_the_new_hash
- test_a_coach_beyond_the_per_coach_burst_is_refused
- test_one_coach_s_spending_does_not_charge_another
- test_a_dead_cookie_s_401_does_not_charge_the_coach
- test_a_drained_shared_budget_still_admits_a_revocation
- test_a_drained_shared_budget_still_admits_a_password_change
- test_the_recovery_budget_still_bounds_the_recovery_routes
- test_a_drained_recovery_budget_leaves_the_ordinary_routes_alone
- test_a_flood_of_unknown_tokens_is_refused
- test_a_drained_unknown_token_tier_still_admits_a_live_session
- test_a_request_carrying_no_cookie_is_never_charged

tests/routers/test_coaches.py — 99/99

- test_signing_up_returns_the_new_coach
- test_signing_up_never_echoes_the_password_or_its_hash
- test_signing_up_stores_a_hash_the_password_verifies_against
- test_two_coaches_sharing_a_password_get_different_hashes
- test_a_mixed_case_email_is_stored_and_read_back_lowercased

- test_signing_up_twice_with_one_email_is_a_conflict
- test_an_email_differing_only_in_case_is_a_conflict
- test_a_transferred_player_keeps_their_notes
- test_an_integrity_error_that_is_not_a_duplicate_email_is_not_a_conflict
- test_signing_up_without_a_password_is_rejected
- test_signing_up_with_too_short_a_password_is_rejected
- test_a_rejected_sign_up_does_not_echo_the_password
- test_signing_up_with_a_password_the_policy_refuses_is_a_400 [mostly one repeated character / consecutive characters / too common / contains name or email]
- test_a_refused_password_leaves_the_verification_to_try_again_with
- test_signing_up_with_a_breached_password_is_a_400
- test_a_breached_password_also_leaves_the_verification_standing
- test_signing_up_survives_a_corpus_that_cannot_answer
- test_a_password_refused_by_policy_is_not_echoed
- test_signing_up_without_a_verification_cookie_is_a_401
- test_signing_up_with_a_cookie_for_another_email_is_a_401
- test_signing_up_with_an_expired_verification_is_a_401
- test_signing_up_with_a_token_verify_otp_never_minted_is_a_401
- test_signing_up_consumes_the_verification
- test_a_spent_verification_cookie_cannot_sign_up_again
- test_a_conflict_does_not_burn_the_verification
- test_sign_up_arrivals_beyond_the_burst_are_refused
- test_sign_up_has_a_total_bound_no_address_escapes
- test_a_sign_up_refused_by_the_shared_tier_spends_no_per_address_token
- test_me_returns_the_signed_in_coach_and_nobody_else
- test_me_without_a_cookie_is_a_401
- test_me_with_a_token_no_login_minted_is_a_401
- test_me_with_an_expired_session_is_a_401_and_discards_it
- test_me_renews_a_session_past_its_half_life
- test_a_renewed_session_re_sends_the_cookie
- test_a_renewal_reaches_the_browser_even_on_an_error_response

- test_a_renewal_reaches_the_browser_even_when_the_route_crashes
- test_me_leaves_a_session_in_its_first_half_alone
- test_an_authenticated_request_may_sweep_all_expired_sessions
- test_a_missed_gc_roll_leaves_expired_sessions_alone
- test_me_never_returns_the_password_hash
- test_updating_the_display_name_stores_the_new_one
- test_updating_the_preferences_stores_them
- test_updating_a_preference_to_null_clears_it
- test_updating_to_a_preference_outside_its_enum_is_rejected [default_surface: sand / default_scoring_system: best_of_3]
- test_me_returns_null_preferences_until_they_are_set
- test_an_update_sending_no_fields_changes_nothing
- test_updating_the_display_name_to_null_is_rejected
- test_updating_to_a_display_name_outside_the_bounds_is_rejected [blank / too long]
- test_an_update_refuses_a_field_the_schema_leaves_out [email / password / password_hash]
- test_updating_without_a_cookie_is_a_401
- test_uploading_a_photo_answers_with_the_coach
- test_an_uploaded_photo_is_served_at_the_url_it_reports
- test_replacing_a_photo_removes_the_one_it_replaced
- test_clearing_a_photo_removes_the_file
- test_clearing_a_photo_that_is_not_there_is_a_no_op
- test_a_file_that_is_not_an_image_is_refused
- test_an_image_with_too_many_pixels_is_refused
- test_an_upload_with_no_file_part_is_a_422
- test_uploading_a_photo_needs_a_session
- test_clearing_a_photo_needs_a_session
- test_clearing_a_photo_answers_401_when_the_coach_is_already_gone
- test_uploads_are_rate_limited_per_coach
- test_closing_an_account_erases_the_coach
- test_closing_an_account_takes_every_photo_under_it
- test_closing_an_account_is_refused_while_a_player_is_active

- test_a_refusal_spends_none_of_the_wrong_guess_budget
- test_closing_an_account_may_take_the_roster_along
- test_closing_an_account_takes_a_match_being_captured_with_it
- test_closing_an_account_ends_every_session_and_clears_the_cookie
- test_a_closure_from_a_session_due_for_renewal_sends_no_renewed_cookie
- test_closing_an_account_deletes_a_live_reset_code
- test_closing_an_account_deletes_a_deleted_match_tombstone
- test_a_transfer_code_survives_the_delete_players_waiver
- test_the_email_can_sign_up_again_afterwards
- test_closing_an_account_leaves_every_other_coach_untouched
- test_a_wrong_password_erases_nothing
- test_a_closure_verified_before_a_password_write_erases_nothing
- test_closing_an_account_needs_a_session
- test_a_rejected_closure_echoes_no_password
- test_wrong_guesses_beyond_the_coach_s_budget_are_refused
- test_a_correct_password_is_never_refused_by_the_coach_tier
- test_closure_arrivals_beyond_the_address_burst_are_refused
- test_an_unauthenticated_flood_spends_none_of_the_closure_budget
- test_closing_an_account_has_a_total_bound_no_address_escapes
- test_the_right_password_verifies_and_leaves_the_account_alone
- test_a_wrong_password_is_refused
- test_verifying_needs_a_session
- test_a_rejected_verification_echoes_no_password
- test_wrong_guesses_here_spend_the_closure_s_budget
- test_a_correct_password_still_verifies_while_someone_guesses
- test_verifications_and_closures_share_one_arrival_budget
- test_an_unauthenticated_flood_spends_none_of_that_budget
- test_coaches_are_not_listable

tests/routers/test_matches.py — 141/141

- test_list_in_progress_matches_returns_the_claimed_match
- test_in_progress_does_not_care_which_device_holds_the_claim
- test_in_progress_puts_the_freshest_claim_first

- test_a_match_nobody_is_capturing_is_not_in_progress
- test_a_lapsed_claim_is_not_in_progress
- test_a_completed_match_is_not_in_progress
- test_in_progress_omits_another_coachs_match
- test_create_match_refuses_an_id_a_player_delete_freed
- test_update_a_deleted_match_is_410
- test_update_a_match_a_player_delete_took_is_410
- test_ingest_into_a_match_a_player_delete_took_is_410
- test_create_match_persists_and_returns_row
- test_create_match_with_only_the_required_fields
- test_create_match_rejects_unknown_player
- test_create_match_against_another_coachs_player_is_404
- test_create_match_for_a_transferred_player_is_404
- test_create_match_rejects_missing_player_id
- test_create_match_rejects_a_malformed_player_id
- test_create_match_rejects_missing_played_on
- test_create_match_rejects_a_malformed_played_on
- test_create_match_rejects_a_future_played_on
- test_create_match_accepts_a_played_on_of_today
- test_create_match_rejects_an_implausibly_old_played_on
- test_create_match_rejects_an_invalid_surface
- test_create_match_rejects_an_invalid_scoring_system
- test_create_match_rejects_an_invalid_result
- test_create_match_rejects_a_whitespace_only_opponent_name
- test_create_match_rejects_an_overlong_opponent_name
- test_create_match_accepts_free_text_at_the_length_limit
- test_create_match_strips_surrounding_whitespace
- test_create_match_stores_a_client_supplied_id
- test_create_match_mints_a_fresh_id_each_time_none_is_sent
- test_create_match_repeated_with_the_same_id_returns_the_stored_match
- test_create_match_repeated_under_another_player_returns_the_stored_match
- test_create_match_refuses_an_id_the_coach_deleted
- test_create_match_allows_an_id_whose_delete_has_aged_out

- test_create_match_with_another_coachs_match_id_is_404
- test_create_match_rejects_a_malformed_id
- test_create_match_rejects_an_id_the_client_could_not_have_minted [nil / v1 / v5]
- test_list_matches_newest_first
- test_list_matches_breaks_same_day_ties_by_creation_order
- test_list_matches_spans_every_player_of_the_caller
- test_list_matches_omits_another_coachs_matches
- test_list_matches_filtered_by_player
- test_list_matches_filtered_by_an_unknown_player_is_empty
- test_list_matches_filtered_by_another_coachs_player_is_empty
- test_list_matches_rejects_a_malformed_player_id
- test_list_matches_with_none_stored_is_an_empty_list
- test_get_match_returns_one
- test_get_match_of_another_coach_is_404
- test_get_match_unknown_id_is_404
- test_get_match_malformed_id_is_422
- test_update_match_changes_only_the_fields_sent
- test_update_match_clears_a_nullable_field_with_an_explicit_null
- test_update_match_records_a_result
- test_update_match_on_a_completed_match_is_409
- test_update_match_cannot_change_a_recorded_result
- test_update_match_resending_the_stored_values_is_a_no_op
- test_update_match_rejects_a_null_played_on
- test_update_match_with_an_empty_body_changes_nothing
- test_update_match_bumps_updated_at
- test_update_match_cannot_move_the_match_to_another_player
- test_update_match_rejects_a_future_played_on
- test_update_match_of_another_coach_is_404
- test_update_match_unknown_id_is_404
- test_delete_match_removes_it
- test_delete_match_takes_its_derived_sets_with_it

- test_delete_match_takes_its_points_with_it
- test_delete_match_leaves_another_matchs_points_alone
- test_delete_match_records_the_id_it_freed
- test_deleting_a_reused_id_starts_its_window_over
- test_delete_match_twice_is_404
- test_delete_match_of_another_coach_is_404
- test_delete_match_unknown_id_is_404
- test_delete_match_malformed_id_is_422
- test_delete_match_is_allowed_on_a_completed_match
- test_delete_match_under_another_devices_claim_is_refused [False / True]
- test_the_capturing_device_may_delete_its_own_match
- test_a_lapsed_claim_does_not_block_the_delete
- test_releasing_the_claim_clears_the_way_for_the_delete
- test_the_capture_refusal_still_stands_over_a_match_with_points
- test_claim_capture_takes_a_free_match
- test_claim_capture_renews_the_claim_of_the_device_holding_it
- test_claim_capture_refuses_a_second_device
- test_claim_capture_takes_over_a_lapsed_claim
- test_claim_capture_on_a_completed_match_is_409
- test_claim_capture_of_another_coachs_match_is_404
- test_claim_capture_unknown_id_is_404
- test_claim_capture_requires_a_device_id
- test_release_capture_frees_the_match_for_another_device
- test_release_capture_on_an_unclaimed_match_is_a_no_op
- test_release_capture_of_another_coachs_match_is_404
- test_recording_the_result_releases_the_capture_claim
- test_ingest_points_stores_the_batch_and_answers_with_the_derived_sets
- test_ingest_stores_every_field_the_client_stamped
- test_ingesting_the_same_batch_again_lands_on_the_same_answer
- test_a_repeated_point_keeps_the_stored_version
- test_an_overlapping_batch_converges

- test_ingest_recomputes_the_sets_a_late_batch_changes
- test_ingest_into_a_completed_match_is_refused
- test_a_sequence_already_held_by_a_different_point_is_refused
- test_a_point_already_stored_against_another_match_is_refused
- test_one_invalid_point_rejects_the_whole_batch
- test_an_empty_batch_is_refused
- test_a_flush_from_the_claim_holder_renews_the_claim
- test_a_flush_from_another_device_is_still_ingested
- test_a_flush_does_not_claim_a_match_nobody_holds
- test_a_flush_does_not_revive_the_lapsed_claim_it_once_held
- test_ingest_into_another_coachs_match_is_404
- test_ingest_into_an_unknown_match_is_404
- test_ingest_into_a_deleted_match_is_410
- test_ingest_into_a_match_whose_delete_has_aged_out_is_404
- test_list_points_orders_by_sequence_regardless_of_insertion_order
- test_list_points_of_a_match_with_none_recorded_is_an_empty_list
- test_list_points_of_another_coachs_match_is_404
- test_list_points_of_an_unknown_match_is_404
- test_list_points_round_trips_every_field_the_ingest_route_stored
- test_list_points_carries_no_ingestion_timestamps
- test_an_integrity_error_that_is_not_a_conflict_is_not_absorbed
- test_the_routes_that_read_the_claim_lock_the_match_row [capture / capture-release / patch / points / delete]
- test_match_routes_require_a_session [post /matches, get /matches, get /matches/{id}, patch /matches/{id}, delete /matches/{id}, post /matches/{id}/capture, post /matches/{id}/capture/release, post /matches/{id}/points, get /matches/{id}/points]

tests/routers/test_players.py — 118/118

- test_delete_player_records_the_ids_of_the_matches_it_freed
- test_delete_player_with_no_matches_records_nothing
- test_a_note_survives_the_write_and_comes_back_on_the_next_read
- test_create_note_files_it_under_the_areas_sent
- test_create_note_defaults_to_no_focus_areas
- test_create_note_cannot_name_a_player_in_the_body
- test_create_note_against_another_coachs_player_is_404

- test_list_notes_is_newest_session_first
- test_list_notes_breaks_a_tie_on_when_the_note_was_written
- test_list_notes_carries_only_this_players_notes
- test_list_notes_with_none_written_is_an_empty_list
- test_list_notes_of_another_coachs_player_is_404
- test_update_note_changes_only_the_fields_sent
- test_update_note_clears_the_focus_areas_with_an_empty_list
- test_update_note_of_another_coachs_player_is_404
- test_a_note_reached_through_the_wrong_player_is_404
- test_delete_note_removes_it_and_leaves_the_rest
- test_delete_note_unknown_id_is_404
- test_delete_player_takes_their_notes_with_them
- test_create_player_persists_and_returns_row
- test_create_player_belongs_to_the_caller
- test_create_player_with_every_optional_field
- test_create_player_without_optional_fields
- test_create_player_cannot_set_coach_id
- test_create_player_mints_the_username_and_numbers_namesakes
- test_create_player_numbers_past_usernames_minted_before_this_request
- test_create_player_returns_a_one_time_password_and_stores_only_its_hash
- test_the_password_never_appears_on_a_read
- test_the_minted_password_never_meets_the_policy
- test_creates_beyond_the_minting_burst_are_refused
- test_create_player_retries_when_a_concurrent_create_takes_the_username
- test_create_player_gives_up_after_three_username_collisions
- test_regenerating_gives_a_backfilled_player_a_password_that_signs_them_in
- test_regenerating_stores_only_a_hash_and_replaces_the_old_password
- test_regenerating_ends_the_player_s_live_sessions_and_no_one_else_s
- test_regenerating_another_coach_s_player_is_the_404_an_unknown_id_gets
- test_a_404_from_the_password_route_spends_no_minting_token
- test_creation_and_regeneration_spend_the_same_minting_budget
- test_create_player_re_raises_an_integrity_error_that_is_not_the_username
- test_create_player_rejects_future_date_of_birth
- test_create_player_rejects_implausibly_old_date_of_birth

- test_create_player_accepts_a_plausibly_old_date_of_birth
- test_create_player_accepts_date_of_birth_of_today
- test_create_player_accepts_an_explicit_null_date_of_birth
- test_create_player_rejects_missing_first_name
- test_create_player_rejects_missing_last_name
- test_create_player_rejects_invalid_dominant_hand
- test_create_player_rejects_invalid_gender
- test_create_player_rejects_overlong_name
- test_create_player_accepts_name_at_the_length_limit
- test_create_player_rejects_overlong_notes
- test_create_player_accepts_notes_at_the_length_limit
- test_create_player_rejects_whitespace_only_name
- test_create_player_strips_surrounding_whitespace
- test_list_players_returns_all_sorted_by_last_then_first_name
- test_list_players_omits_another_coachs_players
- test_get_player_returns_one
- test_get_player_of_another_coach_is_404
- test_list_players_with_none_stored_is_an_empty_list
- test_get_player_unknown_id_is_404
- test_get_player_malformed_id_is_422
- test_update_player_changes_only_the_fields_sent
- test_update_player_clears_a_nullable_field_with_an_explicit_null
- test_update_player_rejects_a_null_name_part [first_name, last_name]
- test_update_player_with_an_empty_body_changes_nothing
- test_update_player_bumps_updated_at
- test_update_player_rejects_coach_id
- test_update_player_rejects_a_future_date_of_birth
- test_update_player_of_another_coach_is_404
- test_update_player_unknown_id_is_404
- test_delete_player_takes_their_matches_with_them
- test_delete_player_takes_their_matches_points_with_them

- test_delete_player_takes_a_match_being_captured_with_it
- test_delete_player_leaves_the_rest_of_the_roster_alone
- test_delete_player_is_refused_while_a_transfer_code_is_outstanding
- test_cancelling_the_code_clears_the_way_for_the_delete
- test_delete_player_leaves_another_coachs_roster_alone
- test_delete_player_unknown_id_is_404
- test_player_routes_require_a_session [post /players, get /players, get /players/{id}, patch /players/{id}, delete /players/{id}, post /players/{id}/transfer-code, delete /players/{id}/transfer-code, post /players/redeem-transfer-code]
- test_a_transfer_code_moves_the_player_and_their_matches
- test_a_pasted_code_may_carry_whitespace
- test_minting_again_replaces_the_previous_code
- test_a_redeemed_code_cannot_be_used_again
- test_redeeming_an_unknown_code_is_400
- test_redeeming_your_own_code_is_409_and_leaves_it_usable
- test_deleting_a_transfer_code_stops_it_working
- test_deleting_a_transfer_code_that_was_never_minted_is_204
- test_transfer_code_for_another_coachs_player_is_404 [post / delete]
- test_a_freshly_created_player_has_no_transfer_code
- test_the_roster_flags_only_the_players_holding_a_code
- test_the_roster_reports_every_flag_in_one_statement
- test_minting_a_code_shows_up_on_the_player
- test_cancelling_a_code_clears_the_flag
- test_redeeming_a_code_clears_the_flag_on_the_response
- test_the_flag_and_the_delete_refusal_agree
- test_uploading_a_photo_answers_with_the_player
- test_an_uploaded_photo_is_served_at_the_url_it_reports
- test_replacing_a_photo_removes_the_one_it_replaced
- test_clearing_a_photo_removes_the_file
- test_clearing_a_photo_that_is_not_there_is_a_no_op
- test_another_coachs_player_is_a_404_and_stores_nothing
- test_a_file_that_is_not_an_image_is_refused

- test_deleting_a_player_removes_their_photo
- test_a_transferred_player_keeps_their_photo

tests/schemas/test_auth.py — 33/33

- test_a_valid_login_parses
- test_the_email_is_stripped_and_validated
- test_a_password_below_the_sign_up_floor_is_accepted
- test_a_password_over_the_ceiling_is_rejected
- test_the_password_is_not_stripped
- test_the_repr_never_contains_the_password
- test_a_valid_otp_request_parses
- test_the_otp_request_email_is_stripped_and_validated
- test_a_valid_verification_parses
- test_a_code_of_the_wrong_length_is_rejected [12345 / 1234567]
- test_a_non_digit_code_is_accepted_by_the_schema
- test_the_repr_never_contains_the_code
- test_a_valid_forgot_password_request_parses
- test_the_forgot_password_email_is_stripped_and_validated
- test_a_valid_reset_code_verification_parses
- test_a_reset_code_of_the_wrong_length_is_rejected [12345 / 1234567]
- test_the_reset_code_verification_repr_never_contains_the_code
- test_a_valid_reset_parses
- test_a_reset_carrying_a_code_is_rejected
- test_a_new_password_below_the_floor_is_rejected
- test_a_new_password_over_the_ceiling_is_rejected
- test_the_new_password_is_not_stripped
- test_the_repr_never_contains_the_new_password
- test_a_valid_change_parses
- test_only_the_new_password_takes_the_sign_up_floor
- test_a_valid_player_login_parses
- test_the_username_is_stripped_and_bounded
- test_the_player_password_has_no_floor_and_keeps_the_ceiling

- test_the_player_login_repr_never_contains_the_password
- test_a_valid_player_change_parses
- test_only_the_player_change_new_password_takes_the_sign_up_floor

tests/schemas/test_coach.py — 33/33

- test_a_valid_sign_up_parses
- test_the_email_is_stripped_and_its_domain_lowercased
- test_a_malformed_email_is_rejected
- test_an_email_over_the_length_bound_is_rejected
- test_the_bounds_reach_the_json_schema
- test_a_password_outside_the_bounds_is_rejected [7 / 129]
- test_a_password_keeps_its_whitespace
- test_a_blank_display_name_is_rejected [empty / whitespace]
- test_coach_create_has_no_password_hash_field
- test_the_password_is_not_in_the_repr
- test_an_update_omitting_the_display_name_dumps_nothing
- test_an_update_carrying_the_display_name_dumps_it_stripped
- test_an_invalid_display_name_is_rejected_on_an_update [empty / whitespace / too long / None]
- test_coach_update_carries_the_name_and_the_preferences_alone
- test_an_update_carrying_a_preference_dumps_it
- test_an_update_may_clear_a_preference_with_null [default_surface, default_scoring_system]
- test_a_preference_outside_its_enum_is_rejected [default_surface: sand / default_scoring_system: best_of_3]
- test_a_closure_carries_the_password_and_the_roster_waiver
- test_a_closure_leaves_the_roster_alone_unless_asked
- test_a_closure_takes_the_ceiling_without_the_floor
- test_verifying_a_password_carries_nothing_else
- test_a_closure_password_is_not_in_the_repr
- test_coach_read_has_no_password_hash_field
- test_reading_a_coach_row_drops_the_password_hash
- test_a_read_carries_the_photo_url_derived_from_the_key

- test_an_update_refuses_a_photo_url

tests/test_auth.py — 7/7

- test_each_token_is_fresh
- test_a_token_is_long_enough_to_be_unguessable
- test_hashing_a_token_is_deterministic
- test_the_hash_is_not_the_token_and_fits_the_column
- test_each_player_password_is_fresh
- test_a_player_password_reads_aloud_without_ambiguity
- test_the_lifetime_is_long_lived

tests/test_breach.py — 19/19

- test_the_vector_is_the_hash_the_api_would_be_asked_for
- test_the_default_checker_is_the_real_one
- test_the_suite_runs_with_the_corpus_switched_off
- test_the_setting_selects_the_real_checker
- test_the_offline_checker_refuses_nothing
- test_only_the_hash_prefix_is_sent
- test_the_answer_is_padded
- test_a_listed_password_is_breached
- test_a_password_the_bucket_does_not_hold_is_not_breached
- test_a_padding_row_is_not_a_sighting
- test_an_answer_of_nothing_but_padding_is_not_a_sighting
- test_the_suffix_comparison_ignores_case
- test_junk_lines_beside_real_rows_are_skipped
- test_a_200_that_is_not_corpus_rows_says_the_corpus_went_unasked
- test_a_refused_lookup_says_the_corpus_went_unasked [429 / 500 / 503]
- test_a_lookup_that_never_answers_says_the_corpus_went_unasked
- test_a_failed_lookup_logs_the_status_and_nothing_of_the_password

tests/test_db.py — 3/3

- test_the_pool_validates_a_connection_before_handing_it_over
- test_get_db_yields_a_usable_session_and_releases_it
- test_the_engine_runs_at_the_isolation_the_otp_attempt_cap_assumes

tests/test_email_templates.py — 25/25

- test_both_parts_carry_the_code [signup_code, password_reset_code]
- test_the_code_is_six_unbroken_digits [signup_code, password_reset_code]
- test_both_parts_say_when_the_code_dies [signup_code, password_reset_code]
- test_the_subject_names_the_app_and_the_code [signup_code, password_reset_code]
- test_the_two_emails_are_told_apart_by_their_subjects
- test_an_unasked_for_code_can_be_ignored [signup_code, password_reset_code]
- test_the_html_is_a_document_a_client_can_read [signup_code, password_reset_code]
- test_every_table_is_marked_presentational [signup_code, password_reset_code]
- test_the_message_has_exactly_one_heading [signup_code, password_reset_code]
- test_the_html_fetches_nothing [signup_code, password_reset_code]
- test_the_brand_blue_dresses_the_message_and_the_data_colors_stay_out [signup_code, password_reset_code]
- test_the_html_answers_a_dark_mode_client [signup_code, password_reset_code]
- test_an_interpolated_value_cannot_carry_markup_into_the_html [signup_code, password_reset_code]

tests/test_errors.py — 11/11

- test_a_secret_field_is_the_only_thing_redacted
- test_a_password_alongside_another_fields_error_is_redacted
- test_a_password_nested_below_the_reported_field_is_redacted
- test_an_unknown_keys_value_is_redacted
- test_an_error_carrying_no_input_is_passed_through
- test_a_field_that_is_not_secret_keeps_its_input
- test_every_secret_field_in_the_schemas_is_named_in_secret_fields
- test_an_unhandled_exception_answers_a_readable_json_body
- test_an_unhandled_exception_tells_the_client_nothing_about_itself
- test_the_500_carries_the_request_id_that_logged_the_traceback
- test_a_failure_above_the_logging_middleware_is_answered_too

tests/test_mail.py — 18/18

- test_the_default_sender_cannot_send
- test_the_selected_sender_is_the_logging_one
- test_the_logging_sender_writes_the_message_to_the_log

- test_the_logging_sender_leaves_the_markup_out
- test_the_resend_sender_posts_the_message_to_the_api
- test_the_resend_sender_logs_the_provider_id
- test_a_refused_send_raises_email_send_error
- test_a_stalled_send_raises_email_send_error
- test_a_rate_limited_send_is_retried_and_succeeds
- test_a_send_rate_limited_twice_still_fails
- test_an_unreadable_success_body_is_still_a_send
- test_a_refused_send_logs_the_provider_s_reason
- test_a_failed_send_keeps_the_code_out_of_the_log
- test_the_file_sender_appends_every_message_as_a_line_of_json
- test_the_setting_selects_the_file_sender
- test_selecting_the_file_sender_without_a_path_fails_loudly
- test_the_setting_selects_the_resend_sender
- test_selecting_resend_without_a_key_fails_loudly

tests/test_migrations.py — 5/5

- test_migrations_upgrade_downgrade_and_upgrade_again
- test_requiring_a_player_owner_refuses_orphans_rather_than_deleting_them
- test_the_username_backfill_numbers_existing_players_deterministically
- test_the_schema_at_head_matches_the_models
- test_dropping_mid_clears_the_placement_rather_than_the_point

tests/test_password_policy.py — 37/37

- test_a_password_that_is_mostly_one_character_is_refused [aaaaaaaa / aaaaa123 / AAAAaaaa / a1a2a3a4a]
- test_a_password_running_through_consecutive_characters_is_refused [12345678 / abcdefgh / 87654321 / hgfedcba / wimbledon12345 / fullwidth digits]
- test_a_common_password_is_refused [password / PASSWORD / matchpoint / password! / Password123! / monkey123 / tennis12345]
- test_a_password_carrying_the_coach_s_own_words_is_refused [the-mike-special / coach-of-the-year / smith-and-wesson / MikeSmith-forever]
- test_an_ordinary_password_is_accepted [a-good-password / correct-horse-battery-st4ple! / wimbledon-in-the-rain / secret-garden-path]
- test_a_player_password_carrying_their_own_words_is_refused

- test_no_email_contributes_no_words
- test_a_short_name_is_not_matched_as_a_substring
- test_the_first_matching_rule_is_the_one_reported
- test_the_refusal_is_a_400_carrying_the_reason
- test_an_accepted_password_raises_nothing
- test_a_password_in_the_corpus_is_refused
- test_a_password_the_corpus_does_not_hold_is_accepted
- test_a_corpus_that_cannot_answer_lets_the_password_through
- test_a_checker_that_raises_anything_else_still_lets_the_password_through
- test_a_locally_refused_password_never_reaches_the_corpus
- test_the_corpus_is_asked_about_the_password_that_gets_hashed

tests/test_rate_limit.py — 17/17

- test_a_key_is_admitted_up_to_the_burst_and_refused_beyond_it
- test_the_scale_setting_widens_a_bucket
- test_the_refusal_says_when_to_retry
- test_tokens_refill_at_the_sustained_rate
- test_an_idle_key_accrues_no_more_than_the_burst
- test_keys_do_not_share_a_bucket
- test_a_refused_request_spends_nothing
- test_a_refunded_request_leaves_the_bucket_as_it_found_it
- test_a_refund_cannot_bank_more_than_the_burst
- test_refunding_a_key_that_never_spent_is_harmless
- test_taking_from_several_buckets_admits_only_when_all_of_them_can_pay
- test_a_bucket_refused_by_another_is_not_charged_for_it
- test_a_slower_per_key_tier_leaves_the_shared_one_to_other_keys
- test_the_refusal_quotes_the_bucket_furthest_from_a_token
- test_the_furthest_bucket_is_quoted_even_when_another_is_refused_first
- test_a_new_key_evicts_buckets_that_have_refilled_to_full
- test_a_new_key_keeps_buckets_still_refilling

tests/test_security.py — 13/13

- test_hashing_the_same_password_twice_gives_different_hashes

- test_the_hash_names_argon2id
- test_the_hash_records_the_chosen_parameters
- test_the_column_storing_the_hash_cannot_truncate_it
- test_verifying_the_right_password_succeeds
- test_verifying_a_wrong_password_fails
- test_verifying_is_case_sensitive
- test_the_same_password_verifies_across_two_encodings_of_it
- test_the_dummy_hash_costs_what_a_real_one_costs
- test_nothing_verifies_against_the_dummy_hash
- test_a_current_hash_does_not_need_a_rehash
- test_a_hash_with_stale_parameters_needs_a_rehash
- test_verifying_against_a_stored_value_that_is_not_a_hash_raises

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Web — vitest (159/159)

src/components/password-input.test.tsx — 3/3

- starts masked
- toggling reveals the password, and toggling again re-masks it
- forwards autoComplete and name so password managers can recognize the field

src/components/password-strength-meter.test.tsx — 4/4

- labels a score of 0 as very weak
- labels a score of 4 as strong
- fills more of the bar for a higher score
- fill color progresses from vermillion to green as score increases

src/features/auth/auth-shell.test.tsx — 1/1

- renders the MatchPoint mark and its children

src/features/auth/forgot-password.test.tsx — 33/33

- renders the email field
- updates the field as the user types
- blocks submit until the email is filled
- links back to sign in
- sends the email to /api/auth/forgot-password

- moves to the OTP step on a 202 response
- shows an inline error, the retry wait, and stays on the email step on a 429 response
- shows an inline error and stays on the email step on a 502 response
- shows a generic error when the request fails outright
- sends one forgot-password request however impatiently Continue is clicked
- renders the OTP field once the OTP step is reached
- updates the OTP field as the user types
- blocks the OTP submit until a code is entered
- blocks submit while the code is the wrong length, with a visible hint
- sends the email and code to /api/auth/verify-reset-code
- shows the server's error and stays on the OTP step on a 400 response
- shows the retry wait and disables submit on a verify-reset-code 429 response
- shows a generic error when the verify-reset-code request fails outright
- sends one verify-reset-code however impatiently Verify is clicked
- requesting a new code from the OTP step returns to the email step with the email preserved
- renders the new password and confirm fields once the password step is reached
- blocks the password submit until matching passwords are entered
- blocks submit while the new password is too short, with a visible hint
- blocks submit while the passwords don't match, with a visible hint
- sends the email and new password to /api/auth/reset-password, no code field
- shows the server's error and stays on the password step on a 400 response
- shows the server's error and stays on the password step on a 401 response
- shows the server's error, the retry wait, and stays on the password step on a 429 response
- retries the password submit alone on failure, without re-verifying the code
- shows a generic error when the reset-password request fails outright
- sends one reset-password request however impatiently Reset Password is clicked
- reaches the confirmation view once both steps succeed, and its link signs back in
- starting over from the password step returns to the email step with the email preserved

src/features/auth/sign-in.test.tsx — 15/15

- renders the email and password fields

- updates the fields as the user types
- blocks submit until both fields are filled
- navigates to /home on a successful sign-in
- sends the credentials to /api/auth/login
- shows the server's error message on a failed sign-in
- falls back to an explicit message when the response has no detail to show
- shows the retry wait and disables submit on a 429 response
- shows a generic error when the request fails outright
- links to the sign-up page
- links to the forgot-password page
- wipes what the previous coach left when a different one signs in
- keeps the queue when the same coach signs in again
- refuses a 200 that names no coach rather than signing in as nobody
- keeps the coach on the form when this device cannot be cleared

src/features/auth/sign-up.test.tsx — 35/35

- renders the email field
- updates the field as the user types
- blocks submit until the email is filled
- links back to sign in
- sends the email to /api/auth/request-otp
- moves to the OTP step on a 202 response
- shows an inline error and stays on the email step on a 409 response
- falls back to an explicit message when the response has no detail to show
- shows the retry wait and disables submit on a 429 response
- shows a generic error when the request fails outright
- sends one request-otp however impatiently Continue is clicked
- renders the OTP field once the OTP step is reached
- updates the OTP field as the user types
- blocks the OTP submit until a code is entered
- blocks submit while the code is the wrong length, with a visible hint
- sends the email and code to /api/auth/verify-otp

- shows the server's error and stays on the OTP step on a 400 response
- shows the retry wait and disables submit on a verify-otp 429 response
- shows a generic error when the verify-otp request fails outright
- sends one verify-otp however impatiently Verify is clicked
- requesting a new code from the OTP step returns to the email step with the email preserved
- renders the display name, password and confirm fields once the password step is reached
- blocks submit until a display name is entered
- blocks submit while the password is too short, with a visible error
- blocks submit while the password is too long, with a visible error
- blocks submit while the passwords differ, with a visible error
- allows submit once the display name is set and the passwords match
- sends the email, password and display name to POST /coaches, no token field
- shows a usable message instead of crashing on a real FastAPI 422 body
- navigates to / on a successful sign-up
- shows the server's error and stays on the password step when verification is missing
- shows the retry wait and disables submit on a sign-up 429 response
- sends one POST /coaches however impatiently Sign Up is clicked
- shows a generic error when the complete-sign-up request fails outright
- starting over from the password step returns to the email step with the email preserved

src/features/players/player-list.test.tsx — 9/9

- renders the players the API returns
- flags a player whose transfer code is still out
- the pending-transfer flag is part of the row, not a control of its own
- requests the roster from the API's own origin
- groups players under their surname initial
- filters the roster by the search term
- reports a failed load instead of rendering an empty roster
- prompts to sign in again on a 401, instead of the generic error
- tapping a player hands it to the caller instead of navigating

src/features/settings/delete-account-sheet.test.tsx — 9/9

- names an outstanding transfer code on the first warning
- says nothing about transfer codes when the roster has none outstanding
- Cancel on the first step closes the sheet
- Back on the password step returns to the roster warning
- the full happy path deletes with delete_players true, purges local data, and navigates home
- a wrong password returns to the password step with the error shown there
- an outstanding transfer code shows its own message on the final step
- a 429 shows a retry countdown and disables the delete button
- says so when the device could not be cleared, and Retry clears the message once storage recovers

src/features/settings/settings-account.test.tsx — 8/8

- links back to settings
- change password asks for confirmation before leaving Settings
- confirming the change-password dialog navigates to the forgot-password flow
- cancelling the change-password dialog makes no navigation happen
- offers Sign Out, grouped with the other session actions
- signing out of all other sessions shows a confirm then a success dialog
- cancelling the sign-out-all-sessions confirm makes no request
- opens the delete-account flow

src/features/settings/settings-app.test.tsx — 4/4

- links back to settings
- Install MatchPoint links to the real install guide
- Notifications is still a stub, not interactive
- Appearance toggles between light and dark, and persists the choice

src/features/settings/settings-match-defaults.test.tsx — 6/6

- links back to settings
- prefills 'No preference' when the coach has no defaults set
- prefills from the coach's current defaults
- choosing a surface PATCHes it straight away
- choosing a scoring system PATCHes it straight away
- choosing 'No preference' PATCHes null

src/features/settings/settings-support.test.tsx — 2/2

- links back to settings
- all three rows are stubs, not interactive

src/features/settings/settings.test.tsx — 6/6

- links back to home
- the Match Defaults row links to its category page
- the Account & Security row links to its category page
- the App row links to its category page
- the Support row links to its category page
- offers Sign Out at the root, for fast single-tap access

src/features/settings/sign-out-action.test.tsx — 5/5

- opens a confirmation dialog when clicked
- Cancel closes the dialog without signing out
- confirming signs out, clears the query cache, and navigates to sign-in directly
- confirming also deletes the points that never synced
- says so with a toast when the device could not be cleared, but still signs out

src/lib/parse-error.test.ts — 10/10

- returns a string detail as-is
- joins a 422 validation array's messages
- falls back when detail is missing
- falls back when the validation array is empty or unrecognisable
- falls back when there is no error at all
- reports a network failure distinctly from an API error
- reads Retry-After off a 429 response
- returns null when there is no Retry-After header
- returns null when the response itself is undefined
- returns null for a non-numeric or non-positive Retry-After

src/lib/password-strength.test.ts — 4/4

- scores an empty password as 0
- scores an obviously weak password below the submit-gating threshold
- scores a long, random passphrase above the submit-gating threshold

- penalizes a password for containing a supplied user input

src/lib/use-countdown.test.ts — 5/5

- returns null when given null
- returns the initial value immediately
- ticks down by one every second
- stops at zero instead of going negative
- restarts from a new value when given one

Backlog Item 4

Test Results Summary

Web (vitest): 359/359 passed across 8 files

Web — vitest (359/359)

src/features/capture/point.test.ts — 95/95

• Point.create structural validation > accepts a minimal valid point

• Point.create structural validation > rejects non-object input

• Point.create structural validation > rejects null input

• Point.create structural validation > rejects an empty id

• Point.create structural validation > rejects an empty matchId

• Point.create structural validation > rejects a non-finite timestamp

• Point.create structural validation > rejects a sequence of -1

• Point.create structural validation > rejects a sequence of 1.5

• Point.create structural validation > rejects an unknown server

• Point.create structural validation > courtSide > accepts DEUCE

• Point.create structural validation > courtSide > accepts AD

• Point.create structural validation > courtSide > rejects an unknown courtSide

• Point.create structural validation > courtSide > rejects a null courtSide — unlike deuceOrAd, every point has one

• Point.create structural validation > serves > rejects a serves array of length 3

• Point.create structural validation > serves > rejects a serves array of length 4

• Point.create structural validation > serves > rejects a serve with the wrong number for its position

• Point.create structural validation > serves > rejects an unknown serve outcome

• Point.create structural validation > serves > rejects a non-null location on an OPPONENT serve

• Point.create structural validation > serves > accepts a null location on an OPPONENT serve

• Point.create structural validation > serves > rejects a location on a TRACKED serve that went in — a landed serve has none

• Point.create structural validation > serves > accepts a null location on a TRACKED serve that went in

• Point.create structural validation > serves > accepts a point with no serve at all — the coach missed it

• Point.create structural validation > serves > accepts a lone faulted serve — the second one went unwatched

• Point.create structural validation > serves > rejects a second attempt that doesn't follow a fault

• Point.create structural validation > serves > rejects a non-FailureMode location on a TRACKED serve that faulted

• Point.create structural validation > serves > rejects a location on an ace — a landed serve carries none

• Point.create structural validation > serves > an ace and its endReason require each other

• Point.create structural validation > serves > rejects a double fault whose endReason isn't DOUBLE_FAULT

• Point.create structural validation > serves > accepts a double fault with endReason DOUBLE_FAULT

• Point.create structural validation > serves > rejects a second serve attempt when the first serve didn't fault

• Point.create structural validation > serves > rejects endReason DOUBLE_FAULT when the serves weren't both faulted

• Point.create structural validation > returnOutcome > rejects a non-null returnOutcome when server is TRACKED

• Point.create structural validation > returnOutcome > accepts a null returnOutcome when server is OPPONENT

• Point.create structural validation > returnOutcome > rejects an unknown returnOutcome

• Point.create structural validation > returnOutcome > accepts a NET returnOutcome — the point ends on the return, no contact fields

• Point.create structural validation > returnOutcome > accepts a LONG returnOutcome — the point ends on the return, no contact fields

• Point.create structural validation > returnOutcome > accepts a WIDE returnOutcome — the point ends on the return, no contact fields

• Point.create structural validation > rejects a non-boolean won

• Point.create structural validation > rejects an unknown endReason

• Point.create structural validation > unusualReason / unusualNote > rejects a non-null unusualReason when endReason isn't UNUSUAL

• Point.create structural validation > unusualReason / unusualNote > rejects a non-null unusualNote when endReason isn't UNUSUAL

• Point.create structural validation > unusualReason / unusualNote > accepts an UNUSUAL point with no reason or note — the escape hatch stays optional

• Point.create structural validation > unusualReason / unusualNote > accepts an UNUSUAL point with a reason and note

• Point.create structural validation > unusualReason / unusualNote > rejects an unknown unusualReason

• Point.create structural validation > note > defaults to null

• Point.create structural validation > note > accepts a coach note on an ordinary (non-UNUSUAL) point

• Point.create structural validation > note > rejects an empty string — null is how 'no note' is spelled

• Point.create structural validation > note > truncates a note past what the API accepts rather than losing the point

• Point.create structural validation > wing / shotType > accepts both set together

• Point.create structural validation > wing / shotType > rejects wing set without the other

• Point.create structural validation > wing / shotType > rejects shotType set without the other

• Point.create structural validation > wing / shotType > rejects a non-null wing/shotType on a ACE point

• Point.create structural validation > wing / shotType > rejects a non-null wing/shotType on a DOUBLE_FAULT point

• Point.create structural validation > wing / shotType > rejects a non-null wing/shotType on a UNUSUAL point

• Point.create structural validation > wing / shotType > rejects a non-null wing/shotType on a return-error point

• Point.create structural validation > ballIn / ballLocation > accepts a CourtZone location when ballIn is true

• Point.create structural validation > ballIn / ballLocation > accepts a FailureMode location when ballIn is false

• Point.create structural validation > ballIn / ballLocation > rejects a FailureMode location when ballIn is true

• Point.create structural validation > ballIn / ballLocation > rejects a CourtZone location when ballIn is false

• Point.create structural validation > ballIn / ballLocation > accepts a null ballIn on a FORCED/MISTAKE point — the coach missed where it landed

• Point.create structural validation > ballIn / ballLocation > rejects a ballLocation with no ballIn beside it

• Point.create structural validation > ballIn / ballLocation > rejects a non-null ballIn/ballLocation on a ACE point

• Point.create structural validation > ballIn / ballLocation > rejects a non-null ballIn/ballLocation on a DOUBLE_FAULT point

• Point.create structural validation > ballIn / ballLocation > rejects a non-null ballIn/ballLocation on a UNUSUAL point

• Point.create structural validation > ballIn / ballLocation > rejects a non-null ballIn/ballLocation on a return-error point

• Point.create structural validation > rallyLengthBucket > accepts a valid bucket

• Point.create structural validation > rallyLengthBucket > accepts null — never silently defaulted

• Point.create structural validation > rallyLengthBucket > does not override a provided bucket even for an ACE

• Point.create structural validation > rallyLengthBucket > rejects an unknown bucket

• Point.fromJSON / toJSON > round-trips through JSON

• immutability > mutating the input serves array after create() does not affect the Point

• immutability > mutating toJSON()'s output does not affect the Point

• immutability > mutating toJSON()'s ballLocation object does not affect the Point

• isAce > true when endReason is ACE

• isAce > false otherwise

• isDoubleFault > true when both serves faulted

• isDoubleFault > false on a single serve

• isDoubleFault > isAce and isDoubleFault are mutually exclusive

• context > is required — a point always knows where in the match it sat

• context > rejects a non-positive setNumber

• context > rejects a non-positive gameNumber

• context > rejects a non-boolean inTiebreak

• context > rejects a non-boolean isBreakPoint

• context > rejects a non-boolean isGamePoint

• context > rejects a non-boolean isGameEnd

• context > rejects a non-boolean isSetPoint

• context > rejects a non-boolean isSetEnd

• context > rejects a non-boolean isMatchPoint

• context > is copied out by toJSON rather than shared

• firstServeIn > true with a single serve

• firstServeIn > false once a second serve was needed

• firstServeIn > null when the serve was never recorded — not the same as it having missed

• firstServeIn > false when the first serve faulted and the second went unwatched

• lastShotPlayer > TRACKED when the tracked player won

• lastShotPlayer > OPPONENT when the tracked player lost

src/features/capture/match.test.ts — 25/25

• createMatch > builds a Match from the given fields

• createMatch > generates a fresh matchId on every call

• createMatch > still runs Match.create's validation, e.g. rejects opponentId equal to trackedPlayerId

• createMatch > rejects an unknown surfaceType

• createMatch > rejects a datetime string, not just a date — playedOn must match the backend's date-only played_on

• createMatch > rejects an unknown scoringSystem

• createMatch > defaults > scoringSystem defaults to traditional when not given

• createMatch > defaults > surfaceType defaults to unspecified when not given

• createMatch > un-rostered opponent > given an opponentId, opponentName is null — the name lives on that Player row

• createMatch > un-rostered opponent > given an opponentName instead, generates a stand-in opponentId

• createMatch > un-rostered opponent > generates a fresh stand-in opponentId on every call

• Match.create opponentName validation > rejects an empty opponentName

• Match.create firstServer validation > accepts trackedPlayerId

• Match.create firstServer validation > accepts opponentId — the opponent can serve first

• Match.create firstServer validation > rejects a firstServer that isn't a participant, e.g. the old "player"/"opponent" role label

• Match.create surfaceType validation > accepts unspecified as a real surfaceType, not just hard/clay/grass/carpet

• outcomeOverride > defaults to null — a newly-started match hasn't ended

• outcomeOverride > createMatch accepts one — a back-filled match already known to have ended early

• outcomeOverride > Match.create accepts retirement as a reason

• outcomeOverride > Match.create accepts walkover as a reason

• outcomeOverride > Match.create accepts default as a reason

• outcomeOverride > Match.create rejects an unknown reason

• outcomeOverride > Match.create rejects a winner that isn't a participant

• Match.fromJSON / toJSON > round-trips through JSON

• Match.fromJSON / toJSON > reads a womens_grand_slam payload back as traditional, the way the API's migration did

src/features/capture/match-set.test.ts — 26/26

• MatchSet.create structural validation > accepts a minimal valid set

• MatchSet.create structural validation > rejects non-object input

• MatchSet.create structural validation > rejects null input

• MatchSet.create structural validation > rejects a setNumber of 0

• MatchSet.create structural validation > rejects a setNumber of -1

• MatchSet.create structural validation > rejects a setNumber of 1.5

• MatchSet.create structural validation > rejects an empty matchId

• MatchSet.create structural validation > rejects an empty trackedPlayerId

• MatchSet.create structural validation > rejects an empty opponentId

• MatchSet.create structural validation > rejects trackedPlayerId equal to opponentId

• MatchSet.create structural validation > rejects a negative trackedPlayerGames

• MatchSet.create structural validation > rejects a negative opponentGames

• MatchSet.create structural validation > rejects a non-integer trackedPlayerGames

• MatchSet.create structural validation > rejects a non-integer opponentGames

• MatchSet.create structural validation > accepts a null winner — the set is still in progress

• MatchSet.create structural validation > rejects a winner that isn't a participant

• MatchSet.create tiebreak validation > accepts a null tiebreak — the set wasn't decided by a breaker

• MatchSet.create tiebreak validation > accepts a valid tiebreak score

• MatchSet.create tiebreak validation > accepts the match_tiebreak_10_points degenerate case: 0/0 games, tiebreak decides it

• MatchSet.create tiebreak validation > rejects a non-object, non-null tiebreak

• MatchSet.create tiebreak validation > rejects a negative tiebreak.trackedPlayerPoints

• MatchSet.create tiebreak validation > rejects a negative tiebreak.opponentPoints

• MatchSet.create tiebreak validation > rejects a tiebreak.target of 0

• MatchSet.create tiebreak validation > rejects a tiebreak.target of -1

• MatchSet.create tiebreak validation > rejects a tiebreak.target of 1.5

• MatchSet.fromJSON / toJSON > round-trips through JSON

src/features/capture/state.test.ts — 56/56

• maxSetsFor > traditional's max sets is 3

• maxSetsFor > mens_grand_slam's max sets is 5

• maxSetsFor > no_ad_scoring's max sets is 3

• maxSetsFor > match_tiebreak_10_points's max sets is 3

• maxSetsFor > short_sets's max sets is 3

• maxSetsFor > pro_set's max sets is 1

• maxSetsFor > legacy's max sets is 3

• matchRulesFor > traditional needs 2 sets to win, out of 3 playable

• matchRulesFor > mens_grand_slam needs 3 sets to win, out of 5 playable

• matchRulesFor > no_ad_scoring needs 2 sets to win, out of 3 playable

• matchRulesFor > match_tiebreak_10_points needs 2 sets to win, out of 3 playable

• matchRulesFor > short_sets needs 2 sets to win, out of 3 playable

• matchRulesFor > pro_set needs 1 sets to win, out of 1 playable

• matchRulesFor > legacy needs 2 sets to win, out of 3 playable

• isAdScoringFor > traditional is ad scoring: true

• isAdScoringFor > mens_grand_slam is ad scoring: true

• isAdScoringFor > no_ad_scoring is ad scoring: false

• isAdScoringFor > match_tiebreak_10_points is ad scoring: true

• isAdScoringFor > short_sets is ad scoring: false

• isAdScoringFor > pro_set is ad scoring: true

• isAdScoringFor > legacy is ad scoring: true

• setRulesFor > traditional's non-deciding set: 6 games, tiebreak at 6, target 7

• setRulesFor > no_ad_scoring's non-deciding set: 6 games, tiebreak at 6, target 7

• setRulesFor > short_sets's non-deciding set: 4 games, tiebreak at 3, target 5

• setRulesFor > pro_set's non-deciding set: 8 games, tiebreak at 8, target 7

• setRulesFor > mens_grand_slam's regular set uses a 7-point breaker

• setRulesFor > mens_grand_slam's decider uses a 10-point breaker

• setRulesFor > set 3 of a best-of-5 mens_grand_slam match is NOT the decider

• setRulesFor > legacy's regular set tiebreaks normally at 6-6

• setRulesFor > legacy's decider has no tiebreak at all

• setRulesFor > match_tiebreak_10_points' regular set plays normally

• setRulesFor > match_tiebreak_10_points' decider is a single breaker — no games phase

• setRulesFor > validation > rejects a setNumber of 0

• setRulesFor > validation > rejects a setNumber of -1

• setRulesFor > validation > rejects a setNumber of 1.5

• setRulesFor > validation > rejects a maxSets of 0

• setRulesFor > validation > rejects a maxSets of -1

• setRulesFor > validation > rejects a maxSets of 1.5

• setRulesFor > validation > rejects setNumber greater than maxSets

• matchStateFor > no sets is 0-0 with no winner

• matchStateFor > counts sets won by each side

• matchStateFor > an in-progress set doesn't count toward either side

• matchStateFor > traditional (best-of-3, setsToWin: 2) > 2-0 is a win

• matchStateFor > traditional (best-of-3, setsToWin: 2) > 2-1 is a win

• matchStateFor > traditional (best-of-3, setsToWin: 2) > 1-1 is not decided yet

• matchStateFor > traditional (best-of-3, setsToWin: 2) > the opponent can win too

• matchStateFor > mens_grand_slam (best-of-5, setsToWin: 3) > 3-1 is a win

• matchStateFor > mens_grand_slam (best-of-5, setsToWin: 3) > 2-2 is not decided yet — traditional's setsToWin (2) would say otherwise

• matchStateFor > pro_set (setsToWin: 1) is decided by a single set

• matchStateFor > consistency checks > rejects sets that don't match the given participants

• finalWinnerFor > retirement override wins regardless of matchSets

• finalWinnerFor > walkover override wins regardless of matchSets

• finalWinnerFor > default override wins regardless of matchSets

• finalWinnerFor > no override, decided by sets, matches matchStateFor's own winner

• finalWinnerFor > no override, not yet decided -> null

• finalWinnerFor > uses the match's own scoringSystem rules, not a hardcoded best-of-3 assumption

src/features/capture/score.test.ts — 71/71

• folding points into a game > a new match is love-all in set 1, nothing observed yet

• folding points into a game > four straight points win a game, and the game resets to love-all

• folding points into a game > ad scoring needs two clear points from deuce

• folding points into a game > no-ad scoring ends the game on the first point past 40-40

• folding games into a set > six games to love takes the set and opens the next one

• folding games into a set > 5-5 is not a set — two clear games are needed

• folding games into a set > games-all starts the breaker, and the breaker's points are not games

• folding games into a set > a breaker won 7-0 records the set as 7-6

• folding games into a set > a breaker needs two clear points, same as a game

• folding games into a set > short sets run to four games with a breaker at 3-3, first to five

• folding games into a set > legacy's decider has no breaker and plays on past 6-6

• folding games into a set > the match-tiebreak decider is a breaker from its first point and credits no game

• folding sets into a match > two sets decide a best-of-three, and the board stops there

• folding sets into a match > a pro set is one set, so winning it wins the match

• folding sets into a match > three sets do not decide a best-of-five

• folding sets into a match > a point past a decided match leaves the board alone

• rotation and court side > serve alternates every game from whoever served first

• rotation and court side > the opponent serving first flips every answer

• rotation and court side > rotation carries across a set boundary rather than restarting

• rotation and court side > a breaker takes one slot, and serve inside it alternates every two points

• rotation and court side > court side alternates by points played, in a game and in a breaker alike

• rotation and court side > a fresh game is always the deuce side

• interpretSnapshot > reads completed sets, the live set and the game

• interpretSnapshot > accepts a completed traditional set of 6-0

• interpretSnapshot > accepts a completed traditional set of 6-4

• interpretSnapshot > accepts a completed traditional set of 7-5

• interpretSnapshot > accepts a completed traditional set of 7-6

• interpretSnapshot > refuses 7-4 as a completed set

• interpretSnapshot > refuses 8-6 as a completed set

• interpretSnapshot > refuses 6-6 as a completed set

• interpretSnapshot > names the set that is wrong

• interpretSnapshot > a completed last set that doesn't decide the match opens the next one

• interpretSnapshot > a decided match has no current set and no game

• interpretSnapshot > refuses a set played after the match was already won

• interpretSnapshot > refuses a set still in progress ahead of a later one

• interpretSnapshot > refuses a game score that can't happen, and one past its format

• interpretSnapshot > refuses a game alongside a breaker, and one after the match is over

• interpretSnapshot > a set at games-all carries its breaker score instead

• interpretSnapshot > refuses a tiebreak score on a set that never had one, and a decided one still running

• interpretSnapshot > refuses a tiebreak score that disagrees with who won the set

• interpretSnapshot > the match-tiebreak decider plays no games

• interpretSnapshot > refuses more sets than the format plays

• interpretSnapshot > a board round-trips through its own snapshot

• checkpoints > applying one moves the score to what the coach declared

• checkpoints > declared games are never counted as observed

• checkpoints > games already watched stay watched when a later checkpoint keeps them

• checkpoints > a correction that shrinks the score can't leave more observed than played

• checkpoints > the serving answer is cross-checked against the score

• checkpoints > on an empty log the answer defines who served first instead

• checkpoints > an explicit order change re-anchors the rotation rather than refusing

• checkpoints > a checkpoint that changes nothing is refused

• the fold over a whole log > points and checkpoints fold together in sequence order

• the fold over a whole log > an event out of order, or from another match, is ignored rather than thrown

• the fold over a whole log > points past a decided match are ignored, not thrown

• the fold over a whole log > a checkpoint the fold can't read is ignored, and the board survives

• stamps > a point mid-game is none of them

• stamps > 40-0 on serve is a game point, not a break point

• stamps > 0-40 on serve is a break point, not a game point

• stamps > no-ad deuce is both at once

• stamps > ad scoring at deuce is neither — two points are still needed

• stamps > serving for the set stamps a set point, and winning it stamps the end

• stamps > a set point for the receiver is still a set point

• stamps > the last point of a match stamps a match point and the set end

• stamps > inside a breaker the stamps answer for the breaker as its own game

• scoreboardFor > a brand-new match is love-all: one games column, one call column

• scoreboardFor > the call column reads deuce and advantage

• scoreboardFor > a completed set keeps its own column and the live set gains one

• scoreboardFor > a breaker shows its points in the call column

• scoreboardFor > a match-tiebreak decider shows its breaker in brackets once complete

• scoreboardFor > a declared set is marked as not fully watched

• coverage > counts watched games against played ones across the whole match

src/features/capture/checkpoint.test.ts — 11/11

• ScoreCheckpoint.create > accepts a well-formed checkpoint

• ScoreCheckpoint.create > rejects a missing id

• ScoreCheckpoint.create > rejects a missing matchId

• ScoreCheckpoint.create > rejects a negative or fractional sequence

• ScoreCheckpoint.create > rejects an unknown servesNext

• ScoreCheckpoint.create > rejects a non-boolean orderChanged

• ScoreCheckpoint.create > requires at least one set

• ScoreCheckpoint.create > rejects a set score that isn't two whole numbers

• ScoreCheckpoint.create > accepts a breaker score, and a set with none

• ScoreCheckpoint.create > does not check whether the score is reachable

• ScoreCheckpoint.create > round-trips through toJSON, copying the snapshot rather than sharing it

src/features/capture/capture-flow.test.ts — 63/63

• startPoint > TRACKED starts on the first serve, with the given courtSide attached

• startPoint > OPPONENT starts on the first serve — courtSide is attached the same way, not tracked-only

• serve — tracked player > in → RALLY — one tap says it landed, not where inside the box

• serve — tracked player > fault on the first serve → SERVE number 2, the fault is kept not overwritten

• serve — tracked player > fault on the second serve → READY, DOUBLE_FAULT, no rally to ask about

• serve — tracked player > ace → READY, won by the server, and carrying no placement

• serve — opponent > ace → READY, won by the opponent (our player never touched it)

• serve — opponent > fault on the second serve → READY, DOUBLE_FAULT, won by the tracked player

• serve — opponent > return in → RALLY, the serve is recorded as IN with no location

• serve — opponent > return error → READY, the no-rally shortcut — returnOutcome already says where it went

• serve — opponent > rejects a return error with no endReason — forced vs mistake must be chosen

• serve — opponent > rejects a return-in with an endReason — nothing to classify yet

• let > service let on the first serve is a no-op — nothing was recorded yet

• let > service let on the second serve keeps the first-serve fault, doesn't reset to first serve

• let > rally let resets the whole point to first serve, clearing serves and history

• let > rally let preserves courtSide unchanged — the same point, replayed

• let > rejects a let from a step other than serve or rally

• note > adds a note to the draft without advancing the step

• note > doesn't push onto history — an undo skips straight past it

• note > a second note appends onto a new line rather than overwriting the first

• note > blank or whitespace-only text is a no-op — no empty note recorded

• note > trims surrounding whitespace off a real note

• note > survives an ordinary step transition

• note > a rally let discards the note along with the rest of the draft

• rally screen > won=true endReason=FORCED → SHOT_DETAIL, length first then outcome

• rally screen > won=false endReason=FORCED → SHOT_DETAIL, length first then outcome

• rally screen > won=true endReason=MISTAKE → BALL_LOCATION, length first then outcome

• rally screen > won=false endReason=MISTAKE → BALL_LOCATION, length first then outcome

• rally screen > won=true endReason=FORCED → SHOT_DETAIL, outcome first then length — order doesn't matter

• rally screen > won=false endReason=FORCED → SHOT_DETAIL, outcome first then length — order doesn't matter

• rally screen > won=true endReason=MISTAKE → BALL_LOCATION, outcome first then length — order doesn't matter

• rally screen > won=false endReason=MISTAKE → BALL_LOCATION, outcome first then length — order doesn't matter

• rally screen > an unforced error commits with no shot recorded, which Point.create allows

• rally screen > picking a length alone is sticky — it writes into the draft without leaving RALLY or touching history

• rally screen > picking an outcome alone is sticky too — same non-advancing update, symmetric with length

• rally screen > a rally let clears a picked length along with the rest of the draft

• rally screen > a rally let clears a picked outcome too, for the same reason

• full happy path — serve through commit > a FORCED win threads through every screen and commits to a valid Point

• unusual point > from the first serve, no serve is recorded rather than one assumed

• unusual point > from the second serve, the fault already recorded survives

• unusual point > a point skipped from the serve commits with its winner and nothing else

• unusual point > reachable from the rally screen, win/loss is mandatory, reason is optional

• unusual point > an unusual point's rally length is never auto-filled, unlike ACE/DOUBLE_FAULT

• undo > undoes one step back to the exact prior draft

• undo > restores the right draft even when two different paths reach the same step

• undo > undoing with no history discards the draft

• undo > undoing twice from a two-step-deep history exhausts the stack, lands on ABORTED

• commitDraft > a valid READY draft for each endReason produces a Point that passes Point.create

• commitDraft > a return error reached via transition() commits — the no-rally shortcut is a real Point

• commitDraft > an ending nobody saw commits — endReason null is an observation hole, not a break

• commitDraft > a broken draft (READY reached with no winner) throws via Point.create, not silently

• didn't see > from the serve, no serve is recorded and the rally is still ahead

• didn't see > from the second serve, the fault already watched survives

• didn't see > from the rally it still needs the winner, and leaves the ending unknown

• didn't see > from the rally, a length already picked is kept

• didn't see > from the rally without a winner it throws rather than guessing

• didn't see > from the shot screen, the stroke is left null and the ball still asked for

• didn't see > from the ball screen, where it landed is left null and the point commits

• didn't see > undo steps back into the screen that was skipped

• didn't see > a point with every observation missing still commits with its winner

• action validity guards > rejects an action that doesn't belong on the current step

• action validity guards > rejects TAP_SERVE_IN when the opponent is serving — no such tap on that screen

• action validity guards > rejects TAP_RETURN when the tracked player is serving — no return tap on that screen

src/features/capture/match-flow.test.ts — 12/12

• startNextPoint > brand-new match: sequence, server and court side all come off the board

• startNextPoint > mid-match: the server follows the rotation and the side follows the game

• startNextPoint > the sequence is the caller's to supply, not derived from the log

• undoLast > drops the newest event and hands it back

• undoLast > drops a checkpoint just the same — the tail is the tail

• undoLast > an empty log has nothing to undo, and says so rather than throwing

• undoLast > never mutates the log it was given

• retireMatch > awards the match to the player who didn't retire

• retireMatch > rejects a player who isn't in the match

• isMatchOver > an override ends it whatever the board says

• isMatchOver > true once the sets decide it

• isMatchOver > false mid-match

Backlog item 5

Test Results Summary

Web (vitest): 212/212 passed across 17 files

Web — vitest (212/212)

src/features/capture/court-diagram.test.tsx — 5/5

• renders one tap target per zone, named for the coach

• a tap flashes before the flow advances

• a second tap during the flash does not advance twice

• reports pending while a tap's flash is in flight

• chips the net band only when labelled

src/features/capture/match-header.test.tsx — 13/13

• marks the server of the point being captured

• shows the live game score from the folded board

• sizes the score for the whole match before a point is played

• keeps that width once sets have been played

• back greys out where there is nothing behind, rather than vanishing

• back steps the capture flow back

• tapping skip asks who won before recording the point

• the skipped point can go to the opponent

• holding skip opens the score editor instead of skipping the point

• a long-press's swallowed click doesn't leave a keyboard activation stuck

• tapping the score opens the editor — the most discoverable way in

• a set the coach declared rather than watched says so

• tapping the note button opens the text sheet where the browser can't transcribe, and adds the note

src/features/capture/match-setup-screen.test.tsx — 8/8

- names the player the match is for
- a match cannot start without an opponent
- starting hands back everything the match needs
- the opponent can serve first
- whitespace alone is not an opponent
- the coach's saved defaults are what the form opens on
- what setup was left holding wins over the saved defaults
- back steps out of setup

src/features/capture/serve-screen.test.tsx — 13/13

- titles the attempt and names the court the point is played from
- a tap in the live service box records a serve in, with no placement
- the live box follows the court side the point is played from
- a tap on Long faults with location LONG
- a tap on Long (ad side) faults with location LONG
- a tap on Wide (left) faults with location WIDE
- a tap on Wide (right) faults with location WIDE
- a tap on Net faults with location NET
- a tap on Ad service box faults with location WIDE
- the ace button records an ace with no placement
- ace and let are disabled while a court tap's flash is pending
- the second attempt says why the first one faulted
- a let reports itself and leaves the attempt where it was

src/features/capture/return-screen.test.tsx — 12/12

- asks where the return landed
- a return into the safe area goes straight to the rally
- a return Long asks whether it was forced before recording it
- a return Wide (left) asks whether it was forced before recording it
- a return Wide (right) asks whether it was forced before recording it

• a return Net asks whether it was forced before recording it

• a returnable ball missed is recorded as a mistake

• an ace against us carries no placement

• the opponent's fault carries no placement either

• ace, fault and let are disabled while a court tap's flash is pending

• the second attempt says a second serve is coming, without a reason

• a let reports itself and leaves the attempt where it was

src/features/capture/rally-screen.test.tsx — 12/12

• Win, forced error dispatches the outcome

• Win, unforced error dispatches the outcome

• Loss, forced error dispatches the outcome

• Loss, unforced error dispatches the outcome

• neither question is required first — the outcome cards are tappable before a length is picked

• once an outcome is already picked, the prompt asks only for the length

• once a length is already picked, the prompt asks only for the outcome

• picking a rally length reports it without ending the point

• the chosen rally length shows as chosen, and re-tapping keeps it

• the chosen outcome shows as chosen, the same way the chosen length does

• an unusual ending hands off to the flow's own unusual path

• a let replays the point from the first serve

src/features/capture/final-shot-screen.test.tsx — 7/7

• offers every stored shot on both wings, and labels the columns

• Backhand drive records that wing and shot type

• Forehand drive records that wing and shot type

• Backhand volley records that wing and shot type

• Forehand drop shot records that wing and shot type

• Backhand lob records that wing and shot type

• Forehand lob records that wing and shot type

src/features/capture/final-ball-screen.test.tsx — 6/6

• offers the four zones players asked for

• Deuce deep records that zone, always as a ball that landed

• Deuce short records that zone, always as a ball that landed

• Ad deep records that zone, always as a ball that landed

• Ad short records that zone, always as a ball that landed

• the zones sit in the half the ball landed in

src/features/capture/unusual-point-sheet.test.tsx — 7/7

• asks who won first, naming both players

• /Win.*Rivera/ awards the point

• /Loss.*Okoye/ awards the point

• cancelling on the first step leaves the flow alone

• offers every stored reason once the winner is known

• the point cannot be recorded until a reason is picked

• back on the reason step steps the flow back, it does not dismiss

src/features/capture/score-flash.test.tsx — 9/9

• a point won=true flashes Point won

• a point won=false flashes Point lost

• a game won=true flashes Game won

• a game won=false flashes Game lost

• a set won=true flashes Set won

• a set won=false flashes Set lost

• it clears itself, and never blocks the screen it covers

• a game holds longer than a point

• a re-render doesn't restart the timer

src/features/capture/score-sheet.test.tsx — 18/18

• opens pre-filled from the board, so a correction only touches what is wrong

• a set each, the opponent taking the first, is enterable

• a breaker row appears at the boundary, and its score is optional
• a 7-6 can carry the breaker score it was won on
• a breaker in progress replaces the game chips with its points
• no-ad scoring offers no advantage chip
• advantage is offered where the format plays it, and stands against the other's 40
• the match-tiebreak decider shows a breaker and no games
• legacy's decider is labelled as playing no tiebreak, and takes an 8-6
• a refusal names the cell it belongs to, and holds the button
• survives 1.5 typed into a cell, refusing it per cell
• survives -1 typed into a cell, refusing it per cell
• a fractional tiebreak score is refused the same way
• a score that changes nothing is refused
• on an empty log the serving answer pins who served first, and says so
• with points already recorded, a disagreeing answer names both causes
• the order-changed escape records the deviation rather than refusing it
• ending the match still asks why and who won

src/features/capture/note-button.test.tsx — 28/28

• no SpeechRecognition on the window: tapping opens the text sheet directly
• SpeechRecognition present: tapping starts recording instead of opening the sheet
• holding opens the text sheet without ever reaching for the mic
• a hold on a touch screen opens the sheet, though its release fires pointerleave first
• a hold is spent by its own click, so the next activation dictates again
• holding while recording stops and adds the note, rather than opening the sheet
• holding while a session the service ended is pending adds the note once, not the sheet
• a touch long-press's contextmenu is prevented, so the click it precedes survives
• the typed sheet is reachable by keyboard, which can't perform the hold
• stopping a recording adds the transcript as a note
• a note goes to the caller's latest handler, not the one recording started with
• the last words before Stop are part of the note, not cut off
• a dictated note longer than the API accepts is capped, not sent whole

• the typed note field stops at the same length the API accepts
• stopping with no speech heard adds nothing
• a session the service ends by itself keeps recording, and the next tap adds the note once
• a session that ends by itself with nothing heard adds nothing
• a recognition that refuses to start falls back to the text sheet, not a stuck button
• a recognition that can't be constructed at all falls back to the text sheet
• a recognition error falls back to the text sheet, pre-filled with what was heard so far
• an errored session never auto-adds what it heard, since the sheet already holds it
• a superseded session's late error doesn't open the sheet over a live recording
• a superseded session's late end doesn't stand down a live recording
• after a recognition error the next tap records afresh, adding only the new note
• navigating away mid-dictation still adds what was heard
• navigating away after a recognition error adds nothing, since the sheet held it
• cancel closes the sheet without adding a note
• the sheet's Add note button is disabled until there's text to add

src/features/capture/result-screen.test.tsx — 9/9

• a match won in straight sets reads as a win, with the score from its points
• a lost match names the winner first and reads the score from their side
• a retirement beats the set count and says so
• a walkover with no points played drops the score, not just its numbers
• reports coverage in games, so a filled-in set never reads as watched
• offers to undo the last point only while there is one to take back
• no undo is offered when there is nothing to unwind
• finishing the match is its own explicit action
• offers the way home and the way to the match's own results

src/features/capture/live-match.test.tsx — 40/40

• setup comes first, and starting it opens the serve screen
• back on setup exits the match entirely
• the opponent serving first opens the return screen instead

- a Grass match paints the court that surface
- a Clay match paints the court that surface
- a Carpet match paints the court that surface
- a hard court keeps the default green, with nothing to override it
- a point threads serve to final ball and lands on the scoreboard
- the rally screen's two questions can be answered in either order
- an unforced error skips the shot screen
- a let after picking a rally length doesn't carry it into the replayed point
- a point the coach only half-saw is kept, not discarded
- a missed stroke still records where the ball landed
- the skip sheet points at the editor when the gap is bigger than one point
- back on the match's first screen returns to setup, still holding what was typed
- nothing is queued while setup is still reachable
- back walks the point's screens, and stops at the serve
- once the match is queued, undoing back to the serve screen leaves rather than reaching setup
- back past a recorded point offers to undo it, and asks first
- undoing takes the point back off the score and queues its delete
- a mis-tapped match point can be undone from the result screen itself
- a tab that dies mid-match reopens at the same score
- a resumed match keeps counting from where its sequence left off
- a logged event this build cannot read leaves the coach on setup, told why
- a finished match is not resumed into
- the score the coach enters is queued as its own event, not as invented points
- an out-of-turn game on court is recordable rather than a dead end
- finishing a match won on sets queues its completion, which nothing else does
- the committed point is queued behind its match, and never awaited
- a skipped point is recorded winner-only and moves the score
- recording a point as a Win flashes Point won
- recording a point as a Loss flashes Point lost
- the flash clears itself rather than sitting over the next point
- a point that takes the game flashes the game, not the point

• a point that takes the set flashes the set

• the point that ends the match doesn't flash over the result screen

• skipping ahead resumes from the score the coach entered

• ending the match shows the result, with the reason it ended

• declaring a walkover before the first point still queues the match and its outcome

• retiring after points were played still queues the outcome, as its own row

src/features/matches/new-match.test.tsx — 5/5

• hands the player and coach to capture, and starts at setup

• starts a match with the coach unreachable

• opens the form on the defaults the coach saved

• says so instead of starting a match no drain can send

• says not found instead of an unnamed form when the player isn't the coach's

src/features/outbox/match-log.test.ts — 11/11

• events > reads back in sequence order, whatever order they were written in

• events > only ever returns the match asked for

• events > rewriting an event under its own id overwrites rather than duplicating

• events > removing one leaves the rest, which is what undo needs

• events > removing one that was never there is not an error

• matches > saves and reads back the facts the events can't carry

• matches > saving again updates in place

• matches > lists a coach's unfinished matches, most recently touched first

• matches > finds the newest unfinished match for one player, and skips a finished one

• matches > another player's match is never offered for this one

• matches > removing a match takes its events with it, and leaves other matches alone

src/features/sync/match-outcome-payload.test.ts — 9/9

• toMatchOutcomeUpdate > produces a body the API's own contract accepts

• toMatchOutcomeUpdate > maps the tracked player winning to a win result

• toMatchOutcomeUpdate > maps the tracked player losing to a loss result

- toMatchOutcomeUpdate > does not send the reason — the API has nowhere to put it yet
- toMatchOutcomeUpdate > rejects a payload that is not an object
- toMatchOutcomeUpdate > rejects a payload that is null
- toMatchOutcomeUpdate > rejects a payload that is missing trackedPlayerWon
- toMatchOutcomeUpdate > rejects a payload that is an invalid reason
- toMatchOutcomeUpdate > rejects a payload that is a non-boolean trackedPlayerWon

Backlog item 6

Test Results Summary

API (pytest): 34/34 passed across 1 file
Web (vitest): 153/153 passed across 18 files
Total: 187/187 passed

API — pytest (34/34)

tests/routers/test_analytics.py — 99/99

- test_routes_require_a_session
- test_routes_hide_another_coachs_player
- test_overview_is_empty_before_any_data
- test_overview_records_and_form
- test_overview_point_metrics
- test_overview_ratio_without_unforced_errors_shows_the_count
- test_serve_is_empty_before_any_points
- test_serving_metrics_and_breakdown
- test_returning_metrics_and_breakdown
- test_serving_fault_modes_split_by_serve_number
- test_the_returning_side_reports_no_fault_modes
- test_serve_outcome_shares_split_by_serve_number
- test_return_outcomes_split_by_serve_number
- test_serve_trend_runs_oldest_to_most_recent
- test_serve_trend_keeps_only_the_most_recent_matches
- test_rally_is_empty_before_any_points
- test_rally_stats
- test_rally_length_buckets_are_reported_whole
- test_point_ender_styles
- test_serve_narrows_to_
- test_rally_narrows_to_one_set
- test_phase_counts_onlyp
- test_serve_narrows_to_a_phase
- test_tiebreak_phase_leher_than_zero
- test_returning_breakdown_narrows_with_the_rest_of_the_payload
- test_set_and_phase_nar
- test_overview_ignores_a_scope_it_does_not_offer
- test_filters_reject_a_
- test_opponents_lists_each_matchup_most_met_first
- test_opponents_is_empt
- test_the_opponent_axis_counts_a_match_without_a_result
- test_serve_narrows_to_
- test_rally_narrows_to_one_opponent
- test_overview_narrows_

- test_opponent_set_and_phase_narrow_together
- test_an_opponent_key_rme
- test_an_unknown_opponent_key_is_refused
- test_a_blank_opponent_
- test_match_routes_require_a_session
- test_match_routes_hide
- test_match_serve_stats_leave_the_players_other_matches_out
- test_match_rally_statshes_out
- test_squad_members_requires_a_session
- test_squad_members_is__
- test_squad_members_covers_a_player_with_no_matches
- test_squad_members_is__
- test_squad_members_counts_the_period_and_the_streak
- test_squad_members_streats
- test_squad_members_days_since_the_last_match
- test_squad_members_trabaseline
- test_squad_members_skip_a_match_a_stat_cannot_answer
- test_squad_members_omi
- test_squad_members_baseline_stops_at_ten_matches
- test_squad_members_ratnominator
- test_squad_members_baseline_skips_a_match_with_no_unforced_errors
- test_squad_members_basw_points
- test_squad_members_current_skips_a_match_of_too_few_points
- test_squad_members_sernt_floor
- test_squad_members_omit_a_stat_left_with_one_countable_match
- test_squad_members_perthe_year
- test_squad_members_period_record_ignores_a_match_without_a_result
- test_squad_members_serut_a_serve
- test_squad_trend_requires_a_session
- test_squad_trend_rejec
- test_squad_trend_answers_the_weeks_without_a_roster
- test_squad_trend_buckenday
- test_squad_trend_stops_at_the_windows_first_monday
- test_squad_trend_stops
- test_squad_trend_leaves_an_unresulted_week_null
- test_squad_trend_pools
- test_squad_trend_pools_a_weeks_points_rather_than_its_players
- test_squad_trend_has_nminator
- test_squad_trend_first_serve_pools_the_weeks_serves

Web — vitest (153/153)

src/components/breakdown-bars.test.tsx — 2/2

- renders a zero-width fill, not a dropped style, when the group's ceiling is zero

- still fills proportionally against a real ceiling

src/components/court-heatmap.test.tsx — 5/5

- renders the court diagram and the heat legend
- labels each zone with its share of the total
- labels the single Wide zone once, not on both sidelines
- handles every zone at zero without dividing by zero
- flip swaps net and long between the top and bottom bars

src/components/rally-tab-error.test.tsx — 1/1

- shows a load-failure message instead of an indefinite skeleton

src/components/rally-tab.test.tsx — 9/9

- shows the point-ender style cards, none open by default
- tapping a style card opens its drill
- tapping the open card again collapses it
- tapping a different style card swaps the drill
- switching You won / They won leaves the drill closed
- switching sides while a drill is open closes it, rather than silently swapping its data
- switching sides changes the radar's title between Your and Their
- the Forced/Unforced explanation defines them by pressure, not ball placement
- shows the rally-length distribution, uncaptioned in the middle

src/components/serve-tab-error.test.tsx — 1/1

- shows a load-failure message instead of an indefinite skeleton

src/components/serve-tab.test.tsx — 7/7

- both serve cards render, headline and stack visible, drills collapsed
- the headline % in includes aces, not just the plain-in outcome bucket
- tapping a card's header opens its own drill without affecting the other card
- both cards' drills can be open at the same time
- their serve's drill also shows your return outcome
- switching 1st/2nd serve changes the fault-mode numbers shown
- switching 1st/2nd serve also updates the headline and outcome stack above it

src/components/stat-hero.test.tsx — 1/1

- renders the value and label beside the icon

src/features/home/home.test.tsx — 31/31

- renders the welcome heading
- greets the coach by name once loaded
- links to the profile page
- links to the settings page
- shows the signed-in coach's name once loaded
- the profile link shows the coach's initials once loaded
- the profile link falls back to a generic icon before the coach loads
- shows no welcome-back line when the coach can't be loaded
- shows the install nudge for a signed-in coach
- shows no install nudge before the coach is known
- shows the add-photo nudge for a signed-in coach with no photo
- shows no add-photo nudge once the coach has a photo
- shows no results list until the user types
- filters players as the user types
- shows an empty state when no player matches
- renders a bubble with initials for each player
- clicking a player bubble offers viewing them or starting a match
- choosing View player opens the profile and records them as last-viewed
- 'See all' links to the players roster
- 'Add' opens the add-player sheet directly
- shows no continue-with box when nothing has been viewed yet
- shows the last-viewed player once one has been viewed
- 'View full stats' opens the last-viewed player's profile
- tapping the continue-with player row opens the view/start-match sheet, not straight to stats
- shows the squad's player count
- ranks best and worst form, and shows nothing at all for a legitimate empty needs-attention state
- shows a dismissible card when a player is flagged
- dismissing the attention card hides it

- best/worst form show last-5 and forced:unforced as icon+number pairs, like Continue with
- tapping a squad-snapshot row goes straight to that player's stats page
- renders the bottom nav with Home active

src/features/matches/match-list.test.tsx — 13/13

- renders the matches the API returns
- falls back to a placeholder for a match whose player can't be resolved
- falls back to a placeholder for a match with no opponent name
- marks a finished match with its result, as an icon rather than a hard-to-read letter
- shows an empty state when there are no matches
- reports a failed load instead of rendering an empty list
- tapping a match row navigates to its stats page
- the edit icon hands the match to the caller instead of navigating
- the delete icon hands the match to the caller instead of navigating
- the edit icon is hidden once the match has a result
- filters by opponent, surface, result and date range
- filters by which of the coach's players played the match
- clearing filters restores the full list

src/features/matches/matches-page.test.tsx — 11/11

- renders the page heading and the match list
- offers starting a match when none is in progress
- tapping a match row navigates to its stats page
- the edit icon is hidden once the match has a result
- the edit icon opens the edit sheet, pre-filled, and PATCHes only what changed
- Cancel closes the delete sheet without deleting
- Delete match asks for confirmation before sending anything
- confirming deletes the match and it leaves the list
- a failed delete says so and keeps the match
- the delete names this device, so its own capture claim cannot block it
- a match being captured on another device offers to release and delete anyway

src/features/players/player-details-card.test.tsx — 3/3

- shows date of birth, dominant hand and notes
- shows a dash for every field the player hasn't filled in
- Edit lives in this card and fires the handler

src/features/players/player-list.test.tsx — 9/9

- renders the players the API returns
- flags a player whose transfer code is still out
- the pending-transfer flag is part of the row, not a control of its own
- requests the roster from the API's own origin
- groups players under their surname initial
- filters the roster by the search term
- reports a failed load instead of rendering an empty roster
- prompts to sign in again on a 401, instead of the generic error
- tapping a player hands it to the caller instead of navigating

src/features/players/player-notes-sheet.test.tsx — 7/7

- lists the player's notes, theme shown as an icon rather than a colored tag
- renders nothing when the sheet is closed
- Quick note opens a composer, not a disabled button
- saving a quick note adds it to the list with its chosen theme
- Save note is disabled until something is typed
- deleting a note asks for confirmation before removing it
- Cancel on the delete confirmation keeps the note

src/features/players/player-profile.test.tsx — 20/20

- renders the player's name and avatar
- says not found for a player id that isn't the coach's
- sends the coach back to sign in on a 401, not to the roster
- falls back to a generic message for an unexpected failure
- announces a load failure as an alert, for every branch
- links back to home
- links to the match setup screen
- shows the all-time record and player details side by side
- the Notes button opens the notes sheet

- opens on the Serve tab by default, with the scope panel and recent form above it
- switches to Rally when its toggle is clicked, scope panel and recent form stay put
- Edit opens the sheet pre-filled with the player's current values
- saving Edit PATCHes only the field that changed
- saving Edit with nothing changed just closes, no request sent
- Delete player asks for confirmation before sending anything
- Cancel on the delete step returns to the form without deleting
- confirming Delete removes the player and returns to the roster
- a player already flagged as pending transfer skips straight to the explanation, no request sent
- a code created since the page loaded is caught by the request's own 409, same explanation
- an unexpected delete failure shows the server's own message

src/features/players/player-record-card.test.tsx — 5/5

- shows the all-time win/loss record, no season row
- shows recent form in the same card, all-matches by default
- each result dot is announced as Won or Lost, not just shown by fill
- scopes recent form to the chosen opponent
- hides recent form for an opponent with no fixture, rather than mislabelling the all-matches record

src/features/players/players-page.test.tsx — 10/10

- renders the page heading and the roster
- tapping a player offers viewing them or starting a match
- choosing View player opens the profile and records them as last-viewed
- filters the roster as the coach searches
- renders the bottom nav with Players active
- the labelled row opens the redeem-transfer-code sheet
- the floating action adds a player rather than starting a match
- submitting the add-player form posts every required field
- refuses to submit until date of birth and gender are given
- rejects a date of birth in the future before sending it

src/features/players/stats-scope-panel.test.tsx — 9/9

- starts collapsed, showing the current filters as a summary
- tapping the trigger expands the panel, not a popup
- selecting chips doesn't commit until Apply
- Apply commits the draft, updates the summary, and closes the panel
- Apply commits both Set and Phase, and the summary shows both together
- the opponent picker only shows once scope is 'vs opponent', grouped by most played
- Apply is disabled for 'vs opponent' until an opponent is actually picked
- searching the opponent list filters it
- picking an opponent and applying sets opponentId and echoes their name

src/lib/analytics/fixtures.test.ts — 9/9

- the Rally tab's style-card counts sum to won and lost
- the hero ratio matches the combined forced and unforced counts
- the rally-length histogram sums to won plus lost
- rallyStatsFor returns the base fixture for the default filter
- rallyStatsFor scales won/lost/hero together, so they stay reconciled
- rallyStatsFor never produces a negative count from a small scale
- serveStatsFor returns the base fixture for the default filter
- serveStatsFor's outcome triples still sum to 100 after scaling
- serveStatsFor is deterministic per opponent, not random

Backlog item 7

Test Results Summary

Web (vitest): 88/88 passed across 5 files

Web — vitest (88/88)

src/features/outbox/outbox.test.ts — 34/34

- stamps the payload version rows are written with
- append > stores the entry as pending, stamped with the payload version
- append > keeps the payload opaque — it is stored, not inspected
- append > hands out increasing enqueue order in call order
- append > keeps enqueue order increasing after earlier rows are drained
- append > keeps the first payload for an id already queued
- append > leaves a parked row parked when its entity is appended again
- append > survives concurrent appends of the same id
- append failures > surfaces a quota error as a typed failure instead of rejecting
- append failures > finds a quota error Dexie has wrapped in an aborted transaction
- append failures > distinguishes any other write failure from a quota one
- append failures > reports a payload the browser cannot clone as a write failure
- append failures > keeps its promise to the tap when a listener throws
- append failures > stops notifying a listener that has unsubscribed
- readPending > returns only this coach's pending rows, oldest first
- readPending > is empty for a coach with nothing queued
- readByMatch > returns one match's rows in enqueue order, failed ones included
- remove > drops only the given ids
- remove > ignores an id that is not queued
- remove > drops nothing when given no ids
- removeByMatch > clears one match's rows, parked ones included, and counts them
- removeByMatch > answers zero for a match with nothing queued
- markFailed > parks a match and its dependent points in one call
- markFailed > records why the rows were parked
- markFailed > parks nothing when given no ids
- retryFailed > returns this coach's failed rows to the queue and counts them
- retryFailed > clears the recorded failure along with the status
- deleteOutboxDatabase > takes every coach's rows with it and leaves the queue writable
- watchCounts > emits this coach's pending and failed counts as the queue changes
- watchCounts > ignores writes belonging to another coach
- persistent storage > asks for eviction exemption once, on the first append
- persistent storage > appends fine where the browser offers no storage manager
- persistent storage > appends fine when the request is refused
- persistent storage > appends fine when the storage manager throws instead of returning

src/features/sync/drain.test.ts — 24/24

- flushing > creates the match first, then flushes its points in enqueue order

- flushing > groups interleaved rows by match, matches in first-queued order
- flushing > flushes points straight away for a match created on an earlier run
- flushing > chunks a long queue at the batch ceiling
- flushing > closes a batch on the body limit before the count ceiling
- flushing > tolerates a row deleted while the drain is mid-flight
- flushing > sends nothing for a match purged while the drain is mid-flight
- refusals > parks a refused match and its points, with the reason, without blocking others
- refusals > parks a refused point batch with the 422's own words, never retrying it
- refusals > skips every batch of a match whose earlier rows are parked
- refusals > parks a payload this app version cannot map, keeping its cause
- refusals > parks an unmappable match with its own points, before any request
- rows that can never land > discards a match the server says was deleted, without parking anything
- rows that can never land > discards the points of a deleted match and flushes the next one
- rows that can never land > parks a match whose player the server will not accept
- rows that can never land > parks the points of a match the server cannot find
- halting and retrying > halts everything on a 401 and keeps the queue intact
- halting and retrying > halts on a 401 from point ingest too, with no match row ahead of it
- halting and retrying > retries a network failure on the backoff schedule, then succeeds
- halting and retrying > gives up after the schedule and leaves the rows pending
- halting and retrying > gives up on a failing match create and leaves the match and its points pending
- halting and retrying > waits out a 429's Retry-After when it outlasts the backoff step
- halting and retrying > retries the sweep when another tab's upgrade closes the database
- the drain lock > skips, sending nothing, while another drain holds the lock

src/features/sync/flush-triggers.test.ts — 13/13

- registerFlushTriggers > drains at launch, for the recorded owner as this device
- registerFlushTriggers > never drains while the device has no recorded owner
- registerFlushTriggers > drains when connectivity returns
- registerFlushTriggers > drains when the app becomes visible, not when it hides
- registerFlushTriggers > a queue write while online nudges a drain
- registerFlushTriggers > a queue write while offline waits for the online event
- registerFlushTriggers > a burst of triggers coalesces into one drain
- registerFlushTriggers > re-arms after a drain that found the lock held
- registerFlushTriggers > unregistering removes every trigger, the pending debounce included
- retryLostAppend > re-appends an entry lost to a closed database
- retryLostAppend > leaves other write failures alone
- retryLostAppend > gives up once the schedule runs out
- retryLostAppend > a repeat failure for an entry already mid-retry starts no second chain

src/features/sync/match-payload.test.ts — 16/16

- toMatchCreate > produces a body the API's own contract accepts
- toMatchCreate > sends the id the match was minted with

- toMatchCreate > maps the tracked player, not the opponent, to player_id
- toMatchCreate > passes played_on through unchanged — Match already stores it date-only
- toMatchCreate > sends a concrete surface through unchanged
- toMatchCreate > maps an unspecified surface to null
- toMatchCreate > maps a free-text opponent name through unchanged
- toMatchCreate > maps a rostered opponent, which has no name, to null
- toMatchCreate > sends the traditional scoring system through unchanged
- toMatchCreate > sends the mens_grand_slam scoring system through unchanged
- toMatchCreate > sends the womens_grand_slam scoring system through unchanged
- toMatchCreate > sends the no_ad_scoring scoring system through unchanged
- toMatchCreate > sends the match_tiebreak_10_points scoring system through unchanged
- toMatchCreate > sends the short_sets scoring system through unchanged
- toMatchCreate > sends the pro_set scoring system through unchanged
- toMatchCreate > sends the legacy scoring system through unchanged

src/features/sync/point-payload.test.ts — 1/1

- toPointCreate > throws until the capture model carries the score-context stamps

Backlog item 8

Test Results Summary

Web (vitest): 222/222 passed across 32 files

Web — vitest (222/222)

src/components/add-photo-nudge.test.tsx — 4/4

- shows the nudge for a coach with no photo
- shows nothing once the coach already has a photo
- hides itself once dismissed, and stays hidden across a remount
- choosing a photo uploads it and closes the picker

src/components/add-player-body.test.ts — 4/4

- the body the form builds satisfies the generated contract
- rejects a date of birth that isn't a calendar date
- rejects names over 100 characters and notes over 2000
- rejects an unknown gender

src/components/add-player-sheet.test.tsx — 3/3

- adds a player with no photo
- adds a player with a photo, uploaded after creation
- a photo-upload failure still creates the player and closes the sheet

src/components/bottom-nav.test.tsx — 3/3

- renders all four tabs
- marks the current tab active and no other
- switches tabs on click

src/components/breakdown-bars.test.tsx — 2/2

- renders a zero-width fill, not a dropped style, when the group's ceiling is zero
- still fills proportionally against a real ceiling

src/components/install-app-nudge.test.tsx — 4/4

- shows the nudge when not dismissed and not installed
- hides itself once dismissed, and stays hidden across a remount
- never shows once the app is already installed
- 'See how' links to the full install guide

src/components/not-found-page.test.tsx — 2/2

- renders for a URL that matches no route
- links back to home

src/components/photo-avatar.test.tsx — 6/6

- renders initials when no photo is set
- renders a circle for the profile shape
- renders a square for the default (roster) shape
- renders the photo, sized and shaped the same way, once one is set
- shows no camera badge by default
- shows a camera badge when editable, over initials or a photo alike

src/components/photo-picker-sheet.test.tsx — 4/4

- offers Remove photo only when one is set
- Choose photo opens the file picker, and a picked file reaches onChoose
- Remove photo calls onRemove
- Cancel closes the sheet without choosing or removing anything

src/components/player-avatar.test.tsx — 6/6

- renders the player's initials
- picks the same color for the same id
- renders the photo instead of initials once one is set
- shows no camera badge by default
- shows a camera badge when editable
- falls back to initials if the photo fails to load

src/components/root-layout.test.tsx — 3/3

- supplies a QueryClient every route can use, with no provider of its own
- renders a read from the persisted cache with no connection
- discards a snapshot an earlier app version persisted

src/features/home/home.test.tsx — 31/31

- renders the welcome heading
- greets the coach by name once loaded
- links to the profile page
- links to the settings page

- shows the signed-in coach's name once loaded
- the profile link shows the coach's initials once loaded
- the profile link falls back to a generic icon before the coach loads
- shows no welcome-back line when the coach can't be loaded
- shows the install nudge for a signed-in coach
- shows no install nudge before the coach is known
- shows the add-photo nudge for a signed-in coach with no photo
- shows no add-photo nudge once the coach has a photo
- shows no results list until the user types
- filters players as the user types
- shows an empty state when no player matches
- renders a bubble with initials for each player
- clicking a player bubble offers viewing them or starting a match
- choosing View player opens the profile and records them as last-viewed
- 'See all' links to the players roster
- 'Add' opens the add-player sheet directly
- shows no continue-with box when nothing has been viewed yet
- shows the last-viewed player once one has been viewed
- 'View full stats' opens the last-viewed player's profile
- tapping the continue-with player row opens the view/start-match sheet, not straight to stats
- shows the squad's player count
- ranks best and worst form, and shows nothing at all for a legitimate empty needs-attention state
- shows a dismissible card when a player is flagged
- dismissing the attention card hides it
- best/worst form show last-5 and forced:unforced as icon+number pairs, like Continue with
- tapping a squad-snapshot row goes straight to that player's stats page
- renders the bottom nav with Home active

src/features/matches/match-list.test.tsx — 13/13

- renders the matches the API returns

- falls back to a placeholder for a match whose player can't be resolved
- falls back to a placeholder for a match with no opponent name
- marks a finished match with its result, as an icon rather than a hard-to-read letter
- shows an empty state when there are no matches
- reports a failed load instead of rendering an empty list
- tapping a match row navigates to its stats page
- the edit icon hands the match to the caller instead of navigating
- the delete icon hands the match to the caller instead of navigating
- the edit icon is hidden once the match has a result
- filters by opponent, surface, result and date range
- filters by which of the coach's players played the match
- clearing filters restores the full list

src/features/matches/match-setup.test.tsx — 4/4

- renders the setup form for the player
- the start match button is disabled — nowhere for it to go yet
- cancel returns to the player profile
- says not found instead of an unnamed form when the player isn't the coach's

src/features/matches/matches-page.test.tsx — 11/11

- renders the page heading and the match list
- offers starting a match when none is in progress
- tapping a match row navigates to its stats page
- the edit icon is hidden once the match has a result
- the edit icon opens the edit sheet, pre-filled, and PATCHes only what changed
- Cancel closes the delete sheet without deleting
- Delete match asks for confirmation before sending anything
- confirming deletes the match and it leaves the list
- a failed delete says so and keeps the match
- the delete names this device, so its own capture claim cannot block it
- a match being captured on another device offers to release and delete anyway

src/features/players/edit-player-body.test.ts — 5/5

- a body with a single changed field satisfies the generated contract

- an empty body — nothing changed — is still valid
- rejects a date of birth that isn't a calendar date
- rejects a name over 100 characters and notes over 2000
- rejects an unknown gender

src/features/players/edit-player-sheet.test.tsx — 7/7

- shows initials when the player has no photo
- choosing a photo uploads it and shows it in place of initials
- removing a photo restores initials
- no Remove photo option when there is no photo to remove
- generating a transfer code shows it once, with a copy button
- shows 'Manage transfer code' and lets the coach cancel a pending one
- generating a new code while one is pending

src/features/players/player-list.test.tsx — 9/9

- renders the players the API returns
- flags a player whose transfer code is still out
- the pending-transfer flag is part of the row, not a control of its own
- requests the roster from the API's own origin
- groups players under their surname initial
- filters the roster by the search term
- reports a failed load instead of rendering an empty roster
- prompts to sign in again on a 401, instead of the generic error
- tapping a player hands it to the caller instead of navigating

src/features/players/player-profile.test.tsx — 20/20

- renders the player's name and avatar
- says not found for a player id that isn't the coach's
- sends the coach back to sign in on a 401, not to the roster
- falls back to a generic message for an unexpected failure
- announces a load failure as an alert, for every branch
- links back to home
- links to the match setup screen
- shows the all-time record and player details side by side

- the Notes button opens the notes sheet
- opens on the Serve tab by default, with the scope panel and recent form above it
- switches to Rally when its toggle is clicked, scope panel and recent form stay put
- Edit opens the sheet pre-filled with the player's current values
- saving Edit PATCHes only the field that changed
- saving Edit with nothing changed just closes, no request sent
- Delete player asks for confirmation before sending anything
- Cancel on the delete step returns to the form without deleting
- confirming Delete removes the player and returns to the roster
- a player already flagged as pending transfer skips straight to the explanation, no request sent
- a code created since the page loaded is caught by the request's own 409, same explanation
- an unexpected delete failure shows the server's own message

src/features/players/players-page.test.tsx — 10/10

- renders the page heading and the roster
- tapping a player offers viewing them or starting a match
- choosing View player opens the profile and records them as last-viewed
- filters the roster as the coach searches
- renders the bottom nav with Players active
- the labelled row opens the redeem-transfer-code sheet
- the floating action adds a player rather than starting a match
- submitting the add-player form posts every required field
- refuses to submit until date of birth and gender are given
- rejects a date of birth in the future before sending it

src/features/players/redeem-transfer-code-sheet.test.tsx — 4/4

- redeeming a valid code shows the transferred player and links to their profile
- shows the server's message when the code is unknown or already used
- shows the server's message when the player is already on this coach's roster
- a 429 shows a retry countdown

src/features/profile/profile.test.tsx — 8/8

- renders the coach's name and email

- shows initials when no photo is set
- links back to home
- editing the name PATCHes only display_name and updates live
- saving with nothing changed closes without a request
- choosing a photo uploads it and shows it in place of initials
- Remove photo is offered only once one is set, and clears it
- no Remove photo option when there is no photo to remove

src/features/pwa/install-guide.test.tsx — 6/6

- shows iOS steps on an iPhone
- shows Android steps on an Android phone
- shows a generic note on a device that isn't iOS or Android
- shows a confirmation instead of steps once already installed
- always shows what happens if signal is lost
- links back to settings

src/features/settings/settings.test.tsx — 6/6

- links back to home
- the Match Defaults row links to its category page
- the Account & Security row links to its category page
- the App row links to its category page
- the Support row links to its category page
- offers Sign Out at the root, for fast single-tap access

src/features/stats/stats-page.test.tsx — 6/6

- shows the squad headline with an icon per stat
- leaderboard defaults to Record order, sub-threshold player dimmed at the bottom
- switching the leaderboard metric reorders the ranked rows
- search filters the leaderboard
- the trend card's metric selection is independent of the leaderboard's
- shows the trend caption

src/lib/date-of-birth.test.ts — 6/6

- accepts an ordinary date of birth
- requires a value

- rejects a future date
- accepts today
- rejects more than 120 years ago
- holds the boundary in a timezone ahead of UTC

src/lib/device-platform.test.ts — 9/9

- detects an iPhone from its user agent
- detects an iPad from its user agent
- detects an iPad browsing in desktop mode by its touch support
- does not mistake an actual Mac (no touch) for an iPad
- detects Android
- falls back to other for a desktop browser
- isRunningInstalled is false when neither platform check matches
- isRunningInstalled is true when display-mode is standalone
- isRunningInstalled is true on iOS Safari's navigator.standalone

src/lib/downscale-image.test.ts — 4/4

- returns a smaller file when encoding succeeds at the preferred type
- falls back to JPEG when the canvas cannot encode WebP
- keeps the original file when the canvas cannot get a 2d context
- keeps the original file when the re-encode is not actually smaller

src/lib/last-viewed-player.test.ts — 4/4

- returns null when nothing has been viewed yet
- returns the id most recently set
- records when the player was set as last-viewed
- a pre-timestamp plain-string value reads as null rather than throwing

src/lib/player-form.test.ts — 7/7

- playerFormValuesFrom maps every field, falling back for nullable ones
- validatePlayerForm requires a name, a valid date of birth, and a gender
- buildPlayerUpdateBody is empty when nothing changed
- buildPlayerUpdateBody includes only the fields that changed
- buildPlayerUpdateBody trims names and notes before comparing and sending
- buildPlayerUpdateBody sends null for notes cleared back to empty

- buildPlayerUpdateBody can include several changed fields at once

src/lib/pwa-install-prompt.test.ts — 2/2

- has not been dismissed until dismissInstallPrompt is called
- stays dismissed once set

src/route-tree.test.tsx — 9/9

- /players renders the roster
- /players/$playerId renders that player's profile, not the roster
- redirects to sign-in on a confirmed 401, instead of rendering the protected route
- renders the protected route through a non-401 session-check failure
- renders the protected route when the session check is rate-limited
- renders a protected route from the persisted cache with no connection
- a confirmed 401 leaves no persisted snapshot on the device
- a confirmed session claims a device nothing has claimed yet
- a confirmed 401 keeps the points that never synced

Backlog item 9

Test Results Summary

API (pytest): 59/59 passed across 1 file

API — pytest (59/59)

tests/scripts/test_import_sample_data.py — 59/59

- test_blank_to_none_strips_and_empties (4 cases)
- test_parse_date_reads_an_iso_date
- test_parse_date_treats_a_blank_cell_as_unset
- test_parse_date_refuses_a_malformed_date
- test_parse_bool_reads_every_spelling_of_true [true, TRUE, True, 1, yes, YES]
- test_parse_bool_reads_every_spelling_of_false [false, FALSE, False, 0, no, NO]
- test_parse_bool_refuses_anything_else (5 cases)
- test_a_coach_row_becomes_a_coach
- test_a_coach_email_joins_case_insensitively
- test_a_player_row_carries_its_optional_columns
- test_a_players_blank_columns_land_as_null
- test_a_player_gets_the_username_the_route_would_mint
- test_players_whose_names_fold_alike_get_distinct_usernames
- test_a_match_row_links_to_its_player_and_carries_its_columns
- test_a_matchs_blank_columns_land_as_null
- test_a_match_row_with_no_played_on_names_its_line
- test_a_match_row_with_a_malformed_date_names_its_line
- test_a_match_row_with_an_unknown_enum_value_names_its_line
- test_a_match_row_naming_an_unknown_player_names_its_line
- test_a_match_with_no_match_key_still_imports
- test_a_point_row_becomes_a_point_on_its_match
- test_a_point_row_carries_the_columns_the_other_shape_leaves_blank
- test_a_point_row_breaking_a_schema_rule_names_its_line
- test_a_point_row_with_an_unreadable_boolean_names_its_line
- test_a_point_row_with_an_unknown_match_key_names_its_line
- test_match_sets_are_recomputed_only_for_matches_with_points
- test_an_import_with_no_points_file_writes_no_points
- test_a_points_file_with_only_a_header_writes_no_points
- test_the_summary_counts_what_it_wrote
- test_the_same_csvs_twice_write_one_copy_of_everything
- test_a_re_run_keeps_each_players_username
- test_a_re_run_leaves_a_stored_point_untouched
- test_a_point_whose_sequence_is_taken_is_skipped_not_rewritten
- test_a_re_run_adds_only_the_rows_the_csvs_gained
- test_a_re_run_refreshes_a_corrected_player_and_match
- test_a_re_run_rehashes_the_coaches_it_reused
- test_two_matches_one_day_apart_only_by_opponent_stay_two

- test_a_re_run_restates_the_set_scores_of_a_match_that_gained_points
- test_a_repeated_row_inside_one_points_file_lands_once
- test_more_points_than_one_statement_can_carry_still_land
- test_main_imports_from_the_csv_dir_and_prints_the_password
- test_main_says_nothing_about_points_when_there_is_no_points_file